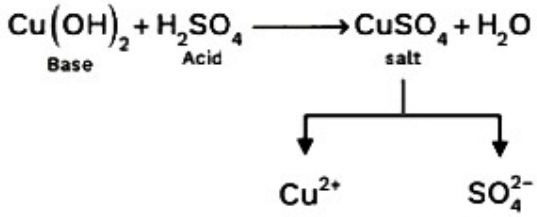
Qualitative Analysis

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Qualitative Analysis

Analysis of substance is carted out in order to establish its qualitative and quantitative chemical composition. Qualitative analysis deals with identifying the components of a substance. Inorganic salts are formed due to neutralization of acids and bases. So a salt comprises of two parts a cation or basic radical contributed by the base and an anion or acidic radical contributed by the base and an anion or acidic radical contributed by an acid. A charged atom or groups of atoms which participates in chemical reactions.



1. Basic radical	2. Acidic radical
Positive radical	**- Basic radical**
Negative radical	**- Acid radical**

1. Basic radical or cation: It is the radical having positive charge. The magnitude of charge depends upon the base in which is in combination with OH^- ions or formally of the basic oxide . $Pb^{+2}, Cu^{+2}, Al^{+3}, Fe^{+3}, Zn^{+2}, Ni^{+2}, Ca^{+2}, Ba^{+2}, Mg^{+2}, NH_4^+$
2. Acid radical or Anion: It is the radical having negative charge. The magnitude of charge depends upon the acid in which it is in combination with hydrogen.
   $CO_3^{2-}, S^{-2}, SO_4^{2-}, NO_2^-, NO_3^-, Cl^-Br^-, I^-$

Qualitative analysis deals with the identification of various constituents present in a chemical mixture. The systematic procedure involves:

1. Preliminary tests
   (a) physical appearance
   (b) Dry heating test
   (c) Charcoal cavity test
   (d) Flame test
   (e) Borax bead test

2. Indicatory test
   (a) Dilute acid tests
   (b) Concentrated acid test

3. Confirmatory tests
   (a) Physical examination
   (b) Gas evolved on heating or by adding any reagent
   (c) Analysis of acid radicals
   (d) Analysis of basic radicals

Types of reactions used in qualitative analysis:

1. **Acid base Reaction**

   Conjugate base of W.A. + $H^+ \longrightarrow$ W.A

   $A^- + H^+ \rightleftharpoons \underset{(W.A)}{HA}$ $\quad \frac{1}{K_a} \uparrow\uparrow$

Example :

$CH_3COO^- + H^+ \longrightarrow CH_3COOH$	$S^{2-} + 2H^+ \longrightarrow H_2S$
$CO_3^{2-} + 2H^+ \longrightarrow H_2CO_3$	$SO_3^{2-} + 2H^+ \longrightarrow H_2SO_3$

Knowledge Booster

Conjugate bases of weak acids are highly inclined towards forming acids. Hence, such reactions have good yield of product.

2. **Precipitation**

   Cation + Anion $\longrightarrow$ Precipitate (sparingly soluble salt)

   $A^+(aq) + B^-(aq) \rightleftharpoons AB(s)$ $\quad \frac{1}{K_{sp}} \uparrow\uparrow$

   Example :

   $Ag^+ + Cl^- \longrightarrow AgCl(s)$

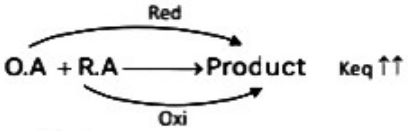
3. **Complexation (formation of complex)**

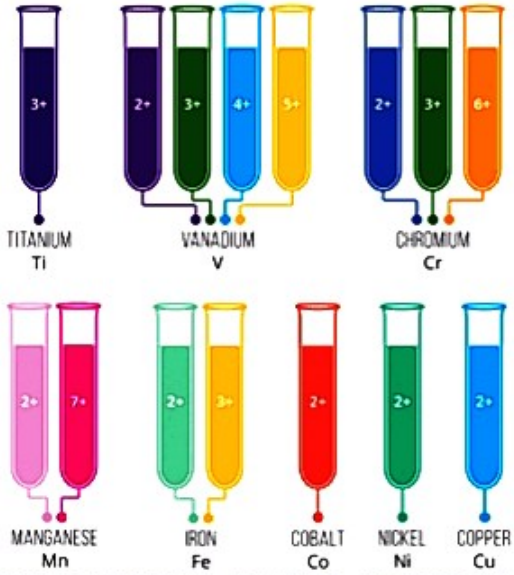
   Cation + Ligands $\longrightarrow$ Complex $\quad K_f \uparrow\uparrow\uparrow$

   Example :

   $Ag^+ + 2NH_3 \longrightarrow [Ag(NH_3)_2]^+$

4. **Redox reaction**





1. **Priliminary test**

   Physical appearance: The physical examination of the unknown mixture involves the study of solubility colour, smell and density.

(A)	Colour	Inference
1.	Light pink or buff colour or dull earth colour	Mn^{2+}
2.	Light green	Fe^{2+}
3.	Dark brown	Fe^{3+}

4.	Pink	Co^{2+}
5.	Bluish – Green	Ni^{2+} or Cu^{2+}
6.	White colour	Absence of Mn^{2+}, Fe^{2+}, Cu^{+}, Ni^{2+}, Cu^{2+}, Cr^{6+}, Fe^{3+}

(B)	**Smell / Order**	**Inference**
1.	Characteristic colour of NH_3	NH_4^+
2.	Smell of rotten eggs H_2S	S^{2-}
3.	Characteristic colour of vinegar (CH_3COOH)	CH_3COO^-

(C)	**Density**	**Inference**
1.	Heavy salt	Pb^{2+} or Ba^{2+}
2.	Light bluffy powder	CO_3^{2-} (Carboante)

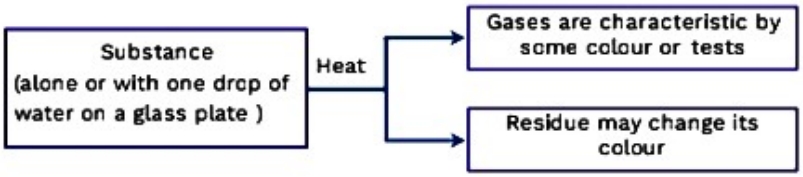
SOLUBILITY CRITERIA USED FOR IDENTIFICATION:

(SOLUBILITY CRITERIA)			
1.	Alkali metal cation & NH_4^+ Generally soluble (M^+)	**1.**	$Cl^-, Br^-, I^-, NO_2^-, NO_3^-, HS^-, HCO_2, HSO_4^-, ClO_4^-$ CH_3COO^- Generally soluble.
2.	Alkali earth metal cation & $Pb^{2+}, Zn^{2+}, Cd^{2+}$ (M^{2+})	**2.**	$CN^-, OH^- \rightarrow$ Generally insoluble
3.	Transition metal cations +2 or +3	**3.**	SO_4^{2-}, CrO_4^{2-} Generally soluble
4.	Cations of pseudo inert gas configuration d^{10}, M^+	**4.**	$CO_3^{2-}, SO_3^{2-}, S^{2-}, C_2O_4^{2-}, PO_4^{3-}$ (A^{-2}) Generally insoluble
Solubility order: $(1) > (2) > (3) > (4)$			

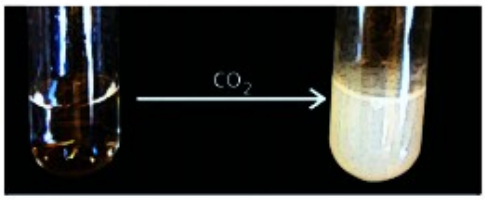
SOLUBILITY CHART

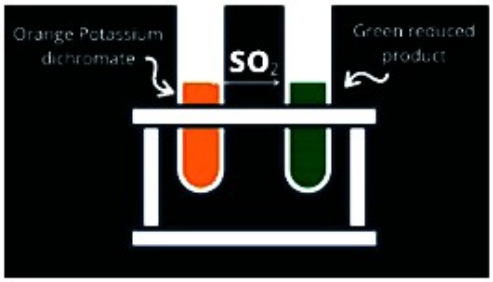
Anion			General solubility	Exception
1.	NO_3^-	Any	Soluble	-------
2.	HS^-, HCO_3^-, HSO_3^-, HSO_4^-	Any	Soluble	-------
3.	NO_2^-	Any	Soluble	$AgNO_3$, is partially soluble
4.	CH_3COO^-	Any	Soluble	CH_3COOAg, $(CH_3COO)_2Hg_2$ & $(CH_3COO)_2Cu_2$ are sparingly soluble
5.	ClO_4^-	Any	Soluble	$KClO_4$, $RbClO_4$, $CsClO_4$, NH_4ClO_4 are sparing soluble.
6.	Cl^-, Br^-, I^-	Any	Soluble	$AgCl, CrCl_3, Hg_2Cl_2$ are insoluble $PbCl_2$ is soluble in not soluble in hot water but insoluble in cold water. $AgBr, CuBr, Hg_2Br_2$ are insoluble $PbBr_2 \rightarrow$ Soluble in hot water but insoluble in cold water. AgI, CuI, HgI, BiI_3 are insoluble. PbI_2 is soluble in hot water but insoluble in cold water.
7.	OH^-, CN^-	Any	Insoluble	Na^+ to Cs^+ & NH_4^+ are soluble.
8.	SO_4^{2-}, CrO_4^{2-}	Any	Soluble	Ag_2SO_4, Hg_2SO_4, $PbSO_4$, $BaSO_4$ $SrSO_4$ are insoluble Ag_2CrO_4, Hg_2CrO_4, $PbCrO_4$, $BaCrO_4$ are insoluble
9.	CO_3^{2-}, $C_2O_4^{2-}$, SO_3^{2-}, S^{2-}	Any	Insoluble	Na^+ to Cs^+ & NH_4^+ are soluble $FeC_2O_4 \rightarrow$ soluble solubility of sulphides of alkali earth metals Al^{3+} & Cr^{3+} is greater than expectation due to Hydrolysis.
10.	PO_4^{3-}	Any	Insoluble	Na^+ to Cs^+ are soluble primary phosphate $(H_2SO_4^-)$ of alkali earth metals are also soluble.

2. Dry heating test:



I	Gases	Inference
	Colourless & Odourless gas ⟶ CO_2 CO_2 : turns lime water milky	CO_3^{2-} (Carbonate)

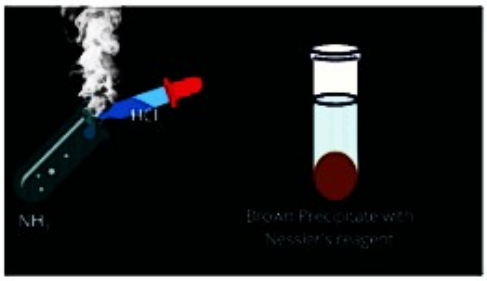




(A) Based on gaseous products and their properties

II	Colourless gas with some odour	Inference
1.	Characteristic odour of NH_3 • NH_3 gives white dense fumes with HCl • NH_3 gas gives brown ppt with Nessler's regent	NH_4^+ (Ammonium)
2.	Smell of rotten eggs • H_2S turns lead acetate paper black	S^{2-}(Sulphide)
3.	Suffocating colour of burning suffer SO_2. SO_2 turns pot dichromate solution green	SO_3^{2-} (sulphite)
4.	Irrataing smell of HCl • HCl gives white dense forms with NH_3 solution • HCl gives white ppt with	Cl^- (Chloride)
5.	Characteristic order of vinegar - CH_3COOH	CH_3COO^- (acetate)





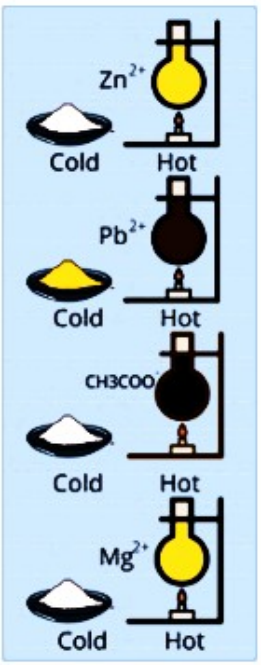
III	Coloured gases	Inference
1.	Greenish-Yellow gas Cl_2 • Cl_2 gives blue – violet colour with starch-iodine paper	Cl^-
2.	Brown vapour → Br_2 • Br_2 gives blue violet colour with starch-iodide paper	Br^- (Bromide)
3.	Violet vapour – I_2 • I_2 gives blue – violet colour with starch paper	I^-
4.	Brown gas-NO_2 • NO_2 gives blue-violet colour with starch-iodide paper immediately • NO_2 turns ferrous sulphate (acidic) solution brown – black	NO_2^- or NO_3^- Nitrite Nitrate

Knowledge Booster

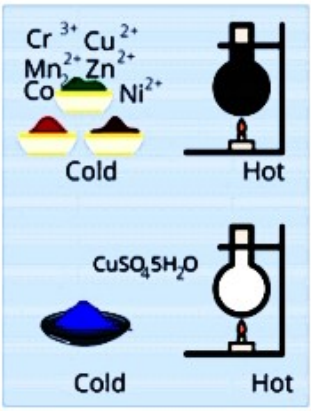
1. Starch paper: Filter paper dipped in starch solution
   Starch + I_2 ⟶ Blue – violet complex
2. Starch – Iodide papers
   Filter paper dipped in starch as well as KI solution
   KI + O.A ⟶ I_2
   Starch + I_2 ⟶ Blue – violet complex
   e.g. $Cl_2 + 2KI \longrightarrow 2KCl + I_2$
   $Br_2 + 2KI \longrightarrow 2KBr + I_2$
3. Starch – Iodate paper
   Filter paper dipped in starch solution & KIO_3 solution
   KIO_3 + R.A ⟶ I_2
   I_2 + Starch ⟶ Blue – violet complex
   e.g. $S^{2-}, SO_3^{2-}, H_2S, SO_2$ etc

(B) Based on residues and their properties

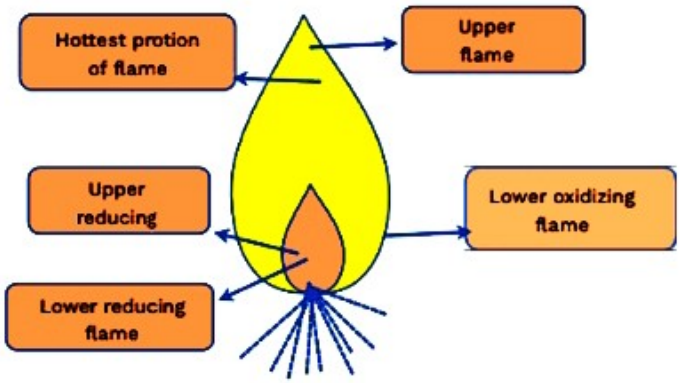
	Residue	Inference
1.	Yellow when hot white when cold	Zn^{2+}
2.	Brown when hot Yellow when cold	Pb^{2+}
3.	White residue becomes black on heating	CH_3COO^-
4.	White residue glucose on heating	Mg^{2+}, Ca^{2+}, Sr^{2+}, Ba^{2+}
5.	Coloured salt becomes brown or black on heating	Mn^{2+}, Fe^{2+}, Co^{2+}, Ni^{2+}, Cu^{2+}, Cr^{3+}, Fe^{3+}



	Residue	Inference
6.	Blue compound becomes white on heating [redacted]	$CuSO_4 5H_2O$ (Cupper sulphate pentahydrate blue vitriol)



(II) Flame Test: (Bunsen Burner)



Luminous flame ⟶ Reducing flame
Non-Luminous flame ⟶ Oxidising flame



Procedure of flame test

1. Make a paste of given compound with one or two drops of HCl
2. Apply the paste over a Pt wire & heat it in the flame & observe change in colour of flame **Reason:** One heating first volatile chlorides are formed. These chlorides volatize off and giving characteristic colour to the flame (according to the cation)

Colour of the flames

Crimson Red	Li^+
Golden Yellow	Na^+
Lilac (or purple)	K^+
Brick-red	Ca^{2+}
Crimson Red	Sr^{2+}
Apple Green	Ba^{2+}
Green with a blue centre	Cu^{2+}

(III) Charcoal Cavity test

Take salt in the cavity (made in Anthracite cool add some solid Na_2CO_3 & 1 or 2 drops of water. Heat it in reducing flame & observe colour of residue.



Knowledge Booster

In s-block Beryllium and magnesium do not give flame test due to high ionized energy and excitation energy.

Salt of Metal + $Na_2CO_3 \xrightarrow{\Delta}$ carbonate of sodium metal $\xrightarrow{\Delta}$ Metallic oxide + CO_2

	Colour of residue	Inference
1	Yellow when hot White when cold	Zn^{2+}
2	White residue glows on heating	Mg^{2+}, Ca^{2+} Sr^{2+}, Ba^{2+}
3	Brown in hot yellow when cold	Pb^{2+}
4	Grey/ Brocons bead	
5	No characteristic colour in residue	Cu^{2+}

Knowledge Booster

Some Equations of the test

Zn^{2+}:

$$ZnSO_4 + Na_2CO_3(s) \xrightarrow[\Delta]{} NaSO_4 + ZnCO_3$$

$$ZnCO_3 \xrightarrow{\Delta} ZnO + CO_3$$

Cu^{2+}:

$$CuSO_4 + Na_2CO_3 \xrightarrow[\Delta]{} CuSO_4 + Na_2SO_4$$

$$\downarrow \Delta$$

$$Cu \leftarrow CuO + CO_2$$

Reddish colour — Reducing flame

Residue which shows cracking noise ⟶ alkali metal halide, NaCl, KCl, KBr, $Pb(NO_3)_2$, SrCl

Residue scrolls up on heating ⟶ Pb_4^{3-}, BO_3^{2-}

(iv) Cobalt nitrate test $Co(NO_3)_2$

This test is used when no clear observations are made in charcoal cavity test Ions which leave white residue in charcoal cavity are tested by this. In case the residue is white during cavity test then one or two drops of cobalt nitrate solution is added and again heated in the oxidizing flame using blow pipe, different metal salts give different coloured mass as given in the table.

	Observation	**Inference**
1	Green residue (Reinmon's green) $CoOZnO$ or $CoZnO_2$	Zn^{2+}
2	Blue residue (Thenard's blue) $CaOAl_2O_3$ or $Co(AlO_2)_2$	Al^{3+}
3	Pink $CoOMgO$ or $CoMgO_2$	Mg^{2+}
4	Bluish – Green $CoOSnO$ or $CoSnO_2$	Sn^{2+}

Knowledge Booster

Some Equations of the test

$ZnSO_4 + Na_2CO_3(s) \xrightarrow[\Delta]{} NaSO_4 + ZnCO_3$

$ZnCO_3 \xrightarrow{\Delta} ZnO + CO_3$

$2Co(NO_3)_2 \xrightarrow{\Delta} 2CoO + 4NO_2 + O_2$

$ZnO + CoO \xrightarrow{\Delta} ZnO.CoO(or\ CoZnO_2)$

(Cobalt zincate) Rinmann's green)

Excess of cobalt nitrate should be avoided due to formation of black cobalt oxide.

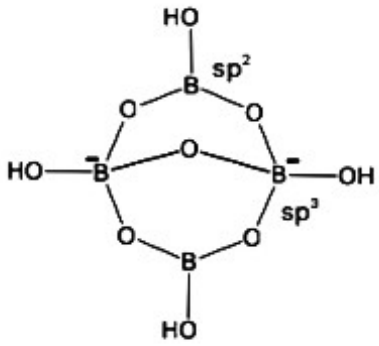
$Co_2O_3(or\ Co_3O_4) \longrightarrow CoO.Co_2O_3$

(V) Borax Bead Test:

Borax ⟶ $Na_2B_4O_4.14H_2O$

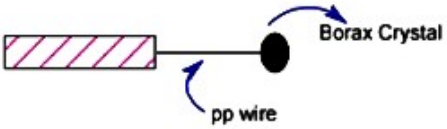
Or

$Na_2[B_4O_5(OH)_4].8H_2O$



Procedure

Take a crystal of borax on a loop of Pt wire



Heat the crystal to form borax bead Touch the bead with salt & heat in flame.

Observe the colour of bead in oxidising as well as reducing flame in hot and cold condition.

Borax $\xrightarrow[\Delta]{}$ Dehydrated Borax + Water

$\downarrow \Delta$

Borax bead

Borax Bead + salt of metal $\xrightarrow[\Delta]{\text{Oxidising flames}}$ Metaborate of characteristic colour

$\downarrow +C$ Reducing flame

Metal $\xleftarrow{+C}$ Lower metaborate

Colour of bead in oxidising flame		Colour of bead in reducing flame		Inference
Hot	**Cold**	**Hot**	**Cold**	
Green	Blue	Colourless	Red Brown Metal	Cu
Yellow	Yellow	Green	Green	Fe
Blue	Blue	Blue	Blue	CO
Green	Green	Green	Green	Cr
Violet	Red	Grey	Grey Metal	Mn
Violet	Brown	Grey	Grey	Ni

Cu

$$Na_2B_4O_7.10H_2O \xrightarrow{\Delta} Na_2B_4O_7 + 10H_2O$$

$$Na_2B_4O_7 \xrightarrow{\Delta} 2NaBO_2 + B_2O_3$$

(Trans barrent / glassy)

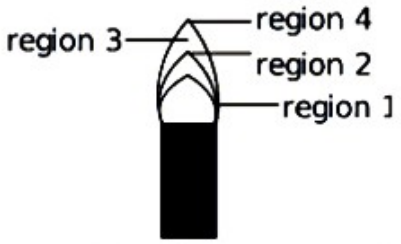
$$CuSO_4 + 2NaBO_2 \longrightarrow Cu(BO_2) + Na_2SO_4$$

Copper(II)metaborate
(Higher metaborate)

Brown
Metal

Q1

The hottest region of Bunsen flame shown in the figure below is:



(A) region 2 **(B) region 3** **(C) region 4** **(D) region 1**

Sol: **A**

Region 2

Q2

Which one of the following ions does not give borax bead test ?

(A) Cr^{3+} **(B) Cu^{2+}** **(C) Mn^{2+}** **(D) Zn^{2+}**

Sol: **D**

Borax bead test is for ions which form coloured salts only. Zinc salts are white so it doesn't give the test

Q3 **Which of the following compound is formed in borax bead test ?**

(A) Orthoborate **(B) Metaborate**

(C) Double oxide **(D) Tetraborate**

Sol: **B**

$$Na_2B_4O_7 \cdot 10H_2O \xrightarrow{\Delta} Na_2B_4O_7 \xrightarrow{\Delta} \underset{\text{Sodium Metaborate}}{2NaBO_2} + B_2O_3$$

On heating, borax loses water of crystallisation and swells up to form fluffy mass. On further heating, it melts to give a clear liquid which solidifies to a transparent glossy bead consisting of sodium metaborate and boric anhydride.

Q4 **Which of the following metal salts gives a red and opaque borax bead in the reducing flame (in cold) ?**

(A) Ni **(B) Fe** **(C) Cu** **(D) Mn**

Sol: **C**

Refer the table given in borax bead test.

The reactions involved are:

$$Na_2B_4O_7 \xrightarrow{\Delta} 2NaBO_2 + B_2O_3$$

$$CuO + B_2O_3 \xrightarrow{\Delta} Cu(BO_2)_2$$

$$2Cu(BO_2)_2 + 4NaBO_2 + 2C \xrightarrow{\Delta} 2Cu(\text{red and opaque}) + 2Na_2B_4O_7 + 2CO\uparrow$$

Q5 **Which one of the following metal salts produces a blue coloured bead in cobalt nitrate charcoal cavity test?**

(A) Zn^{2+} **(B) Mg^{2+}** **(C) Sn^{2+}** **(D) Al^{3+}**

Sol: **D**

Blue residue (Thenard's blue) [redacted] Al^{3+}

$CaOAl_2O_3$ or $Co(AlO_2)_2$

Q6 **If crimson flame is given when an inorganic mixture is tested by flame test, it may be due to the presence of**

(A) potassium **(B) strontium** **(C) barium** **(D) calcium**

Sol: **B**

Colours and ions for flame test

Crimson Red	Li^+
Golden Yellow	Na^+
Lilac (or purple)	K^+
Brick-red	Ca^{2+}
Crimson Red	Sr^{2+}
Apple Green	Ba^{2+}
Green with a blue centre	Cu^{2+}

Q7 **A brick red colour is imparted to Bunsen flame by a:**

(A) Ca salt **(B) Sr salt** **(C) Na salt** **(D) Co salt**

Sol: **A**

Crimson Red	Li^+
Golden Yellow	Na^+
Lilac (or purple)	K^+
Brick-red	Ca^{2+}
Crimson Red	Sr^{2+}
Apple Green	Ba^{2+}
Green with a blue centre	Cu^{2+}

WET TEST:

REACTIONS USED IN WET TESTS:

1. Ion exchange (Acid-Base)

Conjugate base of W.A. + H^+ from S.A. $\longrightarrow$ W.A; $\frac{1}{Ka}$

Conjugate acid of W.B. + OH^- from S.B. $\longrightarrow$ W.B. ; $\frac{1}{K_h}$

Ion exchange for cation:

$$NH_4^+ + OH^- \longrightarrow NH_3 + H_2O$$

C.A. of NH_3 with KOH or NaOH

Ion exchange for anion:

CO_3^{2-}, SO_3^{2-}, S^{2-}, NO_2^-, CH_3COO^- $\Big\}$ $H_2CO_3 \rightarrow H_2O + CO_2\uparrow$; $H_2SO_3 \rightarrow H_2O + SO_2\uparrow$; $\xrightarrow[\text{from S.A.}]{H^+} H_2S\uparrow$; $HNO_2 \rightarrow NO \rightarrow NO_2$; $CH_3COOH\uparrow$ $\Big\}$ H_2SO_4

Cl^-, Br^-, I^-, NO_3^-, SO_4^{2-} $\Big\}$ $\xrightarrow[\text{from S.A.}]{H^+}$ No reaction

2. Redox reaction

Higher Oxidation No. $\xrightarrow{R.A.}$ lower Oxidation no

Higher Oxidation No. $\xrightarrow{O.A.}$ No Oxidation

Lowest Oxidation. No. $\xrightarrow{C.A.}$ Higher Oxidation no.

Lowest Oxidation No. $\xrightarrow{R.A.}$ No Reaction

Stable ion $\xrightarrow{R.A \text{ or } O.A}$ no Reaction.

Redox reactions for anions:

CO_3^{2-} (Stable): O.A → No rxn; R.A → No rxn

$\overset{+4}{SO_3^{2-}}$: O.A → Can be oxidised to $\overset{+6}{SO_4^{2-}}$; R.A → Can be reduced to S ↓

S^{2-} — O.A → Can be oxidised to S ↓; R.A → Can't be reduced

NO_2^- — O.A → Can be oxidised to NO_3^-; R.A → Can't be reduced to NO or N_2

CH_3COO^- Stable — O.A Strong → Can be oxidised to CO_2; R.A → Can't be reduced

Cl^-, Br^-, I^- — O.A → Can be oxidised to X_2 sometimes $\rightarrow IO_3^-$; R.A → Can't be reduced (Behave on R.A.)

$Cl^- < Br^- < I^-$ Behave as R.A.

$Cl_2 > Br_2 > I_2$ Behave as O.A.

NO_3^- — O.A → Can't be oxidised; R.A → Can't be reduced NO or NO_2

SO_4^{2-} — O.A → Can't be oxidised; R.A → Can't be reduced

Redox reactions for cations:

$Hg_2^{2+} \xrightarrow{\text{disproprtionation}} Hg^{2+} + Hg\downarrow$ (Black)

$Cu^{2+} \xrightarrow[S_2O_3^{2-}]{\text{R.A.like}} Cu^+$

$d^9 \qquad\qquad d^{10}$

$Co^{2+} \xrightarrow[\text{agent}]{\text{complexing}} Co^{3+}$

$Cr^{3+} \xrightarrow{\text{Strong OA}} CrO_4^{2-}$ or $Cr_2O_7^{2-}$

$Mn^{2+} \xrightarrow{\text{Strong OA}} MnO_4^{2-} \longrightarrow MnO_4^-$

$Sn^{2+} \xrightarrow{\text{O.A}} Sn^{4+}$

Points to remember

(i) Common Oxidising agents:
Conc. HNO_3
acidic $KMnO_4$
acidic $K_2Cr_2O_7$
(H_2O_2 – OH^-)
Chlorine water
Bromine water
Ozone etc.

(ii) Common Reducing agent:
Some metals Zn, Fe, etc.
SO_2, $SO^{2-}{}_3$, KI, $C_2O_4{}^{2-}$
$S_2O_3{}^{2-}$ etc.

3. PRECIPITION REACTION:

Formation of precipitate (sparing soluble salt) from the ions is known as precipitation.

Any reaction which has more precipitate products will move in forward direction.

$$\underset{(\text{soluble})}{A} + \underset{(\text{soluble})}{B} \rightarrow \underset{(\text{soluble})}{C} + \underset{(\text{soluble})}{D}$$

(Such reactions don't take place)

$$\underset{(\text{soluble})}{A} + \underset{(\text{soluble})}{B} \rightarrow \underset{(\text{Insoluble})}{C} + \underset{(\text{soluble})}{D}$$

(Such reactions move towards product formation)

$$\underset{(\text{soluble})}{A} + \underset{(\text{Insoluble})}{B} \rightarrow \underset{(\text{soluble})}{C} + \underset{(\text{soluble})}{D}$$

(Such reactions don't take place)

Points to remember

All nitrates are soluble in water but nitrates of Bi^{3+} & Sb^{3+} on dilution with excess water gets converted to oxy compounds which will be precipitated.

$$Bi^{3+} + NO_3^- + H_2O \rightleftharpoons \left(BiO^+NO_3^-\right)$$

white ppt.

8

Predict if following reactions will occur or not:

1. $KCl_{(s)} + NaNO_{3(s)} \rightarrow KNO_{3(s)} + NaCl_{(s)}$
2. $KOH_{(s)} + CuCl_2 \rightarrow KCl + Cu(OH)_2$
3. $Na_2SO_2 + K_2CO_3 \rightarrow Na_2CO_3 + K_2SO_4$
4. $Na_2CO_3 + ZnCl_2 \rightarrow ZnCO_3$
5. $K_2SO_4 + BaCl_2 \rightarrow \underset{(l)}{BaSO_4}$
6. $AgNO_3 + KCl \rightarrow AgCl$
7. $Pb(NO_3)_2 + K_2SO_4 \rightarrow PbSO_4$
8. $NiCl_2 + KCN \rightarrow KCl + Ni(CN)_2$
9. $Cu(NO_3)_2 + Na_2S \rightarrow CuS$
10. $Pb(NO_3)_2 + CH_3COONa \rightarrow Pb(CH_3COO)_2 + 2NaNO_3$

Sol:

1. $\underset{s}{KCl} + \underset{s}{NaNO_3} \rightarrow \underset{s}{KNO_3} + \underset{s}{NaCl}$ ×
2. $\underset{s}{KOH} + CuCl_2 \rightarrow KCl + Cu(OH)_2$ ✓
3. $Na_2SO_2 + K_2CO_3 \rightarrow Na_2CO_3 + K_2SO_4$ ×
4. $Na_2CO_3 + ZnCl_2 \rightarrow ZnCO_3$ ✓
5. $K_2SO_4 + BaCl_2 \rightarrow \underset{(i)}{BaSO_4}$ ✓
6. $AgNO_3 + KCl \rightarrow AgCl$ ✓
7. $Pb(NO_3)_2 + K_2SO_4 \rightarrow PbSO_4$ ✓
8. $NiCl_2 + KCN \rightarrow KCl + Ni(CN)_2$ ✓
9. $Cu(NO_3)_2 + Na_2S \rightarrow CuS$ ✓
10. $Pb(NO_3)_2 + CH_3COONa \rightarrow$ ×

Q9 **The carbonate of which of the following cation is soluble in water?**
(A) Na^+ (B) K^+ (C) NH_4^+ (D) Ca^{2+}

Sol: **A,B,C**
Carbonates of first group and Amonium are soluble in nature

ANALYSIS OF ANIONS

Formation of sodium Carbonate extract (SE)

Salt + $NaCO_3$(s)+10 to 20 times water $\xrightarrow{\Delta}$ solution $\xrightarrow{\text{filler}}$ clear solution + sodium extract

$$\underset{\text{salt}}{(M^{x+}A^{y-})} + Na_2CO_3 \rightarrow \underset{\text{ppt}}{(M^{x+}CO_3^{2-})}\downarrow + \underset{\text{soluble}}{(Na^+A^{y-})}$$

filter

Residue ppt $(Mn^+CO_3^{2-})$

Eiltrate sol^n of (Na^+A^{-y})

Classification of anions

Group -I:

This group consists of radical which are detected by dilute H_2SO_4 or dilute HCl.

These are (i) carbonate, (ii) Sulphite, (iii) Sulphide, (iv) Acetate and (v) Nitrite

CO^-, SO^-, S^-, NO^-, $CH\ COO^-$

$H_2CO_3 \rightarrow H_2O + CO_2$

$H_2SO_3 \rightarrow H_2O + SO_2$

$\xrightarrow[\text{S.A.}]{H^+} H_2S$

$HNO_2 \rightarrow NO_2$ (brown)

CH_3COOH

Group II:

This group consists of radicals which are detected by concentrated H_2SO_4

These are (i) Chloride, (ii) Bromide, (iii) Iodide, (iv) Nitrate and

(v) Oxalate

Knowledge Booster

SE is prepared due to two reasons.

(i) Anion is insoluble with given cation so it will be converted to soluble form with sod.in

(ii) Cation interferes with the test of anion so we have to remove the cation (M^{n+})

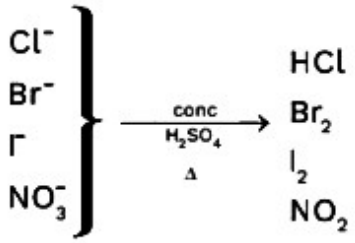
Excess of Na_2CO_3 is neutralised by adding dil. HNO_3 drop by drop.

$CO_3^{2-} + 2H^+ \rightarrow H_2O + CO_2 \uparrow$

Instead of Na_2CO_3 extracted, we can prepare sod. Sulphide, sod sulphite or sod. Phosphate extract.

Points to remember

All the salts of above anions are soluble in dilute HNO_3.



Group III:

The radicals which do not given any characteristic gas with dilute and concentrated H_2SO_4

These are (i) Sulphate, (ii) Phosphate, (iii) Borate and (iv) Fluoride.

These ions are analysed by their characteristic reactions

ANALYSING ANIONS INDIVIDUALLY

(A) Anions with salts soluble in HNO_3

1. CARBONATE (CO_3^{2-})

All the carbonate except alkali metals & NH_4^+ are insolute

All the bicarbonates (HCO_3^-) are soluble

- $CO_3^{2-} \xrightarrow{OA}$ NO reaction
  $\xrightarrow{RA}$ NO reaction

Aq. solution of carbonate is $\xrightarrow{basic}$ due to anionic hydrolysis

$$CO_3^{2-} + H_2O \rightleftharpoons HCO_3^- + OH^-$$

Heating effect

$MCO_3(s) \xrightarrow{\Delta} MO(s) + CO_2(g)$

(M = Be, Mg, Co, Sr, Zn, Pb, Cu etc)

$$Li_2CO_3(s) \xrightarrow{\Delta} Li_2O(s) + CO_2(g)$$

$M_2(CO_3)(s) \xrightarrow{\Delta}$ No reaction

(M = Na to Cs)

$$Ag_2CO_3(s) \xrightarrow{\Delta} Ag_2O(s) + CO_2(g)$$

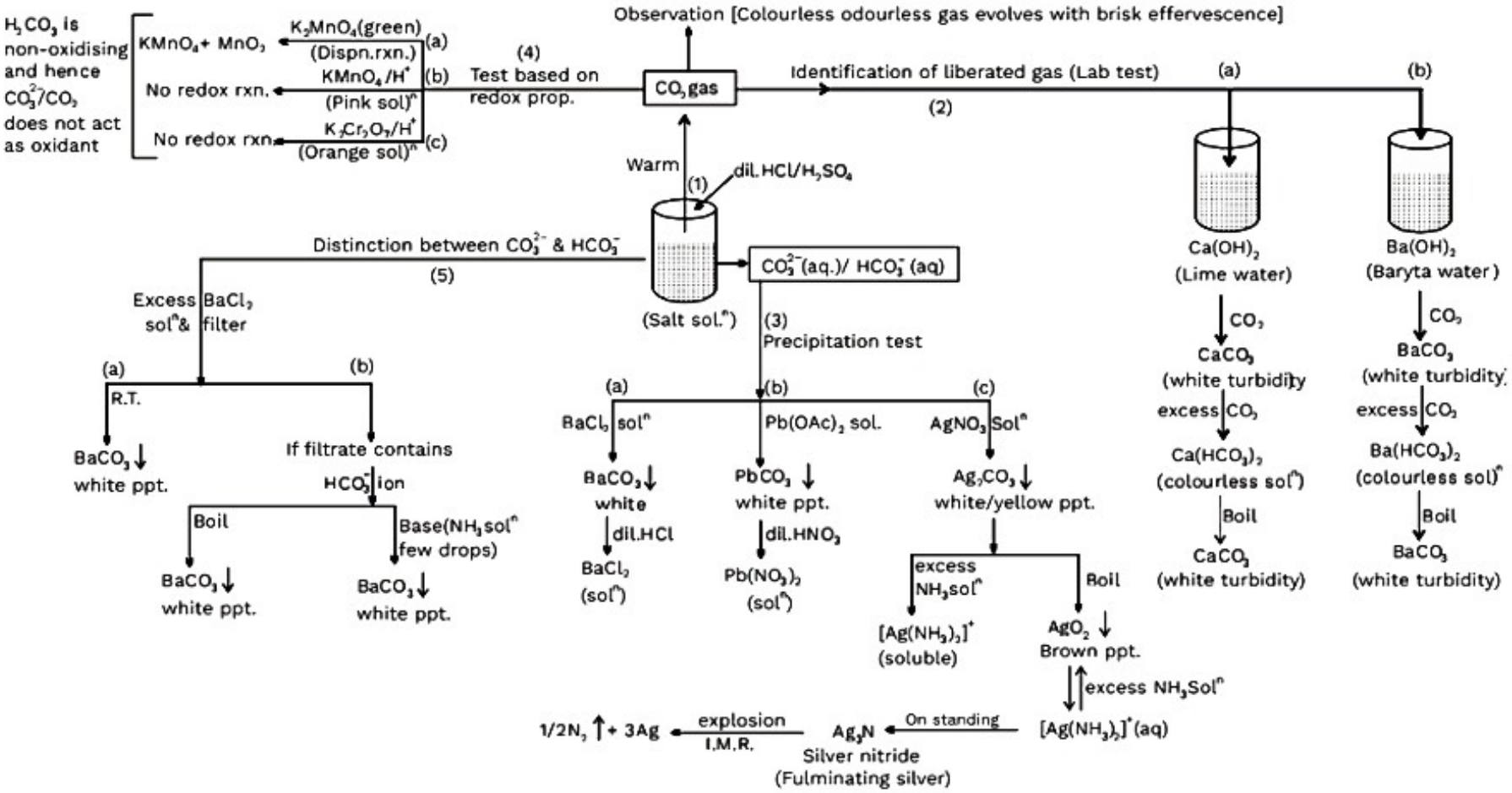
$$2MHCO_3(s) \xrightarrow{\Delta} M_2CO_3(s) + CO_2(g) + H_2O(g)$$

$(M = Na \text{ to } Cs)$ $\quad M^{2+} + 2HCO_3^- \xrightarrow{\Delta} M_2CO_3 \downarrow H_2O + CO_4(g)$

$(M = Mg \text{ to } Ba)$

Carbonate (CO_3^{2-}) / Bicarbonate (HCO_3^-)

1. All CO_3^{2-} salts are water insoluble except alkali metal carbonate salts and $(NH_4)_2CO_3$
2. All metal bicarbonate salts are water soluble. Except $NaHCO_3$(Sparingly soluble)



Note : 1. CO_3^{2-} ion gives ppt. at room temp. whereas HCO_3^- ion forms ppt. of carbonate on boiling.

$$2HCO_3^- \xrightarrow[100°C]{\text{Boil}} CO_3^{2-} + CO_2 + H_2O$$

2. Water insoluble carbonate salts are soluble in soda water (CO_2 + HO_2) and CH_3COOH.

I.M.R. = Inter Molecular Redox

2. SULPHITE (SO_3^{2-})

All sulphites except alkali metals & NH_4^+ are insoluble
All bisulphites are soluble (HSO_3^-)

$$SO_3^{2-} \xrightarrow{OA} SO_4^{2-}$$
$$\downarrow RA \rightarrow Sb$$

Heating effect

$$MSO_3(s) \xrightarrow{\Delta} MO(s) + SO_2(g)$$

$(M = Ca \text{ to } Ba)$

$$M^{2+} + 2HSO_3^- \xrightarrow{\Delta} MSO_3(s) + SO_2 + H_2O$$

$(M = Ca \text{ to } Ca)$

Baryta water:

$$Ba(OH)_2 + SO_2 \rightarrow \underset{\text{white}}{BaSO_3} \downarrow \xrightarrow[So_2 + H_2O]{\text{excess}}$$

$$Ba^{2+} + 2HSO_3 \xrightarrow{\Delta} BaSO_3 + H_2O \text{ (white)}$$

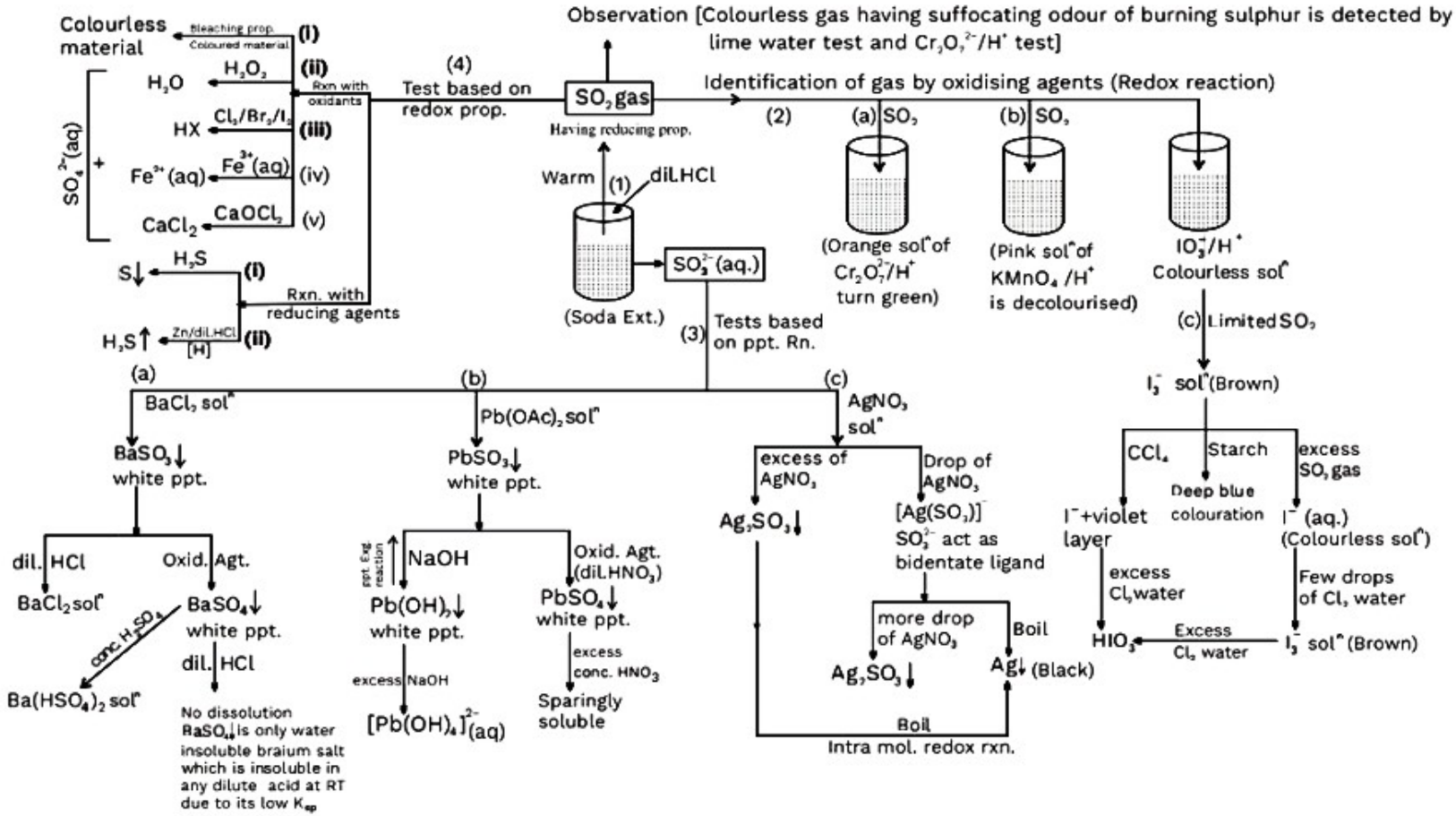
$$BaSO_3 \downarrow \text{(White)} \xrightarrow[\text{or standing}]{[co]} BaSO_4 \downarrow \text{(White)}$$

soluble in dil HNO_3 or dil Hcl

Insoluble in dil HNO_3 or dil HCl

Sulphite (SO_3^{2-})

1. All SO_3^{2-} salt are water insoluble except alkali metal sulphite salts and $(NH_4)_2SO_3$



Note : 1. As SO_2 gas is more soluble in water, therefore effervescence is not observed during evolution of SO_2 gas.

2. Bleaching action due to oxidising property is permanent, whereas bleaching action due to reducing prop. is temporary.
H_2SO_3/SO_2 exhibit better reducing properties and weaker oxidising property.

Q10

SO_2 and CO_2 both turn lime water (X) milky, SO_2 also turns K_2Cr_2O7 /H^+ (Y) green while O_2 is soluble in pyrogallol (Z) turning it black. These gases are to be detected in order by using these reagents. The order is:

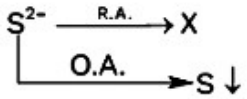
(A) (X), (Y), (Z) **(B) (Y), (X), (Z)**

Sol: **(C) (X), (Z), (Y)** **(D) The correct order cannot be predicted.**

B

SO_2 and CO_2 both turn lime water (X) milky, SO_2 also turns $K_2Cr_2O_7$/H+ (Y) green while O_2 is soluble in pyrogallol (Z) turning it black. These gases are to be detected in order by using these reagents. The order is (Y),(X),(Z). (Y) will absorb SO_2 first. Thus, it will be removed from the mixture. Then (X) will absorb CO_2. Thus, it will also be removed from the mixture. Now (Z) will absorb oxygen.

3. Sulphide (S^{2-}):

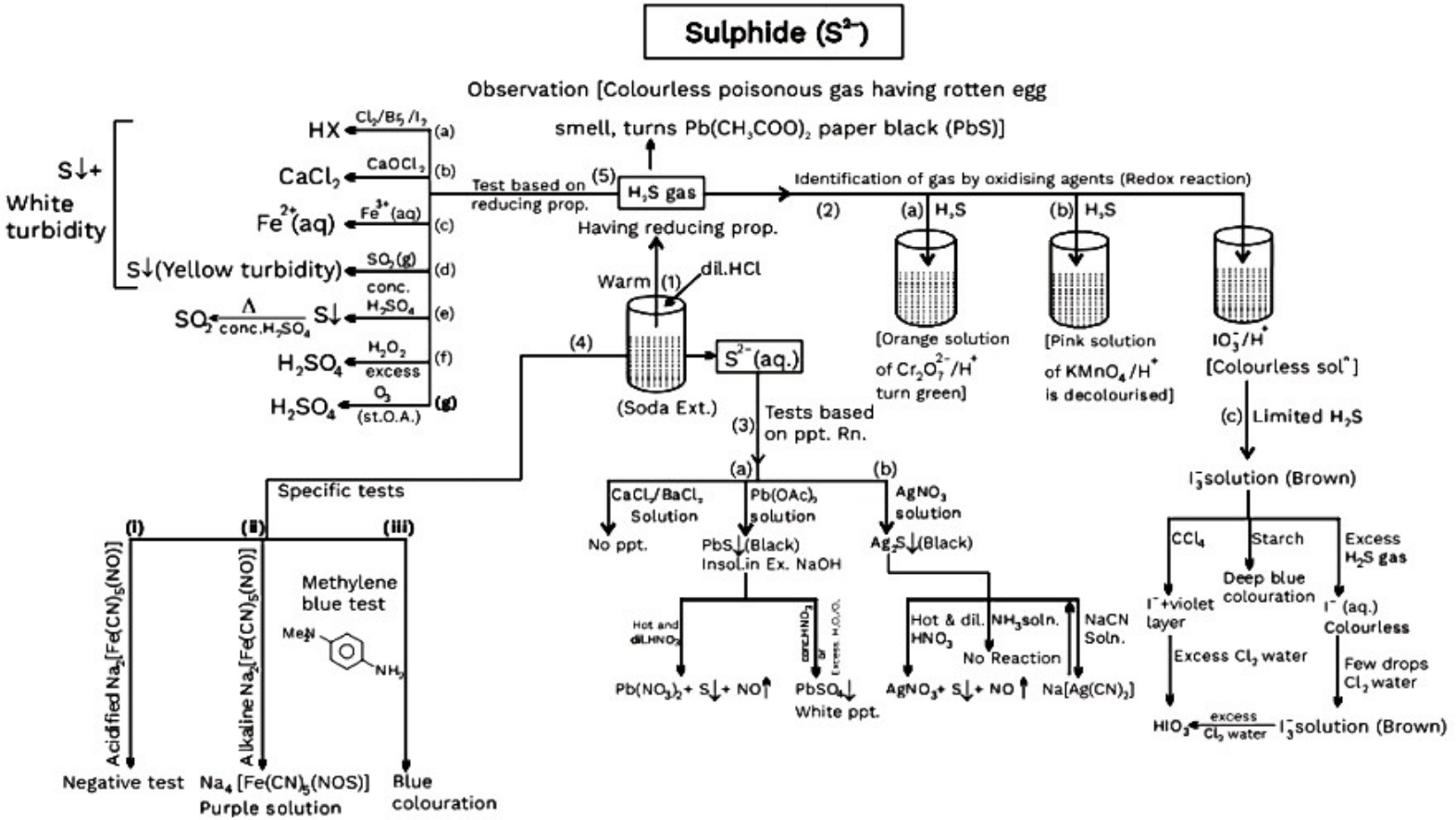


All S^{2-} salts of s-block metals are water soluble and colourless whereas S^{2-} salts of p-block & d-block are water insoluble and usually coloured due to polarisation effect.

S^{2-} salts of Al^{3+}, Cr^{3+} and Mg^{2+} do not exist in aqueous solution because they readily hydrolysed into corresponding metal hydroxide due to their low Ksp, hence they are prepared in dry condition. Fe^{3+} shows redox reaction with S^{2-} ions. [$Fe^{+3} + S^{2-} \longrightarrow Fe^{+2} + S\downarrow$]

FeS↓ is the only black metal sulphide salt decomposed by non-oxidising acid (like dil.HCl/ dil.H_2SO_4) whereas other remaining black sulphide are dissolved in oxidising acid (hot and dil. HNO_3) except HgS↓ (Insoluble in hot and dil. HNO_3 but soluble in aquaregia and S^{2-} salt solution)

$FeS(black) + dil.\ 2HCl\ /\ dil.\ H_2SO_4(non\text{-}oxidising\ acid) \xrightarrow[Kipps\ apparatus]{Rxn.\ occurs\ in} FeCl_2\ /\ FeSO_4 + H2S\downarrow$


Sulphide (S^{2-})
Observation [Colourless poisonous gas having rotten egg smell, turns $Pb(CH_3COO)_2$ paper black (PbS)]
S↓+ White turbidity
HX
$CaCl_2$
Fe^{2+}(aq)
S↓(Yellow turbidity)
SO_2
S↓
H_2SO_4
H_2SO_4
$Cl_2/Br_2/I_2$ (a)
$CaOCl_2$ (b)
Fe^{3+}(aq) (c)
SO_2(g) (d)
conc. H_2SO_4 (e)
H_2O_2 excess (f)
O_3 (st.O.A.) (g)
Test based on reducing prop.
(5) H_2S gas
Having reducing prop.
Identification of gas by oxidising agents (Redox reaction)
(2)
(a) H_2S
[Orange solution of $Cr_2O_7^{2-}/H^+$ turn green]
(b) H_2S
[Pink solution of $KMnO_4/H^+$ is decolourised]
IO_3^-/H^+
[Colourless sol^n]
(c) Limited H_2S
I_3^-solution (Brown)
CCl4
Starch
Excess H_2S gas
I^-+violet layer
Deep blue colouration
I^- (aq.) Colourless
Excess Cl_2 water
Few drops Cl_2 water
HIO_3 ← excess Cl_2 water I_3^-solution (Brown)
Warm (1) dil.HCl
(4)
(Soda Ext.)
S^{2-}(aq.)
(3) Tests based on ppt. Rn.
(a)
(b)
$CaCl_2/BaCl_2$ Solution
No ppt.
$Pb(OAc)_2$ solution
PbS↓(Black) Insol.in Ex. NaOH
$AgNO_3$ solution
Ag_2S↓(Black)
Hot and dil.HNO_3
$Pb(NO_3)_2$ + S↓ + NO↑
conc.HNO_3 or Excess H_2O_2
$PbSO_4$↓ White ppt.
Hot & dil. HNO_3
NH_3soln.
No Reaction
NaCN Soln.
$AgNO_3$ + S↓ + NO↑
$Na[Ag(CN)_2]$
Specific tests
(i)
(ii)
(iii)
Acidified $Na_2[Fe(CN)_5(NO)]$
Alkaline $Na_2[Fe(CN)_5(NO)]$
Methylene blue test
Me_2N
NH_2
Negative test
$Na_4[Fe(CN)_5(NOS)]$ Purple solution
Blue colouration

Note :

1. M^{n+} (p and d-block metal cation) forms metal suphides with H_2S gas depending upon acidic, neutral, basic medium of solution
2. PbS↓ (black) is insoluble in excess NaOH solution or excess NH_4OH solution due to its low K_{sp}
3. Specific test with $Na_2[Fe(CN)_5NO]$ (sodium nitroprusside) requires high conc. of S^{2-} ion, hence neutral H_2S and acidified H_2S gives negative test, whereas alkaline H_2S gives purple colouration.

Points to remember

1. H_2S gas does not give nitroprusside test due to very less availability of S^{2-} ions. But H_2S gas gives nitroprusside test in NH_3(aq.) solution.
2. In presence of SO_3^{2-} ions, red colour compound $Na_4[Fe(CN)_5(NOSO_3)]$ is formed.

4. Nitrite (NO_2^-)

All nitrites, except $AgNO_2$, are soluble

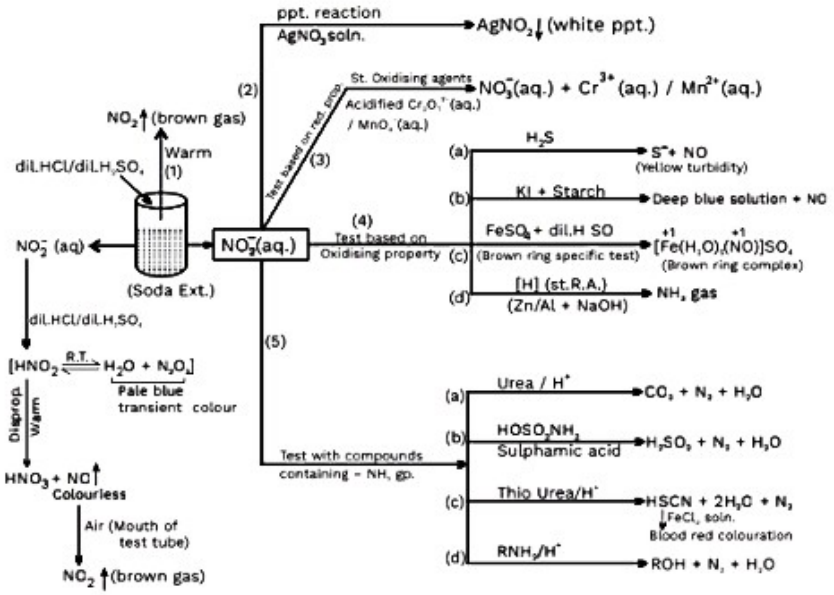
$$NO_2^- \xrightarrow{R.A.} N_2 \text{ or } NO$$
$$\downarrow O.A.$$
$$NO_3^-$$

Heating effect

$$NH_4NO_2(s) \xrightarrow{\Delta} N_2(g) + 2H_2O$$

Nitrite (NO_2^-)

All NO_2^- salts are water soluble except NO_2^- salt of Ag^+, Hg_2^{2+}, Cu^+



Q11 **Colourless salt (A) + dil. H_2SO_4 or CH_3COOH + KI ⟶ blue colour with starch. (A) can be**

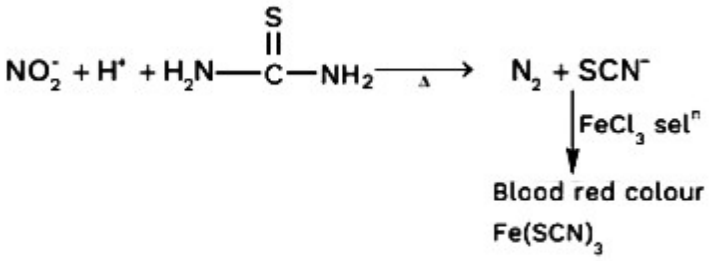
(A) K_2SO_3 **(B) Na_2CO_3** **(C) NH_4NO_2** **(D) NH_4Cl**

Sol: **C**

Refer above table for detailed understanding

Decomposition of NO_2^- :

On heating with urea or thiourea of NH_4Cl or sulphonic acid in acidic medium NO_2^- decomposed to $N_2(g)$



Brown ring Test :

Solution of NO_2 $\xrightarrow[\text{or } CH_3COOH]{\text{add dil } H_2SO_4}$ solution $\xrightarrow{\text{Add freshy prepared, } FeSO_4\text{, solwly by the side of test tube}}$ A brown ring is formed at the Junction of two liquids (Brown)

$$H^+ + NO_2^- \rightarrow HNO_2$$

$$Fe(H_2O)_6^{2+} + NO + SO_4^{2-} \rightarrow \underset{\text{(Brown ring)}}{\left[Fe(H_2O)_5(NO^+)\right]SO_4}$$

H_2O

Q12 **Consider following reaction ; Nitrite + Acetic acid + Thiourea $\longrightarrow N_2 \uparrow + SCN^- + 2H_2O$. Formation of the product in the above reaction can be identified by :**

(A) $FeCl_3$ / dilute HCl, when blood red colour appears.

(B) $FeCl_3$ / dilute HCl, when blue colour appears.

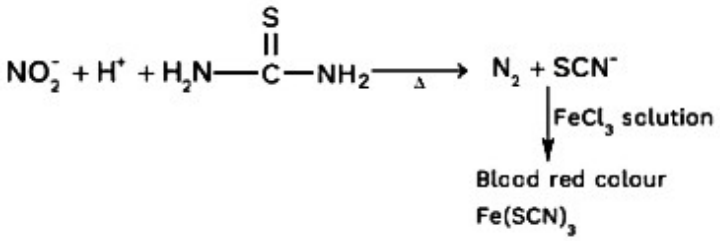
(C) $K_2Cr_2O_7$ / HCl, when green colour appear.

(D) $KMnO_4$ /HCl, when colourless solution is formed

Sol:

A

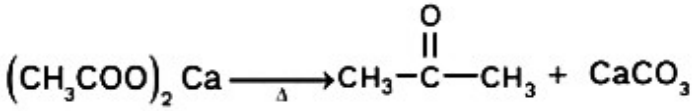
On heating with urea or thiourea of NH_4Cl or sulphonic acid in acidic medium NO_2^- decomposed to $N_2(g)$



5. Acetate (CH_3COO^-)

All acetates, except Ag^+ & $Hg^{2+}$are soluble

Heating effect



Did you Know?

Why we use neutral ferric chloride solution?

Aqueous solution of $FeCl_3$ is acidic due to cationic hydrolysis

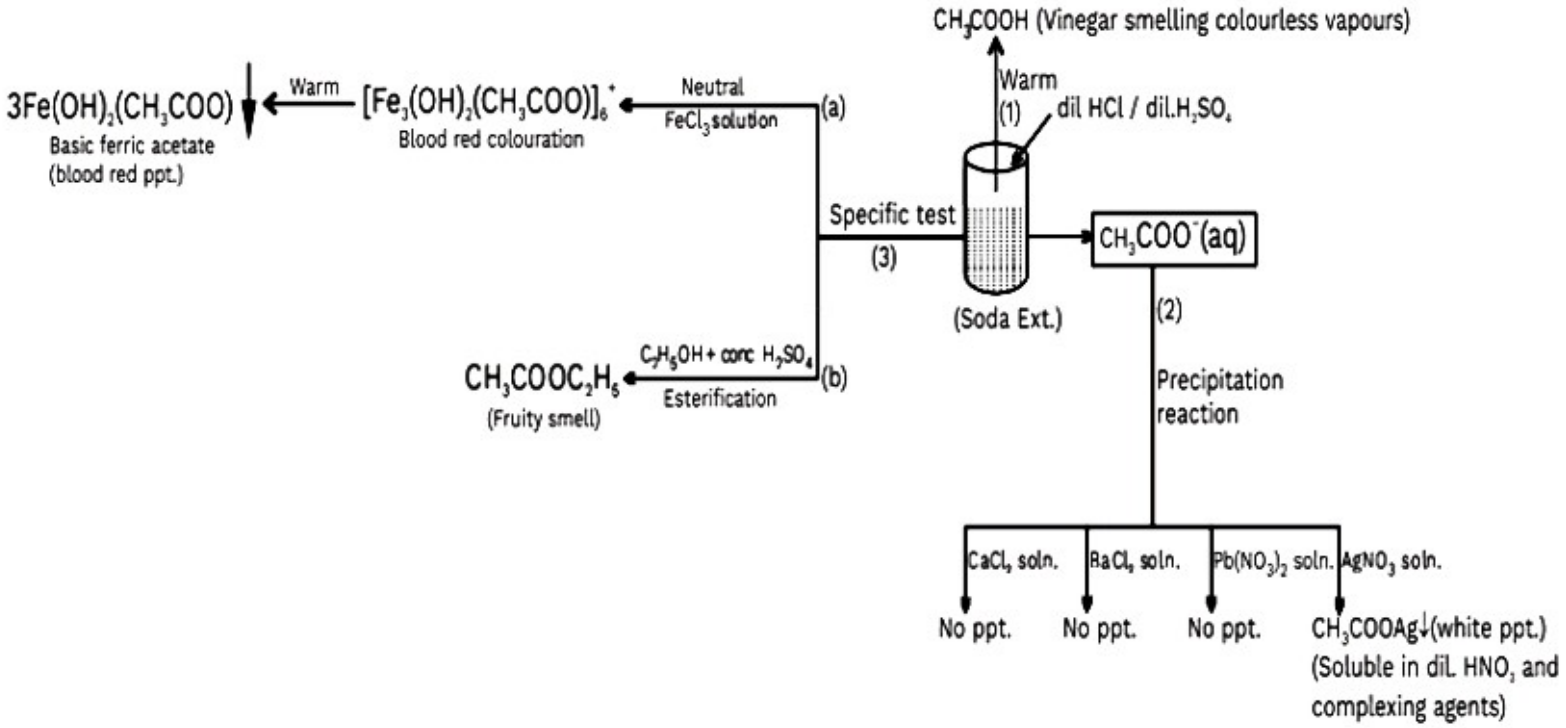
$$Fe^{3+} + H_2O \rightleftharpoons Fe(OH)^{2+} + H^+$$

On adding aq. solution of $FeCl_3$ into $CH_3COO^-H^+$ ions protonate CH_3COO^- so it can't ↓ which is not sufficient to give the test so we have to neu tralise it.

Aq. solution of $FeCl_3$ $\xrightarrow[\text{drop by drop till the turbidity appears \& filter}]{\text{Add } Na_2CO_3 \text{ soln}}$ Neutral ferric chlorides solution.

Acetate (CH_3COO^-)

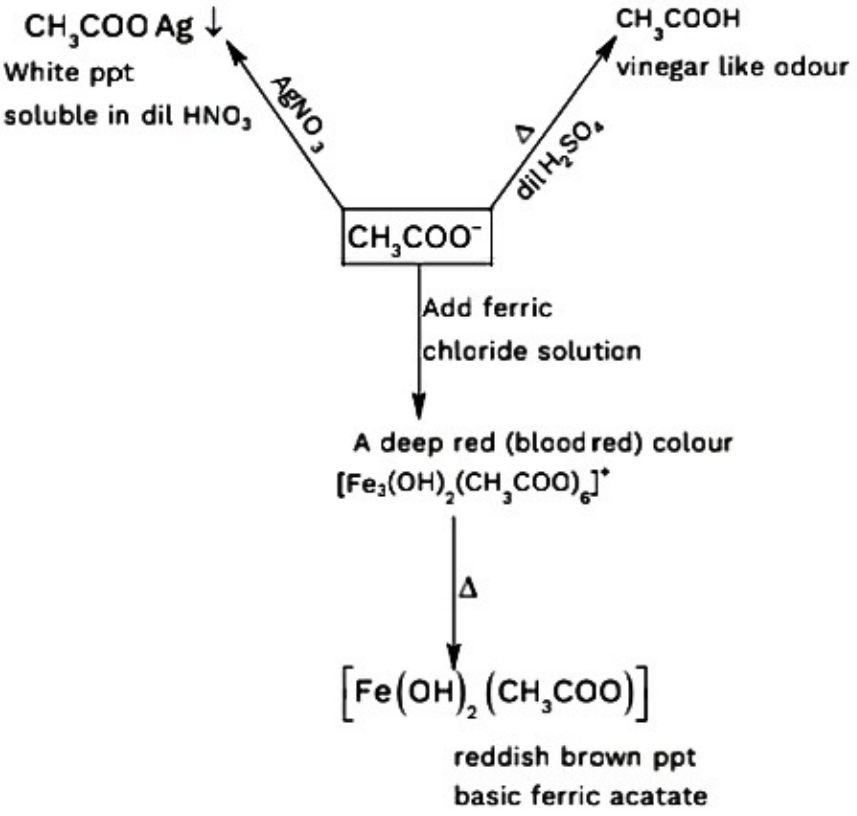
1. All CH_3COO^- salts are water soluble except CH_3COO^- salt of Ag^+ , Hg_2^{2+}, Cu^+



Note : Deep red colouration is obtained by reaction of Fe^{3+} cation with SCN^- (aq) and CH_3COO^- (aq).

Q13 **A mixture when rubbed with dilute acid smells like vinegar. It contains:**
(A) sulphite **(B) nitrate** **(C) nitrite** **(D) acetate**
D

Sol:



Group - III:

- Cl^-, Br^-, I^-, NO_3^- $\xrightarrow[\Delta]{\text{conc } H_2SO_4}$ HCl; Br_2 reddish-brown vapours; I_2 violet vapour; NO_2 reddish brown gas
- $Br^- + H_3PO_4 \xrightarrow[\text{conc.}]{\Delta} HBr$
- $I^- + H_3PO_4 \xrightarrow[\text{conc}]{\Delta} HI$

Points to remember:

AgCl ↓	AgBr ↓	AgI ↓
White soluble in dil NH_3 solution $AgCl + 2NH_3 \downarrow [Ag(NH_3)_2]^+ + Cl^-$	Pale yellow soluble in concentrated NH_3 solution partially soluble in dil NH_3 $AgBr + 2NH_3 \longrightarrow [Ag(NH_3)_2]^+ + Br^-$	Yellow insoluble in concentrated. NH_3 solution

$$AgX + 2CN^- \rightarrow Ag(CN)_2^- + X^-$$

$$AgX + 2S_2O_3^{2-} \rightarrow Ag(S_2O_3)_2^{3-} + x^-$$

6. CHLORIDE (Cl^-)

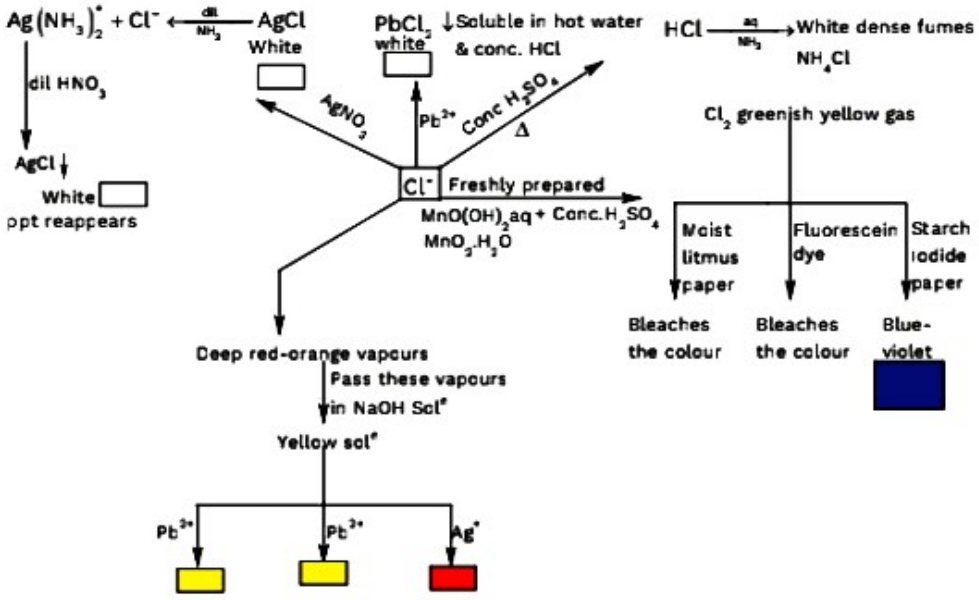
- All chlorides, except AgCl, CuCl, Hg_2Cl, & $PbCl_2$ are soluble:
- $Cl^- \xrightarrow{R.A.} NO$ reaction

$$Cl^- \xrightarrow{O.A.} Cl_2$$

Heating effect

$$2FeCl_G \xrightarrow{\Delta} 2FeCl + Cl$$

$$PbCl_G \xrightarrow{\Delta} PbCl + Cl$$



- AgCl turns yellow by shaking with KI soln^{n} or sod. Assenate soln^{n}

$$AgCl\downarrow + I^- \rightleftharpoons AgI\downarrow + Cl^-$$

White — Yellow

$$AgCl\downarrow + AsO_3^{3-} \longrightarrow Ag_3AsO_3 + 3Cl^-$$

White — Yellow

Chromyl chloride test

Soluble

$$\underset{\text{Soln}}{Cl^-} + \underset{(\text{solid})}{K_2Cr_2O_7} + H_2SO_4 \xrightarrow{\Delta} \underset{\text{deepred-orange}}{CrO_2Cl_2}$$

Vaps. | Pass these vapours in NaOH

$$CrO_2Cl_2 + OH^- \rightarrow \underset{\text{Yellow solution}}{CrO_4^{2-}} + Cl^-$$

Pb^{2+} | Ba^{2+} | Ag^{2+}

$PbCrO_4$ Yellow | $BaCrO_4\downarrow$ Yellow | $Ag_2CrO_4\downarrow$ Brick red

Points to remember

HCl insoluble heavy metal chlorides do not give chromyl chloride test due to unavailability of Cl^- ions. These give chromyl chloride test in the form of S.E. Br^- & I^- do not give chromyl chloride test due to the unstable compounds chromyl bromide. (CrO_2Br_2) & chromyl iodide (CrO_2I_2)

Q14 **Chromyl chloride vapours are dissolved in water and acetic acid and barium acetate solution is added, then:**

(A) the solution will remain colourless.

(B) the solution will become dark green.

(C) a yellow solution will be obtained.

(D) a yellow precipitate will be obtained.

Sol: **D**

$$CrO_2 + OH^- \longrightarrow CrO_4^{2-} \xrightarrow{Ba^{2+}} BaCrO_4\downarrow \text{ (yellow ppt)}$$

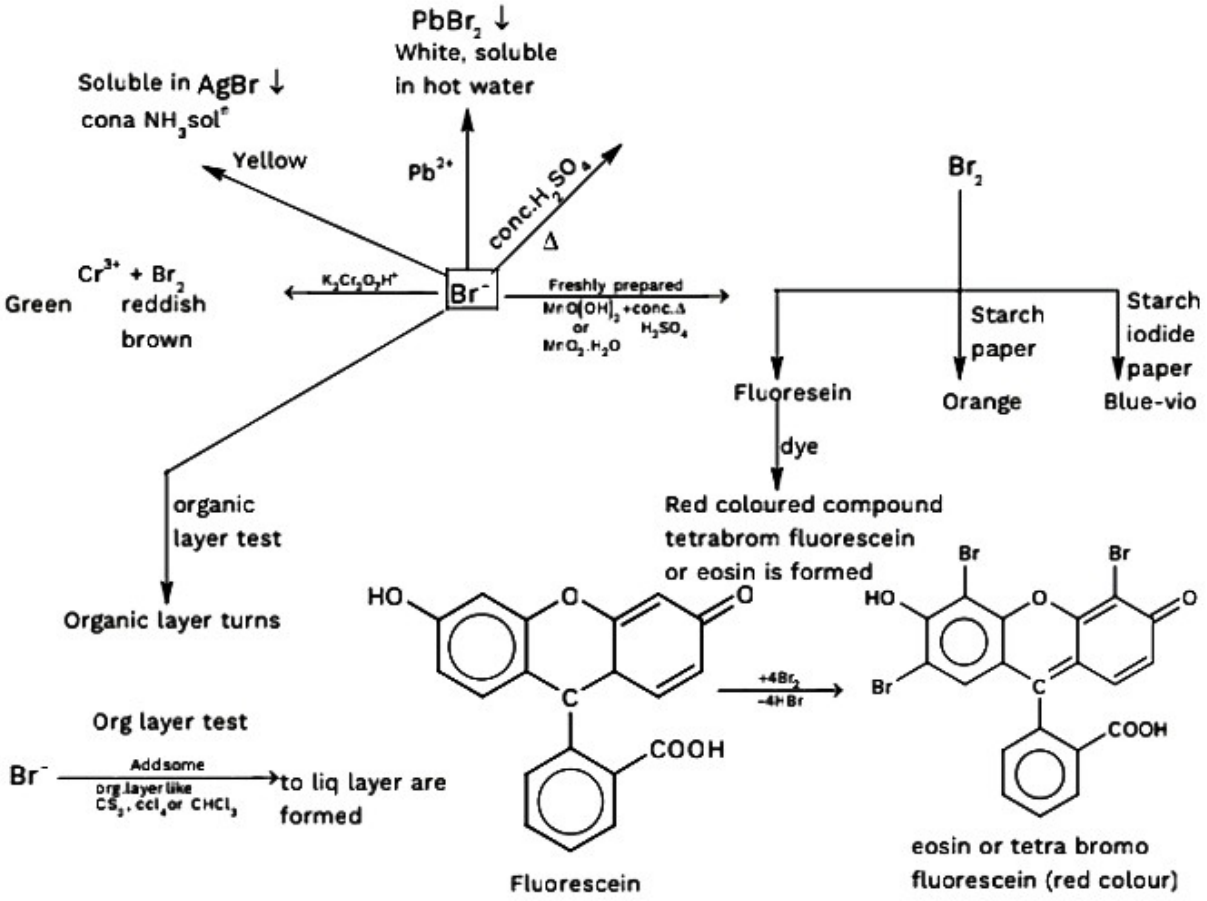
7. BROMIDE (Br^-)

- All bromides except AgBr, CuBr, Hg_2Br_4, $PbBr_2$ are soluble.
- $Br^- \xrightarrow{O.A.} Br_2$

$Br^- \xrightarrow{R.A.}$ NoReaction

Heating effect

$2FeBr_G \xrightarrow[\Delta]{} 2FeBr + Br$



- organic layer turns brown

$Br^- + Cl_2 \rightarrow Br_2 + Cl^-$



Brown covalent compound

- Excess of Cl_2-water should be avoided due to the formation of BrCl

$$Br_2 + Cl_2 \rightarrow 2BrCl$$

8. Iodide (I^-)

- All iodides except AgI, CuI, PbI_2, HgI_2, BiI_3 are soluble.
- $I^- \xrightarrow{R.A.}$ No reaction

$$I^- \xrightarrow{O.A.} I_2 \xrightarrow[\text{paper}]{\text{Starch}} \text{Blue – violet}$$

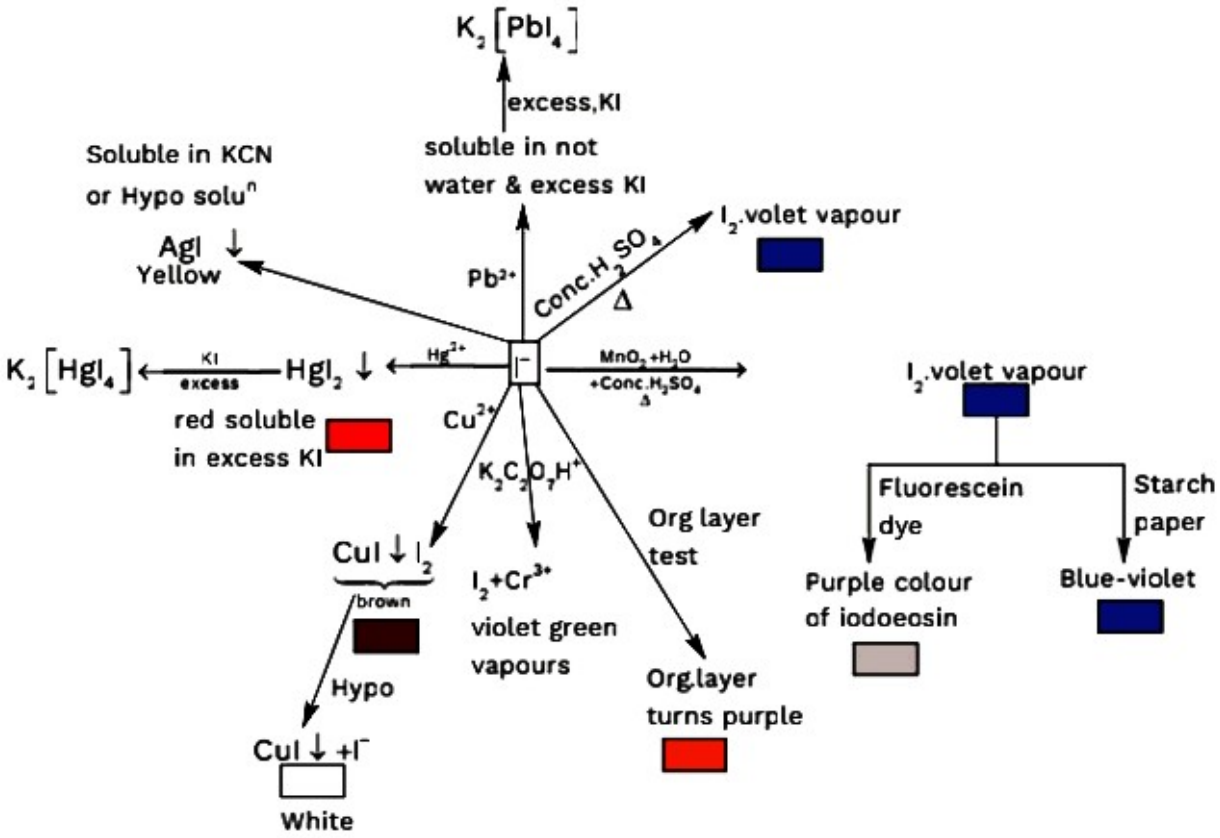
Heating effect

FCl_3, CuI_2, PbI_4 etc are unstable due to reduction.

$$Fe^{3+} + I^- \rightarrow Fe^{2+} + I_2$$

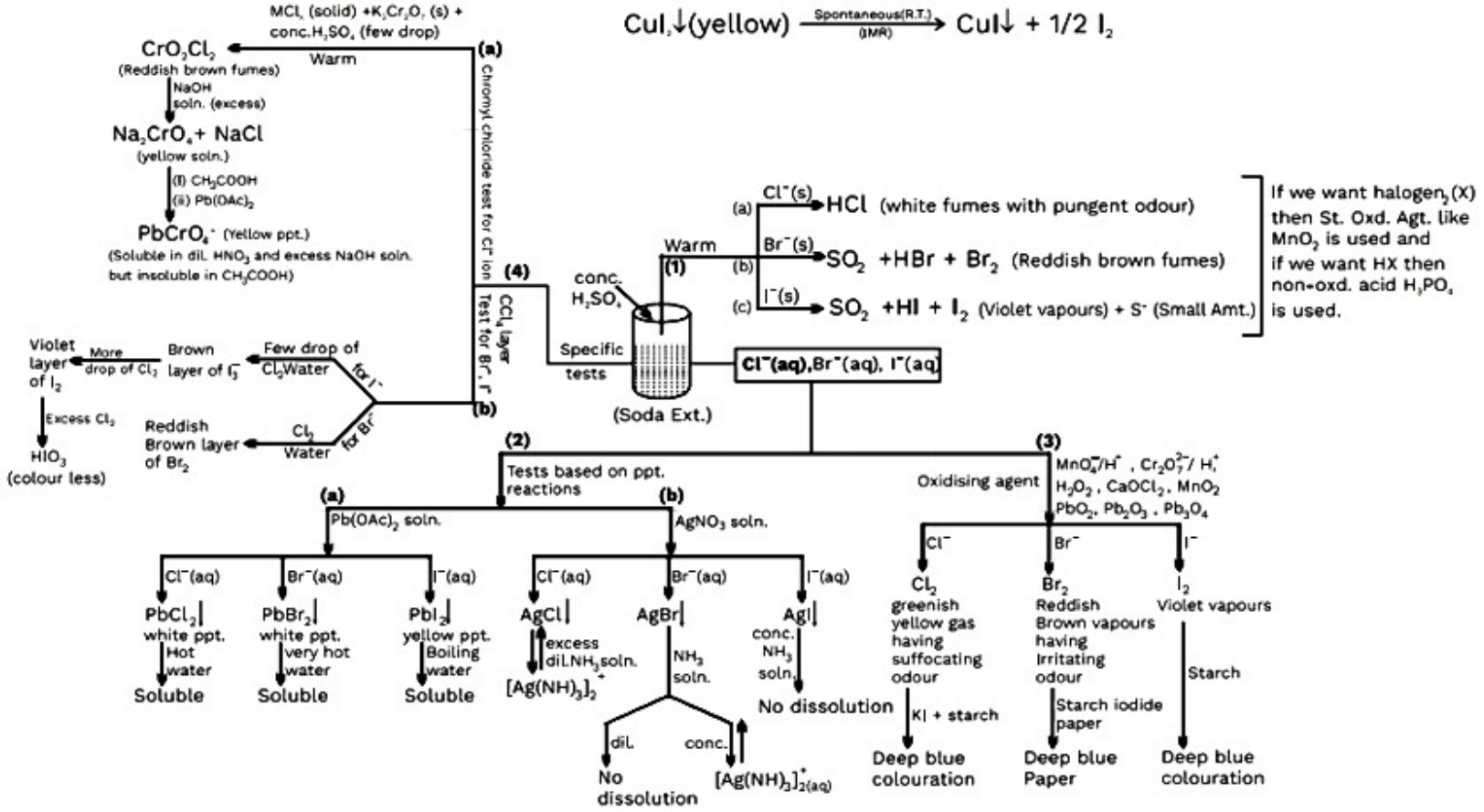
$$Cu^{2+} + I^- \rightarrow \underset{\text{(white)}}{CuI}\downarrow + I_2$$

$$Pb^{4+} + I^- \rightarrow \underset{\text{(yellow)}}{PbI_2}\downarrow + I_2$$



Chloride (Cl^-) , Bromide (Br^-) and Iodide (I^-)

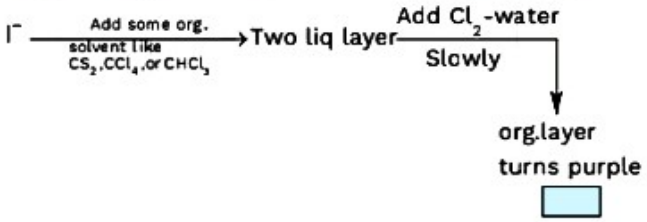
1. All Cl^- salts are water soluble except AgCl ↓, Hg_2Cl_2↓, $PbCl_2$ ↓, CuCl ↓ (all are white ppt.)
2. All Br^- salts are water soluble except AgBr ↓ (pale yellow) $PbBr_2$↓(white), Hg_2Br_2↓(yellow), CuBr↓(white)
3. All I^- salts are water soluble except AgI↓(yellow), PbI_2↓(yellow), HgI_2↓(scarlet red), Hg_2I_2↓(green), BiI_3↓(black), CuI↓(white)



Note : 1. AgCl↓, $PbCl_2$↓, Hg_2Cl_2↓, CuCl↓, $HgCl_2$↓(pred. covt.) $PbCl_4$↓(pred. covt.) $SnCl_4$↓(pred. covt.) do not give chromyl chloride test due to low concentration of Cl^- ion in the solution.

2. CH_3COOH used in chromyl chloride test neutralises excess NaOH otherwise yellow ppt. of $PbCrO_4$ is not formed.

Organic layer test (chloroform layer test)



$I^- + Cl_2 \rightarrow I_2 + Cl_2$

Soluble ion org. layer

Do You Know?

Excess of chlorine water should be avoided due to the Oxidation of I_2 to IO_3^-

$I_2 + Cl_2 + H^+ \rightarrow IO_3^- + Cl^-$

Q15 **When CS_2 layer containing both Br_2 and I_2 (2 : 1) is shaken with excess of chlorine (Cl_2) water, the violet colour due to I_2 disappears and a pale yellow colour appears in the solution. The disappearance of violet colour and appearance of pale yellow colour is due to the formation of :**

(A) I_3^- and Br_2 respectively. **(B) HIO_3 and BrCl respectively.**

(C) ICl and BrCl respectively. **(D) I^- and Br^- respectively.**

Sol: **B**

$5Cl_2 + I_2 + 6H_2O \rightarrow 2HIO_3$ (colourless) $+10HCl$

$Br_2 + Cl_2 \Leftrightarrow 2BrCl$(pale yellow)

Q16 **Which of the following will not give positive chromyl chloride test ?**

(A) Copper chloride, $CuCl_2$. **(B) Mercuric chloride, $HgCl_2$.**

(C) Zinc chloride, $ZnCl_2$. **(D) Anilinium chloride $C_6H_5NH_3Cl$.**

Sol: **B**

Mercuric chloride, $HgCl_2$ will not give positive chromyl chloride test. Chromyl chloride test fails in ease of chlorides of Hg, Sn, Ag, Pb, and Sb.

Q17 **A one litre flask is full of reddish brown bromine fumes. The intensity of brown colour of vapour will not decrease appreciably on adding to the flask some :**

(A) pieces of marble **(B) animal charcoal powder**

(C) carbon tetrachloride **(D) carbondisulphide**

Sol: **A**

A one litre flask is full of brown bromine vapour. The intensity of brown colour of vapour will not decrease appreciably on adding to the flask some pieces of marble because it does not react with it.

9. Nitrate (NO_3^-)

All nitrates are soluble in water

$NO_3^- \xrightarrow{O.A.}$ No rxn

$NO_3^- \xrightarrow{R.A.}$ NO or NO_2

Heating effect

$$2M(NO_3)_2(s) \xrightarrow{\Delta} 2MO(s) + 4NO_2(g) + O_2(g)$$

$(M = Be \text{ to } Ba, Co, Zn, Cu, Pb)$

$$4LiNO_3(s) \xrightarrow{\Delta} Li_2O(s) + 4NO_2(g) + O_2(g)$$

$$2MNO_3(s) \xrightarrow{\Delta} 2MNO_2(s) + O_2(g)$$

$(M = Na, K, Rb, Cs)$

↓

at 500°C

$$4AgNO_3(s) \xrightarrow[\Delta]{} 2Ag_2O(s) + 4NO_2(g) + O_2(g)$$

$$\downarrow \Delta$$

$$4Ag + O_2(g)$$

$$NH_4NO_3(s) \xrightarrow{\Delta} \underset{\substack{\text{laughing} \\ \text{(gas)}}}{N_2O} + 2H_2O$$

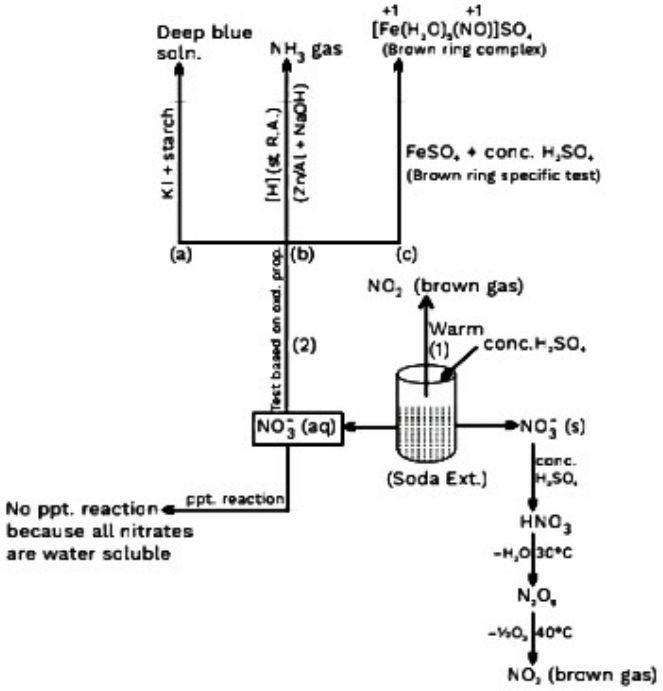
Points to remember

For reduction of nitrate Excellent results are obtained with Devarda's alloy (Zn + Al + Cu) + NaOH

$I_2 + Cl_2 + H^+ \rightarrow IO_3^- + Cl^-$

Nitrite (NO_2^-)

All NO_2^- salts are water soluble.



[With piece of filter paper or copper turnings evolution of reddish brown gas is intensified]

Note: 1. Aq. soln. of $M(NO_3)_2$ / $M(NO_3)_3$ salts (M^{2+} = Ba^{+2}, Sr^{+2}, Pb^{2+}) do not give brown ring test because these cations form insoluble SO_4^{2-} salts with H_2SO_4 used in ring test, formation of ppt. will create turbulence in soln. therefore, brown ring complex is not formed. In such cases, above test can be by first adding Na_2SO_4 soln. Now obtained filtrate can be used for ring test.

2. Ring test for NO_3^- ion in presence of NO_2^-(aq) should not be done however ring test for NO_2^- (aq) ion in presence of NO_3^- (aq) can be performed.

3. HNO_2 exhibits better oxidising character and weak reducing character.

Brown ring test :

$$NO_3^-\ (\text{Sol}^n) \xrightarrow[\text{Sol}^n]{\text{Add freshly prepered } FeSO_4} \text{Solution} \xrightarrow[\text{by the side of the test tube}]{\text{Add conc.} H_2SO_4}$$

$$NO_3^- + H^+ + \underset{(\text{excess})}{Fe^{2+}}(aq) \rightarrow Fe^{3+}(aq) + NO + H_2O$$

Do You Know?

Most of the oxidising agent give blue ring with diphenyl amine.
E.g.

$MnO_4^- - H^+, Cr_2O_7 - H^+, NO_3^- - H^+,$
$NO_2^- - H^+, ClO_3^-, H_2O_2, Fe^{3+}$ etc.

$$Fe(H_2O)_6^{2+} + NO \xrightarrow{SO_4^{2-}} \left[\overset{+}{Fe}(H_2O)_5\left(\overset{+}{NO}\right)\right]SO_4$$

(Brown ring complex)

Properties of brown ring complex

Octahedral complex

sp^3d^2 hybridisation

Unpaired else

Paramagnetic

Its colors is due to electron transfer spectra

10. Sulphate (SO_4^{2-})

All sulphates except Ag^+, Hg_2^{2+}, Pb^{2+}, Ba^{2+}, Sr^{2+} are soluble

All bisulphate HSO_4^- are soluble

$SO_4^{2-} \xrightarrow{O.A.}$ NO reaction

$SO_4^{2-} \xrightarrow{R.A.}$ No reaction

Knowledge Booster

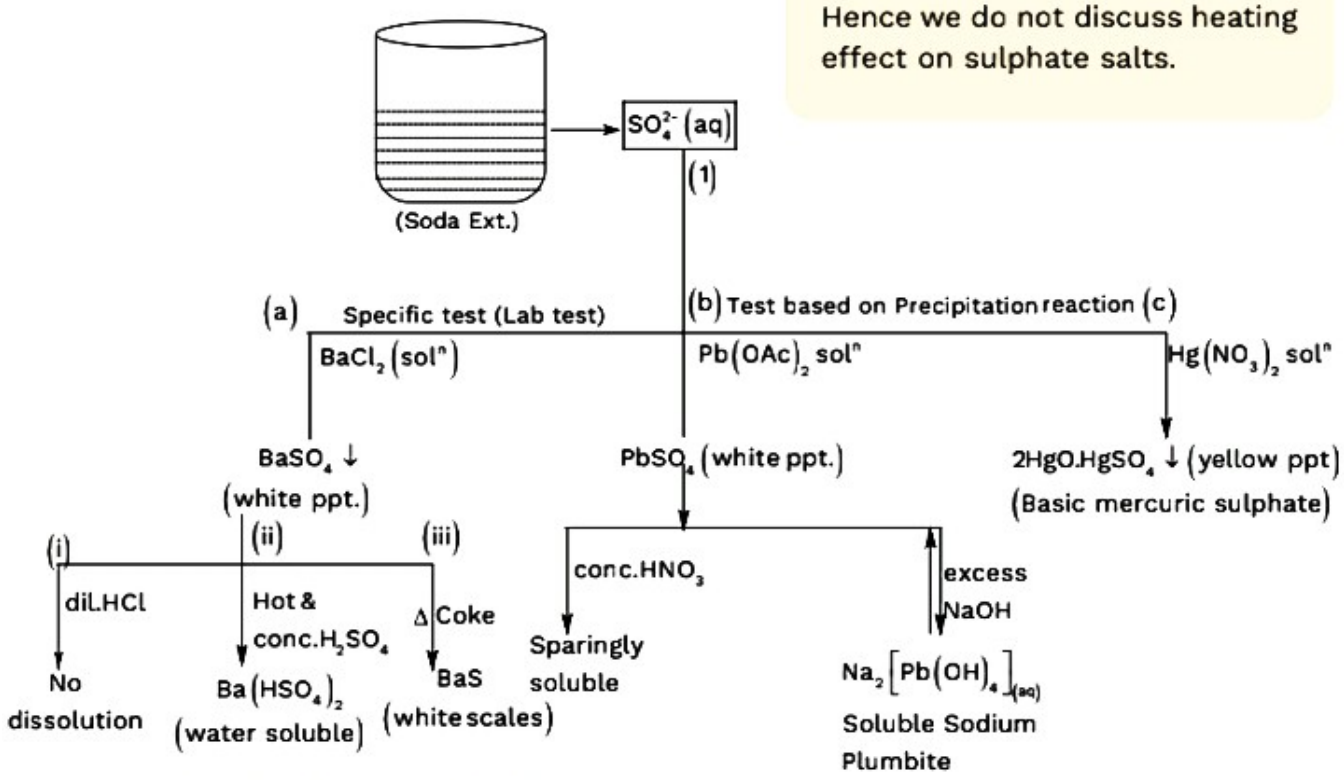
If NO_3^- is present with NO_2^- & we have to carry out brown ring test for NO_3^-.

$$\left.\begin{matrix} NO_3^- \\ + \\ NO_2^- \end{matrix}\right\} \xrightarrow[\Delta]{Urea(s)} \left.\begin{matrix} NO_3^- \\ + \\ N_2(g)\uparrow \end{matrix}\right\} \xrightarrow[test]{Brown\ ring}$$

Urea reduces NO_2^- to Nitrogen gas which escapes and nitrate is left behind for brown ring test.

Do You Know?

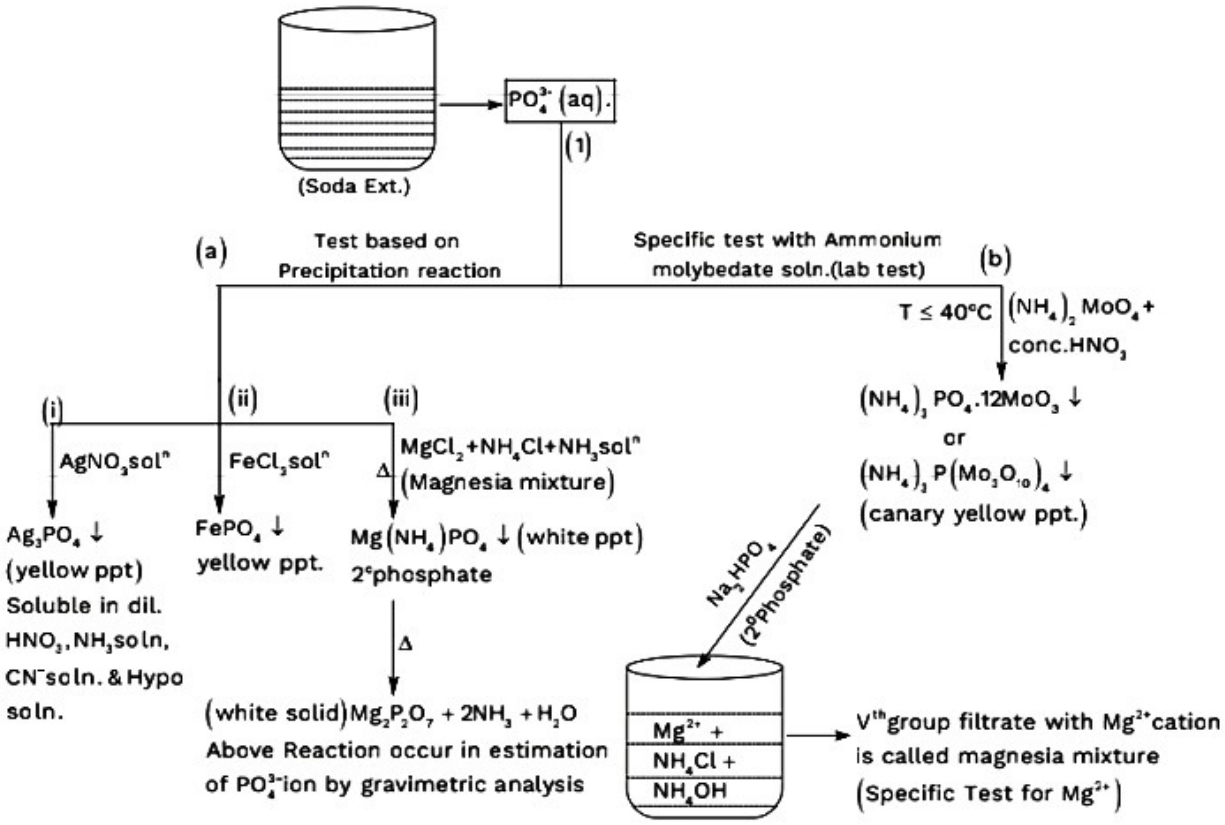
Sulphates are thermally stable & do not decompose on heating. Hence we do not discuss heating effect on sulphate salts.



Note : $BaSO_4\downarrow$ $SeSO_4\downarrow$ and $PbSO_4\downarrow$ salts are insoluble in any dil. acid at room temp. due to their low Ksp.

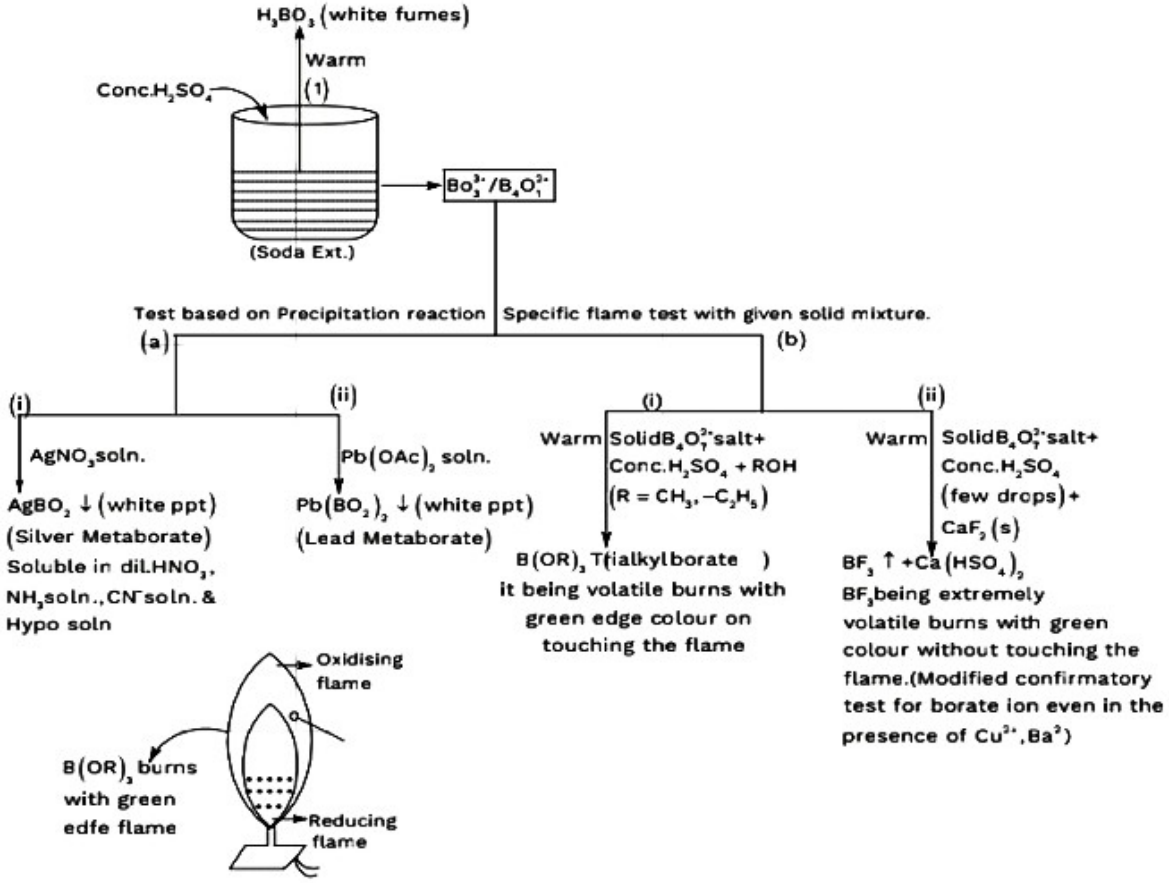
11. Phosphate (PO_4^{3-}):

All the phosphate except Na^{2+} to Cs^{+} are insoluble.



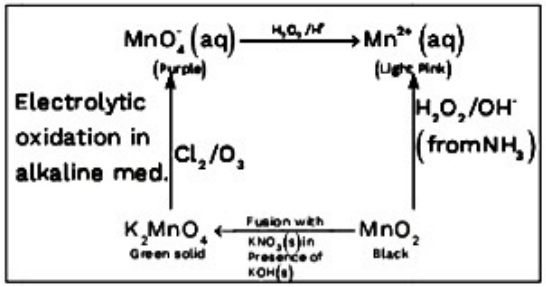
12. Borate (BO_3^{2-})

Borate salts of s-block are water soluble and other borate salts are either water soluble or sparingly water soluble. Very few borate salts derived from boric acid (H_3BO_3) are known, mostly borate salts are derived either from metaboric acid (HBO_2) or tetraboric acid ($H_2B_4O_7$)



13. Permagnate (MnO_4^-)

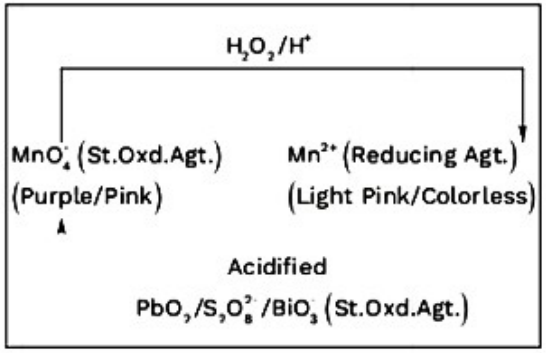
All MnO_4^- (permanganate) salt are water soluble.
$KMnO_4$ decomposes photochemically into MnO_2 hence it is stored in dark bottle in Lab.

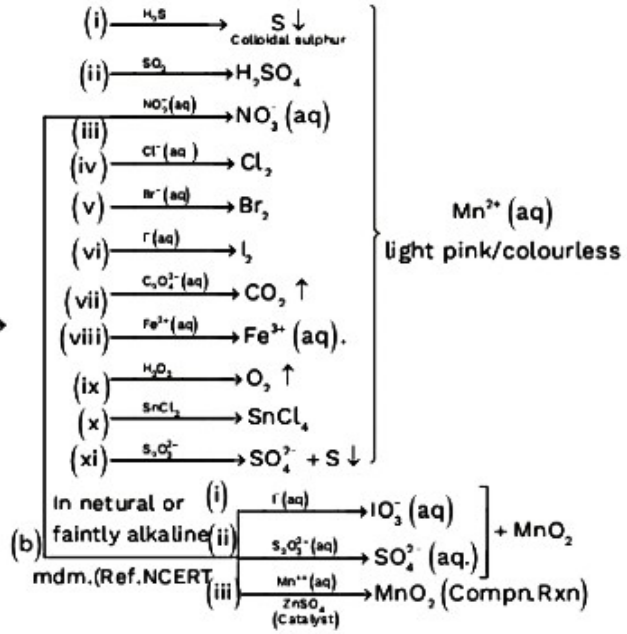




$MnO_4^-(aq)$ (Purple/Pink) (St.Oxd.Agt.)

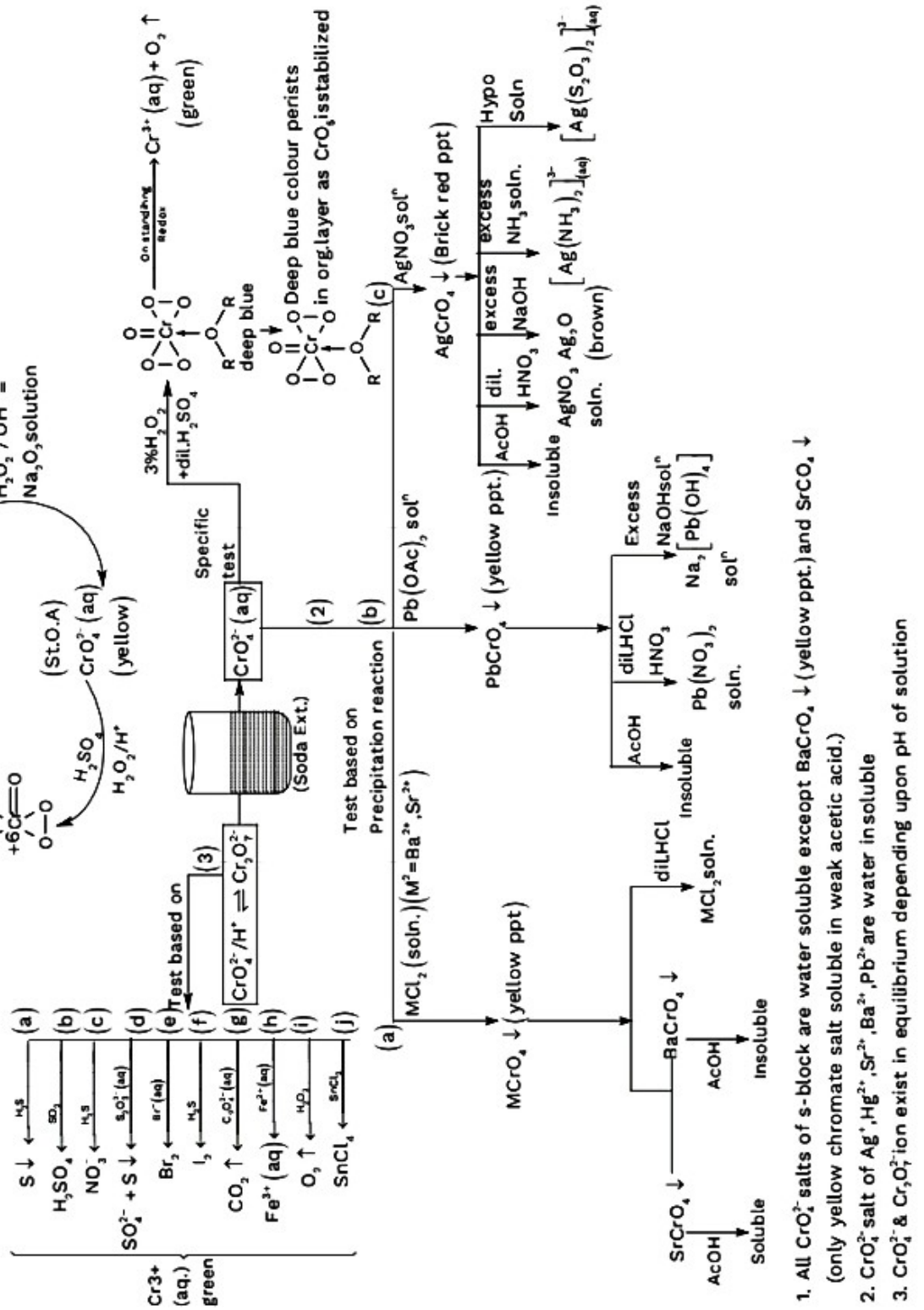
Test best on Oxd.prop (1)





Heating Effect: $2KMnO_4(solid) \xrightarrow{\Delta} K_2MnO_4(s) + MnO_2(s) + O_2(g)$
(Purple) (Green) (Black)

14. Chromate (CrO_4^{2-}) / Dichromate($Cr_2O_4^{2-}$)



1. All CrO_4^{2-} salts of s-block are water soluble exceopt $BaCrO_4 \downarrow$ (yellow ppt.) and $SrCO_4 \downarrow$ (only yellow chromate salt soluble in weak acetic acid.)
2. CrO_4^{2-} salt of $Ag^+, Hg^{2+}, Sr^{2+}, Ba^{2+}, Pb^{2+}$ are water insoluble
3. CrO_4^{2-} & $Cr_2O_7^{2-}$ ion exist in equilibrium depending upon pH of solution

Q18 **Zinc pieces are added to acidified solution of SO_3^{2-}. Gas liberated can :**
(A) turn lead acetate paper black
(B) turn lime water milky
(C) give white precipitate with$AgNO_3$ solution
(D) All are correct

Sol: **A**

$SO_3^{2-} + Zn + 8H^+ \longrightarrow H_2S \uparrow + 3Zn^{2+} + 3H_2O$

$Pb^{2+} + S^{2-} \longrightarrow PbS \downarrow$ (black)

$Ag^+ + S^{2-} \longrightarrow Ag_2S \downarrow$ (black)

Q19 **A mixture when rubbed with dilute acid smells like vinegar. It contains :**
(A) sulphite **(B) nitrate** **(C) nitrite** **(D) acetate**

Sol: **D**

Q20 **A substance on treatment with dilute H_2SO_4 liberates a colourless gas which produces (i) turbidity with baryta water and (ii) turns acidified dichromate solution green. The reaction indicates the presence of :**
(A) CO_3^{2-} **(B) S_2^-** **(C) SO_3^{2-}** **(D) NO_2^-**

Sol: **C**

$BaSO_3 + 2HCl(dil.) \longrightarrow 2NaCl + H_2O + SO_2;$

$K_2Cr_2O_7 + H_2SO_4 + 3SO_2 \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + H_2O$

Q21 **Which of the following combines with Fe(II) ions to form a brown complex?**
(A) N_2O **(B) NO** **(C) N_2O_3** **(D) N_2O_4**

Sol: **B**

$[Fe(H_2O)_6]^{2+} + NO \longrightarrow [Fe(H_2O)_5(NO)]^{2+}$ (brown ring)

Q22 **Which of the following statements is/are incorrect ?**
(A) A filter paper moistened with cadmium acetate solution turns yellow, when brought in contact with H_2S gas.
(B) Both carbonate ions as well as bicarbonate ions in the solutions, give reddish-brown precipitate with mercury(II) chloride.
(C) Sulphites in presence of zinc, reacts with dilute H_2SO_4 to liberate SO_2 gas.
(D) A filter paper moistened with KIO_3 and starch turns blue in contact with SO_2 vapours.

Sol: **B,C**

(A) $Cd^{2+}(aq) + H_2S(g) \rightarrow CdS\downarrow(\text{yellow}) + 2H^+(aq)$
(B) $Co_3^{2-} + 4Hg^{2+} + 3H_2O \rightarrow Hg_4O_3CO_3\downarrow(\text{reddish-brown}) + 6H^+HCO_3^-(aq)$
do not give precipitate.
(C) $SO_3^{2-} + 3Zn + 8H^+ \rightarrow H_2S\uparrow + 3Zn^{2+} + 3H_2O$
(D) $5SO_2 + 2IO_3^- + 4H_2O \rightarrow I_2 + 5SO_4^{2-} + 8H^-$

Q23 **Which of the following reagents turns white precipitate of AgCl yellow ?**
(A) $NaNO_3$ **(B) Na_3AsO_3** **(C) Na_3AsO_4** **(D) NaCN**

Sol: **B**
$3AgCl\downarrow AsO_3^{3-} \rightarrow Ag_3AsO_3\downarrow(\text{yellow}) + 3Cl^-$

Q24 **When a mixture of solid NaCl and solid $K_2Cr_2O_7$ is heated with concentrated H_2SO_4, deep red vapours are obtained. This is due to the formation of :**
(A) chromous chloride **(B) chromyl chloride**
(C) chromic chloride **(D) chromic sulphate**

Sol: **B**
$K_2Cr_2O_7 + 4NaCl + 6H_2SO_4 \rightarrow 2KHSO_4 + 4NaHSO_4 + \underset{\text{deep-red fumes}}{2CrO_2Cl_2\uparrow} + 3H_2O$

Q25 **AgCl dissolves in ammonia solution giving :**
(A) Ag^+, NH_4^+ and Cl^- **(B)** $[Ag(NH_3)]^+$ and Cl^-
(C) $\left[Ag_2.(NH_3^-)\right]^{2+}$ and Cl^- **(D)** $\left[Ag(NH_3)_2\right]^+$ and Cl^-

Sol: **D**
$AgCl + NH_3 \rightarrow \left[Ag(NH_3)_2\right]^{2+}$

Q26 **A mixture upon adding conc. H_2SO_4 gives deep red fumes. It may contain the anions pair :**

(A) $Cr_2O_7^{2-}$ **and** Cl^- **(B)** Br^- **and** $Cr_2O_7^{2-}$

(C) NO_3^- **and** Cl^- **(D)** CrO_4^{2-} **and** NO_3^-

Sol: **A**

$$K_2Cr_2O_7 + 4NaCl + 6H_2SO_4 \rightarrow 2KHSO_4 + 4NaHSO_4 + \underset{\text{deep-red fumes}}{2CrO_2Cl_2 \uparrow} + 3H_2O$$

Q27 **A solution of a salt in concentrated sulphuric acid H_2SO_4 acid produced a deep blue colour with starch iodide solution. The salt may be**

(A) chloride **(B) carbonate** **(C) acetate** **(D) bromide**

Sol: **D**

Bromine gives deep blue colour with starch iodide solution.

Q28 **A colourless solution of a compound gives a precipitate with$AgNO_3$ solution but no precipitate with a solution of Na_2CO_3. The action of concentrated H_2SO_4 on the compound liberates a suffocating reddish brown gas. The compound is :**

(A) $Ba(CH_3COO)_2$ **(B)** $CaCl_2$ **(C) NaI** **(D) NaBr**

Sol: **D**

$Ag^+ + Br \rightarrow AgBr \downarrow$ (pale yellow),

$NaBr + Na_2CO_3 \rightarrow$ No reaction.

$2NaBr + 2H_2SO_4 \rightarrow Br_2 \uparrow$ (reddish brown) $+ SO_2 + SO_4^{2-} + 2Na^+ + 2H_2O$

Q29 **When chlorine (Cl_2) water in excess is added to a salt solution containing chloroform, chloroform layer turns pale yellow. Salt contains :**

(A) Br^- **(B)** I^- **(C)** NO_3^- **(D)** S^{2-}

Sol: **A**

Reactivity $Cl^- > Br^- > I^-$ Hence Cl^- will replace these halide anions from their salt. A pale – yellow solution is formed due to presence of Br_2 in solution as a result of displacement by Cl^-.

Q30 **An aqueous solution of salt containing an acidic radical X^- reacts with sodium hypochlorite in neutral medium. The gas evolved produces blue black colour spot on the starch paper. The anion X^- is :**

(A) CH_3COO^- **(B) Br^-** **(C) I^-** **(D) NO_2^-**

Sol: **C**

$OCl^- + 3I^- + H_2O \rightarrow I_3^- + 2OH^- + Cl^-$

I_3^- + starch $\rightarrow$ blue – black spot on starch paper appears due to the formation of iodine starch adsorption complex.

Q31 **When chlorine water is added to an aqueous solution of potassium halide in the presence of chloroform, a colour is developed but on adding more of chlorine water the colour disappears, and a colourless solution is obtained. This test confirms the presence of the following in aqueous solution.**

(A) Iodide **(B) Bromide**

(C) Chloride **(D) Iodide and bromide**

Sol: **A**

HIO_3 is a colorless compound and adding more chlorine water to solutions confirms formation of this compound and hence Iodide.

Q32 **Nitrate is confirmed by ring test. The brown colour of the ring is due to formation of :**

(A) ferrous nitrite **(B) nitroso ferrous sulphate**

(C) ferrous nitrate **(D) $FeSO_4.NO_2$.**

Sol: **B**

Nitroso ferrous sulphate

$[Fe(H_2O)(NO)]SO_4$

$NO_3^- + Fe^{2+} + 4H^+ \rightarrow 3Fe^{+3} + NO + 2H_2O$

$[Fe(H_2O)_6]^{2+} + NO \rightarrow [Fe(H_2O)_5(NO)]^{2+}$ (brown ring)

Q33 **Nitrates of all the metals except mercury and bismuth are :**

(A) coloured **(B) unstable**

(C) soluble in water **(D) insoluble in water**

Sol: **C**

Q34 **Which of the following reagents can be used for making the distinction between AgCl and AgI ?**

(A) Sodium arsenite solution. **(B) Dilute ammonia solution.**

(C) Potassium cyanide solution. **(D) Dilute HNO_3.**

Sol: **A,B**

Both are soluble in Dilute HNO_3 while both form cyanide complexes with KCN, hence C and D can not be used. Now, AgCl reacts with Dilute ammonia solution to give product while AgI doesn't dissolves with ammonia solution.

Q35 **Precipitate of $PbSO_4$ is soluble in :**

(A) ammonium acetate (6M) **(B) dilute HCl**

(C) dilute H_2SO_4 **(D) none**

Sol: **A**

Forms a soluble complex with acetate ion

$PbSO_4 + CH_3COONH_4^- \rightarrow (NH_4)_2\left[Pb(CH_3COO)_4\right](aq.)$

Q36 **There are four test tubes containing dilute HCl, $BaCl_2$, $CdCl_2$ and KNO_3 solutions. Which of the following reagents will help in the identification of $BaCl_2$?**

(A) NaOH **(B) K_2CrO_4**

(C) $AgNO_3$ **(D) both (B) and (C)**

Sol: **B**

The precipitate is insoluble in dilute acetic acid but readily soluble in mineral acids. $K_2CrO_4 + BaCl_2 \rightarrow BaCrO_4$ (yellow ppt)

ANALYSIS OF CATION

- Cations are divided into some groups for Analysis
- The Reagent which is used to detect all cations of a particular group is known as group reagent:

Group	Cations	Group reagent	Precipitates/gas
Zero	NH_4^+	Heat with NaOH solution	NH_3 gas
Ist group	Pb^{2+}, Ag^+, Hg^{2+}	Dil. HCl or soluble chloride solution	$PbCl_2 \rightarrow$ white AgCl → white $Hg_2Cl_2 \rightarrow$ white
2nd group	Pb^{2+}, Cu^{2+}, Cd^{2+}, Hg^{2+}, Bi^{3+}, As^{3+}, Sb^{3+}, Sn^{2+}	H_2S gas in presence of dil. HCl	PbS, CuS, HgS, $Bi_2S_3 \rightarrow$ Black Cds → Yellow AS_2S_3, SnS_2 →yellow SnS → Brown $Sb_2S_3 \rightarrow$ Orange
3rd group	Fe^{3+}, Al^{3+}, Cr^{3+}	Aq. NH_3 solution in presence of NH_4Cl	$Fe(OH_3)$ Reddis brown $Cr(OH)_3$ Green
4th group	Zn^{2+}, Mn^{2+}, Ni^{2+}, Co^{2+}	H_2S gas in presence of NH_3 solution	ZnS → white MnS → Pink or Buff G, CoS } black
5th group	Ba^{2+}, Sr^{2+}, Ca^{2+}	$(NH_4)CO_3$solution in presence of NH_4Cl & NH_4OH	$BaCO_3$, $SrCO_3$, $CaCO_3$ } white
6th group	Mg^{2+}	Na_2HPO_4(solution) in presence of NH_3 solution	$Mg(NH_4)PO_4$ white

Q37 Why cations are analysed after anions?

Sol:

If soluble chloride is there → 1st group absent

If soluble sulphide is there → 2nd & 4th group are absent

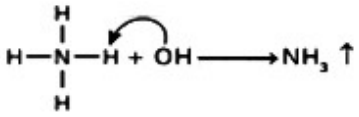
If soluble sulphate in there → 1st group 5th group are absent.

General procedure to carry out group wise analysis of cations

Zero group : Ammonium is highly soluble cation so it cannot be analysed by precipitation.

$$NH_G^+ + OH^- \longrightarrow NH + H\,O$$

Salt of W.B S.B S.B



- **1st group** : K_{sp} value of chlorides of 1st group cation are less as compare to other cations. So no adding dil HCl or Cl^- solution only 1st group cation form ppt & rest are soluble.
- **2nd group** :

(I) Ksp of lead chlorides is not very less so some lead ion remain in solution after analysis of first group. These will be ppt as Pbs in IInd group.

(II) Group reagent of second group is H_2S gas in presence of dil HCl to prevent the ppt. of IVth group sulphide.

$$H_2S \rightleftharpoons H^+ + HS^-$$

$$K_1 = \frac{[H^+][HS^-]}{[H_2S]}$$

$$HS \rightleftharpoons H^+ + S^{2-}$$

$$K_2 = \frac{[H^+][S^{2-}]}{[HS^-]}$$

$$K_1K_2 = \frac{[H^+]^2[S^{2-}]}{[H_2S]}$$

$$[S^{2-}] \downarrow = \frac{[H_2S]K_1K_1}{[H^+]^2 \uparrow}$$

K_{sp} values of sulphites of IIrd group are very less so there sulphides will be ppt with very less sulphide ion conc. since Ksp value of sulphides of IVth group are high so these sulphides will not be ppt with less sulphide conc.

(III) We cannot used H_2SO_4 instead of HCl, due to the ppt. of Vth group sulphites. we cannot HNO_3 in place of HCl, due to oxidation of H_2S_2 into S (Yellow).

- **3rd Group :**
  (1) Add some drops of conc HNO_3 in solution after analysis of IInd group to reoxidised Fe^{2+} to Fe^{3+} which was reduced to Fe^{3+} by H_2S early.
  (2) Group reagent of 3rd group cation is aq. NH_3 is presence of NH_4Cl because in presence of NH_4Cl (due to common ion effect) hydroxide ion conc is less which is sufficient to ppt. IIIrd group & not sufficient to ppt. IVth group hydroxide are less as compare to IVth group hydroxide.
  (3) We cannot use $(NH_4)_2(SO_4)$ instead of NH_4Cl due to ppt of Vth group sulphates.
- **4thGroup :**
  Since KsP values of S^{2-} of IVth group cations are high therefore we need more S^{2-} conⁿ to ppt. these therefore group is H_2S gas in presence of aq. NH_3.
- **5thGroup :**
  In 5th group we use $(NH_4)_2CO_3$ solutionnⁿ in presence of NH_4Cl because in presence of NH_4Cl. Mg $(OH)_2$ will not be ppt.
  Formation of original solution:
  (1) Salt + water $\xrightarrow{\Delta}$ solution (O.S)
  If salt is not soluble
  (2) Salt + dil HCl $\xrightarrow{\Delta}$ Solution (O.S)
  If salt is not soluble.
  (3) Salt + conc. HCl $\xrightarrow{\Delta}$ Solution (O.S)

Analysis of Zero Group

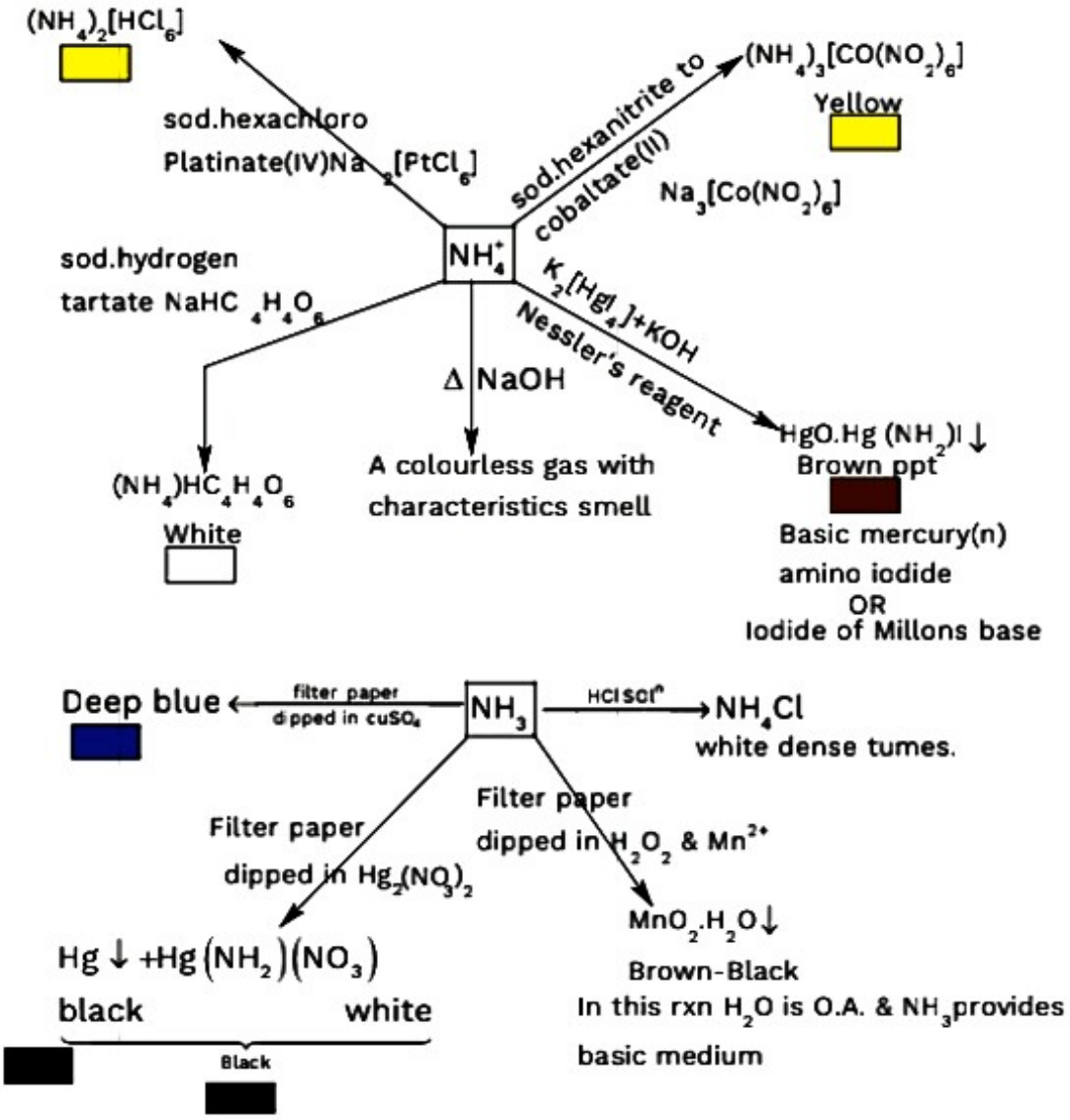
Cation → NH_4^+ (Ammonium)

Group reagent → heating with NaOH or KOH

- Most of the NH_4^+ salts are soluble in water.

Heating effects.

If anion is not oxidising agent	**If anion is oxidising agent**
Salt $\xrightarrow{\Delta}$ NH_3	Salt $\xrightarrow{\Delta}$ N_2
$(NH_4)_2CO_3 \xrightarrow{\Delta} 2NH_3 + CO_2 + H_2O$ $NH_4HCO_3(s) \xrightarrow{\Delta} NH_3 + H_2O + CO_2$ $(NH_4)_2SO_4(s) \longrightarrow NH_3 + NH_4HSO_4$	$NH_4NO_2(s) \xrightarrow{\Delta} N_2 + 2H_2O$ $NH_4NO_3(s) \xrightarrow{\Delta} N_2O + 2H_2O$ laughing gas $(NH_4)_2Cr_2O_7(s) \xrightarrow{\Delta} N_2(g) + Cr_2O_3 + 4H_2O$ (Orange) (Green)
$A(NH_4)_3PO_4(s) \xrightarrow{\Delta} NH_3 + P_4O_{10} + 6H_2O$ $NH_4Cl\ (s) \xrightarrow{\Delta} NH_3 + HCl$	$2NH_4ClO_4 \xrightarrow{\Delta} N_2(g) + Cl_2(g) + 2O_2(g) + 4H_2O$ $2NH_4IO_3(s) \xrightarrow{\Delta} N_2(g) + I_2 + O_2 + 4H_2O$



ANALYSIS OF 1ST GROUP:

This group includes Ag^+, Pb^{2+} and Hg_2^{2+} (ous). The group reagent is dilute hydrochloric acid. The radicals are precipitated as their chlorides because the solubility product of these chlorides (AgCl $PbCl_2$ and Hg_2Cl_2) is less than the solubility of all other chlorides which thus remains in solution.

Cation: Pb^{2+}, $Ag^{+\cdot}$, Hg_2^{2+}

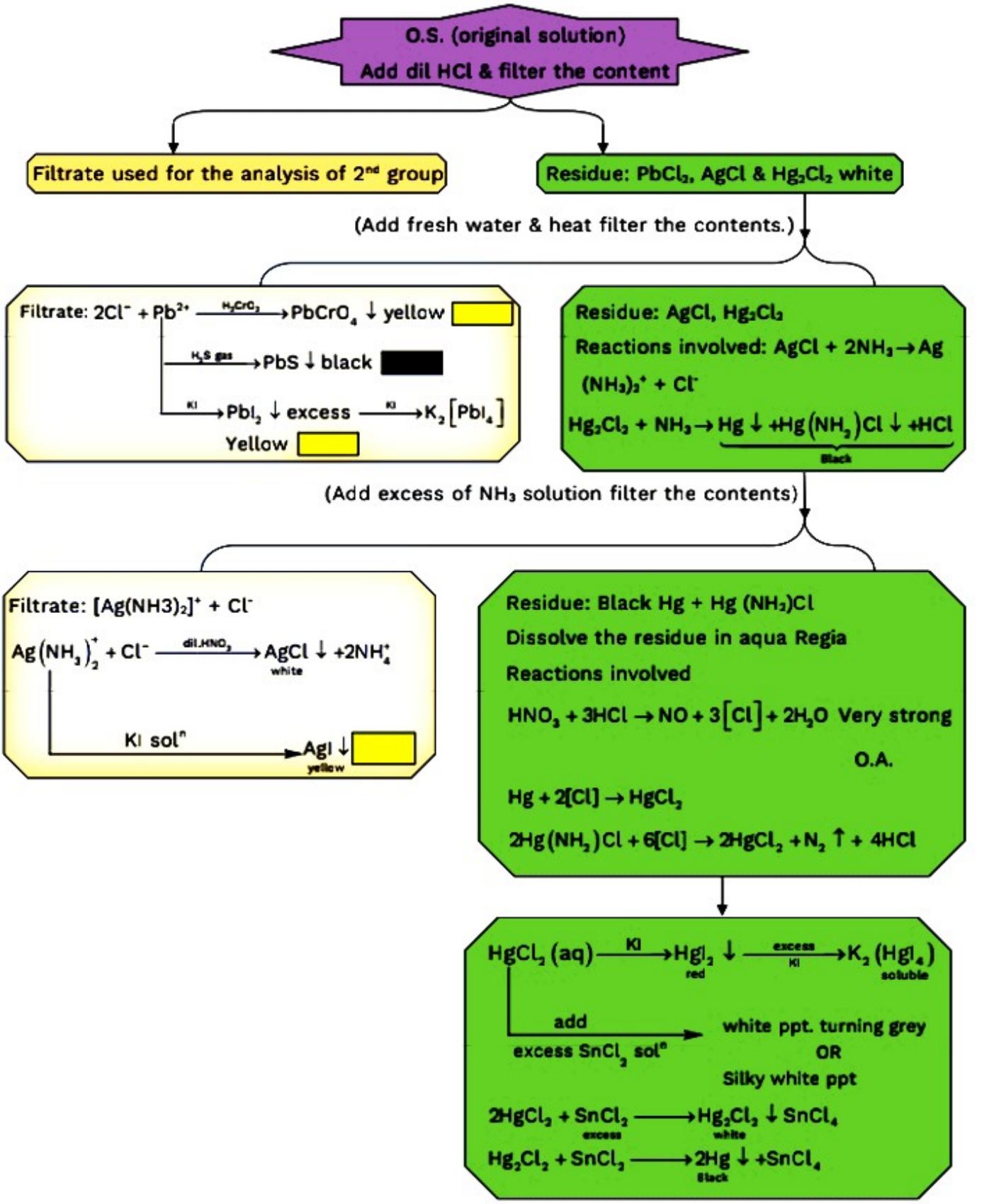
Group reagent → dil. HCl or Cl solution

Reactions Involved

$Pb^{2+} + 2Cl^- \rightarrow PbCl_2\downarrow$

$Ag^+ + Cl^- \rightarrow AgCl\downarrow$

$Hg_2^{2+} + 2Cl^- \rightarrow Hg_2Cl_2\downarrow$


O.S. (original solution)
Add dil HCl & filter the content
Filtrate used for the analysis of 2nd group
Residue: PbCl2, AgCl & Hg2Cl2 white
(Add fresh water & heat filter the contents.)
Filtrate: 2Cl⁻ + Pb2+ → PbCrO4 ↓ yellow
→ PbS ↓ black
→ PbI2 ↓ excess → K2[PbI4]
Yellow
Residue: AgCl, Hg2Cl2
Reactions involved: AgCl + 2NH3 → Ag(NH3)2+ + Cl⁻
Hg2Cl2 + NH3 → Hg ↓ + Hg(NH2)Cl ↓ + HCl
Black
(Add excess of NH3 solution filter the contents)
Filtrate: [Ag(NH3)2]+ + Cl⁻
Ag(NH3)2+ + Cl⁻ → AgCl ↓ + 2NH4+
white
KI sol^n → AgI ↓
yellow
Residue: Black Hg + Hg(NH2)Cl
Dissolve the residue in aqua Regia
Reactions involved
HNO3 + 3HCl → NO + 3[Cl] + 2H2O Very strong O.A.
Hg + 2[Cl] → HgCl2
2Hg(NH2)Cl + 6[Cl] → 2HgCl2 + N2 ↑ + 4HCl
HgCl2 (aq) → HgI2 ↓ → K2(HgI4)
red
excess KI
soluble
add excess SnCl2 sol^n
white ppt. turning grey
OR
Silky white ppt
2HgCl2 + SnCl2 → Hg2Cl2 ↓ + SnCl4
excess
white
Hg2Cl2 + SnCl2 → 2Hg ↓ + SnCl4
Black

Reactions of 1st Group :

	Pb^{2+}	Ag^{+}	Hg_2^{2+}
Solubility of comp.	Most of the Pb2+ compounds are soluble in excess of NaOH & form complex $[Pb(OH)_4]^{2-}$	Most of the Ag^{+} compounds are soluble in aq NH_3 & form $[Ag(NH_3)_2]^{+}$	$Hg_2^{2+} \rightarrow Hg + Hg^{2+}$ soluble in aqua-regia Hg_2^{2+} ions disproportionate in chemical reactions
dil HCl	$PbCl_2$ ↓white • Soluble in hot water • Soluble in excess of NaOH • Soluble in conc. (6M) HCl to form $PbCl_4^{2-}$	AgI ↓white • Soluble in dil NH_3 • Soluble in KCN & Hypo	Hg_2Cl_2 ↓white • Turns black in NH_3 Solution • Hg + $Hg(NH_2)Cl$ soluble in aqua regia.
dil H_2SO_4	$PbSO_4$ ↓white • Soluble in hot ammino-acetate & ammino tartarated • Soluble in excess of NaOH	Ag_2SO_4 ↓white	Hg_2SO_4 ↓white
dil HCl + H_2S gas	PbS ↓ black conc. HNO_3 → $PbSO_4$ white 50% w/w HNO_3 → Pb^{2+} + NO + S↓	Ag_2S black	Hg + HgS ↓ Black Soluble in aq. regia
NH_3(aq)	$Pb(OH)_2$ ↓ white excess NaOH → $Pb(OH)_4^{2-}$ dil.HNO_3 → Pb^{2+}	Ag_2O (brown) excess NH_3 → $[Ag(NH_3)_2]^{+}$	Hg_2O black

	Pb^{2+}	Ag^{+}	Hg_2^{2+}
Na_2CO_3 (aq)	$Pb(OH)_2$, $PbCO_3\downarrow$ white Soluble in excess of NaOH & dil HNO_3	$Ag_2CO_3\downarrow \xrightarrow{\Delta} Ag_2O\downarrow + CO_2$ (brown)	Hg_2CO_3 yellow $\xrightarrow{\Delta} Hg_2O\downarrow + CO_2$ (black)
NaOH (aq)	$Pb(OH)_2\downarrow$ white $\xrightarrow[\text{NaOH}]{\text{excess}} Pb(OH)_4^{2-}$ It is also soluble in HNO_3	$Ag_2O\downarrow$ (brown) Soluble in NH_3 sol	$Hg_2O\downarrow$ Black
K_2CrO_4	$PbCrO_4\downarrow$ yellow excess NaOH → $Pb(OH)_4^{2-}$ HNO_3 → $Pb^{2+} + Cr_2O_7^{2-}$	$Ag_2CrO_4\downarrow$ (brick red) Soluble in excess of NH_3 & HNO_3 NH_3 → $[Ag(NH_3)_2]^+$ HNO_3 → $Ag^+ + Cr_2O_7^{2-}$	$Hg_2CrO_4\downarrow$ red
KI (aq)	$PbI_2\downarrow$ yellow soluble in hot water excess KI ⇄ On dilution PbI_2 reappears $2K^+ + [PbI_4]^{2-}$	$AgI\downarrow$ yellow Soluble in KCN & Hypo	$Hg_2I_2\downarrow$ green excess KI $Hg\downarrow + HgI_4^{2-}$ black
KCN	$Pb(CN)_2\downarrow$ white; soluble in excess NaOH	$AgCN\downarrow$ white; excess KCN $Ag(CN)_2^-$	$Hg\downarrow + Hg(CN)_2$ (Black)
Na_2HPO_4	$Pb_3(PO_4)_2\downarrow$ yellow soluble in excess of NaOH & dil HNO_3 HNO_3 → $Pb^{2+} + H_2PO_4^{3-}$ NaOH → $Pb(OH)_4^{3-} + PO_4^{3-}$	$Ag_3PO_4\downarrow$ yellow soluble in excess of NH_3 & dil HNO_3 HNO_3 → $Ag^+ + H_2PO_4^{3-}$ NH_3 → $[Ag(NH_3)_2]^+ + PO_4^{3-}$	

ANALYSIS OF 2nd GROUP:

This group includes

$Hg^{2+}, Pb^{2+}, Bi^{3+}, Cu^{2+}, Cd^{2+}$ (all in IIA),

$As^{3+}, Sb^{3+}, Sn^{2+}$ and Sn^{4+} (all in IIB).

The group reagent is hydrogen sulphide in presence of HCl. These radicals are precipitated as their sulphides, whereas the sulphides of other metals remain in solution because of their high solubility product.

HgS, PbS, Bi_2S_3 and CuS are black, CdS, As_2S_3 and SnS_2 yellow, SnS brown, and Sb_2S_3 is orange.

The function of HCl is the decrease ionisation of H_2S (due to common ion effect) so that only the solubility product of sulphides of II group radical is exceeded and not that of III, IV and V group. Hence III, IV and V group radicals are not precipitated by H_2S in the presence of HCl.

Points to remember

1. Criteria of separation between IIrd-A & IIrd-B sub groups
   Sulphides of IInd-B sub group are soluble in YAS (Yellow ammonium sulphide) & those of IInd-a sub group are soluble.
   $YAS + AS_2S_3 \rightarrow$ **Thio arsenate**
   $YAS + Sb_2S_3 \rightarrow$ **Thio antimonate**
2. YAS → is prepared by dissolving excess sulfer in $(NH_4)_2S$
   $(NH_4)_2S + S_{excess} \rightarrow (NH_4)_2S_x$
   Colourless YAS
   (yellow solution)
   $S^{2-} + S \rightleftharpoons S_2^{2-}$
   $S^{2-} + 2S \rightleftharpoons S_3^{2-}$
   $S^{2-} + 4S \rightleftharpoons S_4^{2-}$
3. For reaction purpose YAS is consider as $(NH_4)_2S_2$

Pbs and CdS are precipitated only on dilution due to higher $K_{sp.}$

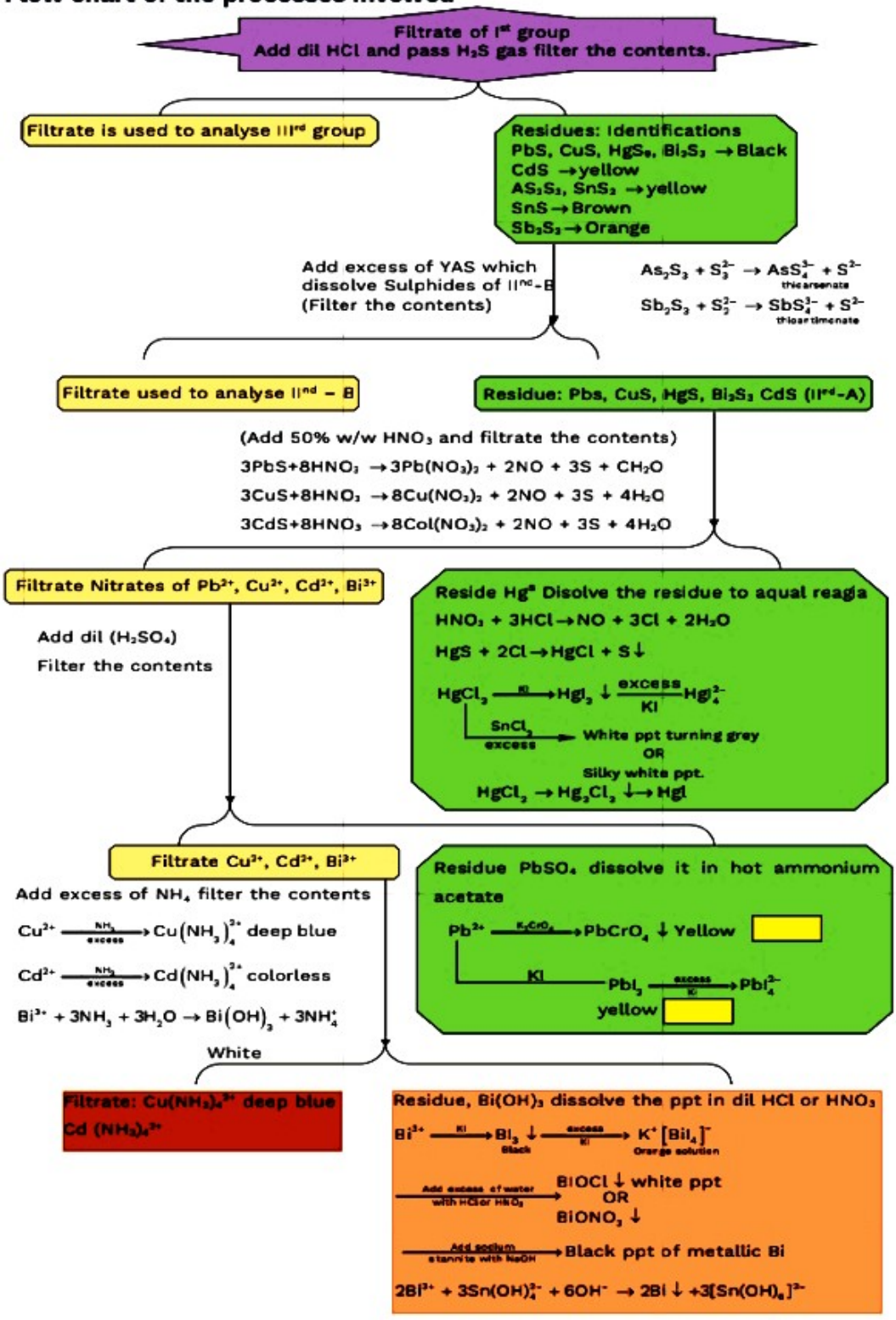
IIA and IIB group sulphides are separated by yellow ammonium sulphide $(NH_4)_2S_x$ in which IIA group sulphides are insoluble whereas that if IIB are soluble forming thiosalts.

Cations: $Pb^{2+}, Cu^{2+}, Cd^{2+}, Hg^{2+}, Bi^{3+} \rightarrow II^{nd}A$

$As^{3+}, Sb^{2+}, Sn^{2+} \rightarrow II^{nd}B$

Group reagent: H_2S gas in presence of dil. HCl

Flow chart of the processes involved



Case – I

If the filtrate is colourless, $Cu2+^{2+}$ must be absent

In this case

$$\text{Filtrate} \xrightarrow[H_2S \text{ gas}]{\text{Pass}} CdS\downarrow$$ Yellow ppt of CdS confirms Col^{2+}

Case – II

If the filtrate blue Cu^{2+} must be present

$$\text{Filtrate} \xrightarrow[\text{of KCN sol}^n]{\text{Ass excess}} \text{colourless} \xrightarrow{H_2S \text{ gas}} \text{yellow ppt of CdS confirms } Cd^{2+}$$

$$Cu(NH_3)_4^{2+} + KCN \rightarrow \underset{\text{Yellow}}{Cu(CN)_2}\downarrow + K^+ \; 4NH_3\uparrow$$

$$2Cu(CN)_2 \rightarrow \underset{\text{White}}{CuCN}\downarrow \quad \underset{\substack{\text{Cyanagen gas}\\ \text{(highly poisonous)}}}{(CN_2);\uparrow}$$

also $CuCN + 3KCN \longrightarrow K_3^+[Cu(CN)_4]$

$$Cd(NH_3)_4^{2+} + 2KCN \rightarrow \underset{\text{White}}{Cd(CN)_2}\downarrow + 2K^+ + 4NH_3$$

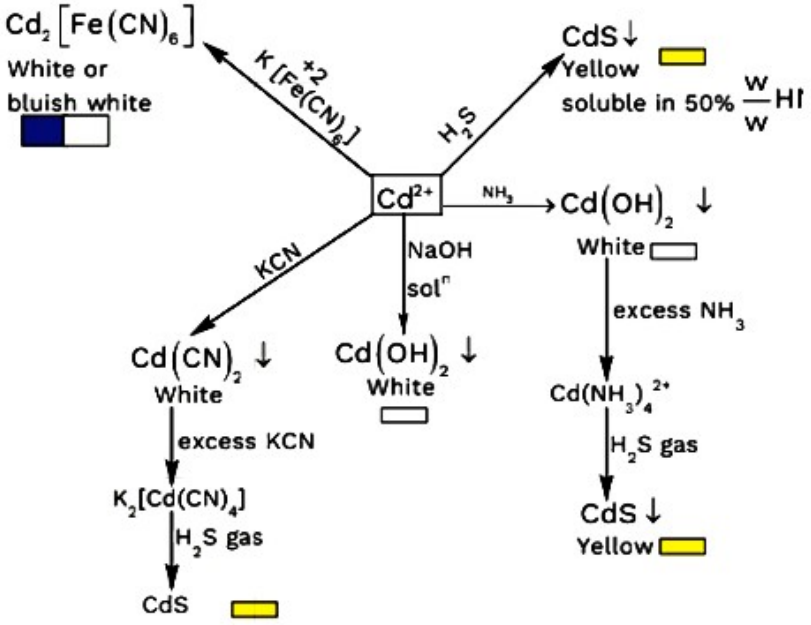
$$Cd(CN)_3 + 2KCN \rightarrow \underset{\substack{\text{colourless complex}\\ \text{(less stable)}}}{K_2[Cd(CN)_4]} ;$$

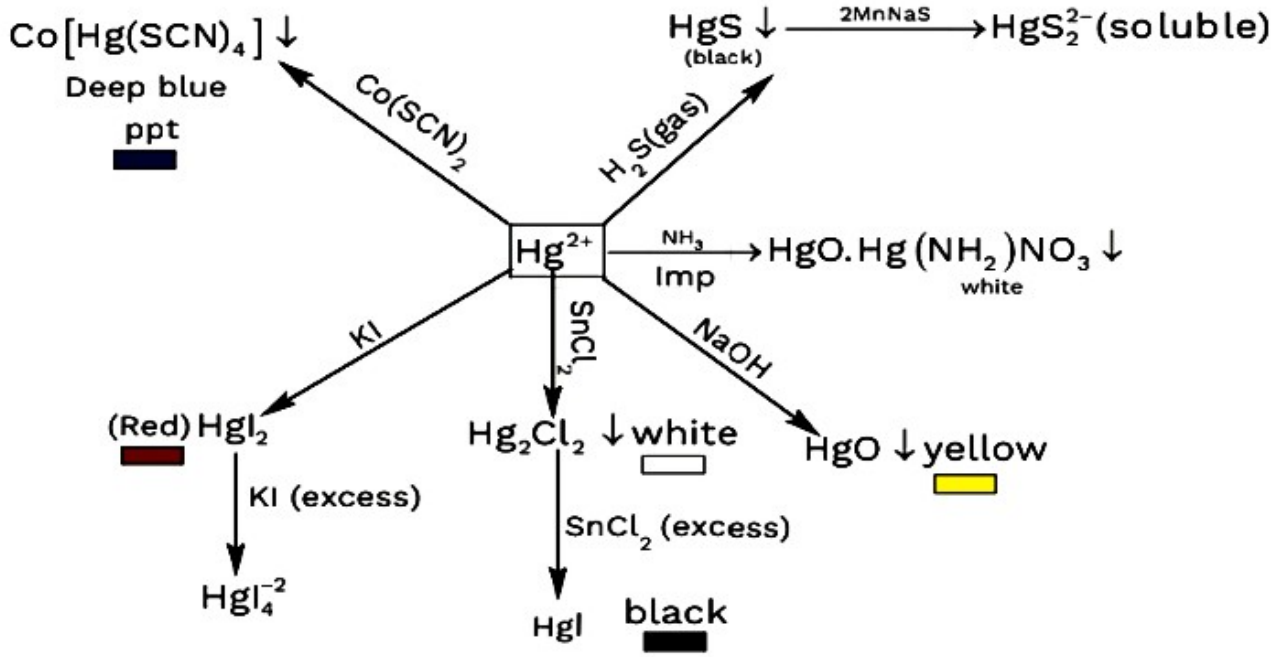
$$K_3[Cu(N)_4] \xrightarrow[\text{gas}]{H_2S} \text{No rxn}$$

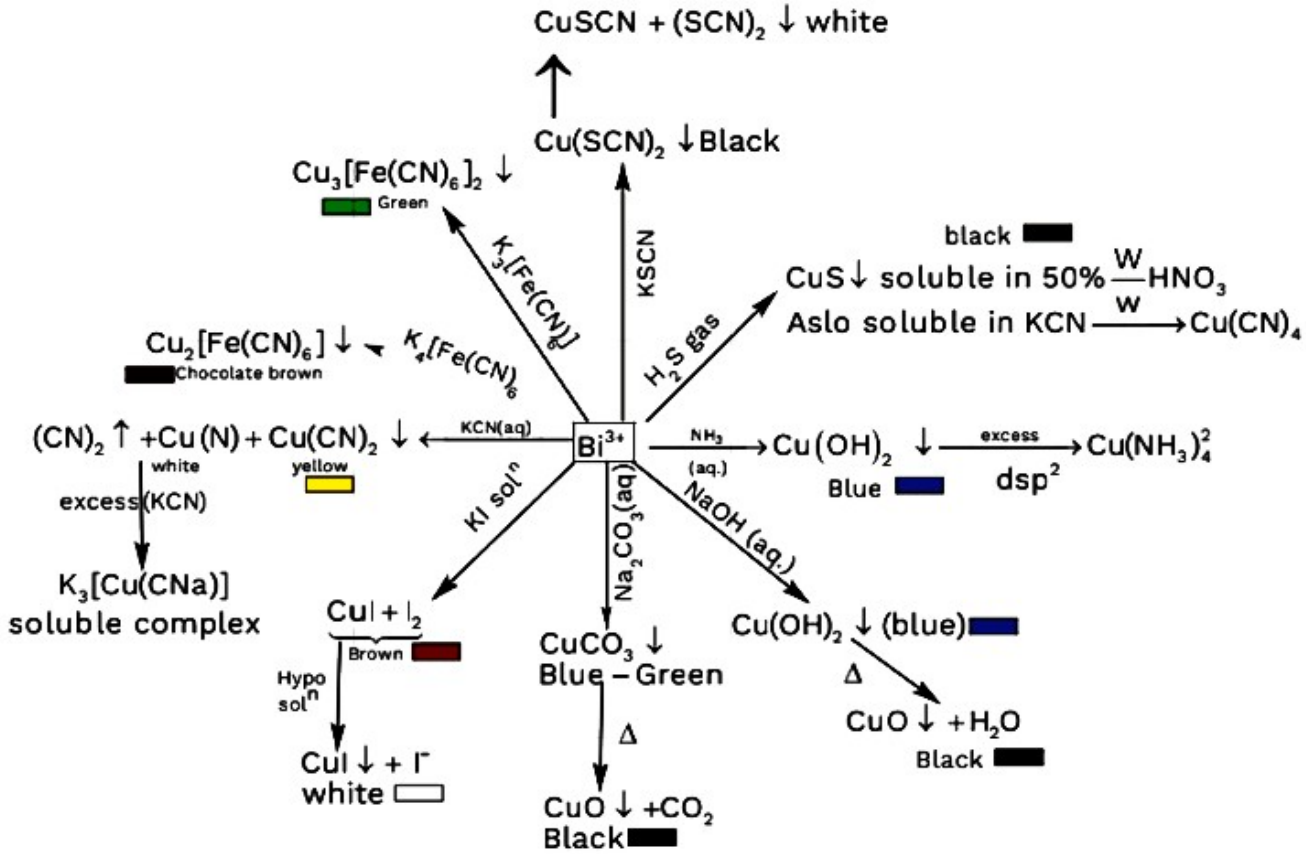
$$K_2[Cd(CN)_4] \xrightarrow[\text{gas}]{H_2S} CdS\downarrow + 3KCN + 2HCN$$

In this test KCN acts as masking agent which masks Cu^{2+} ion in the form of $K_3[Cu(CN)_4]$

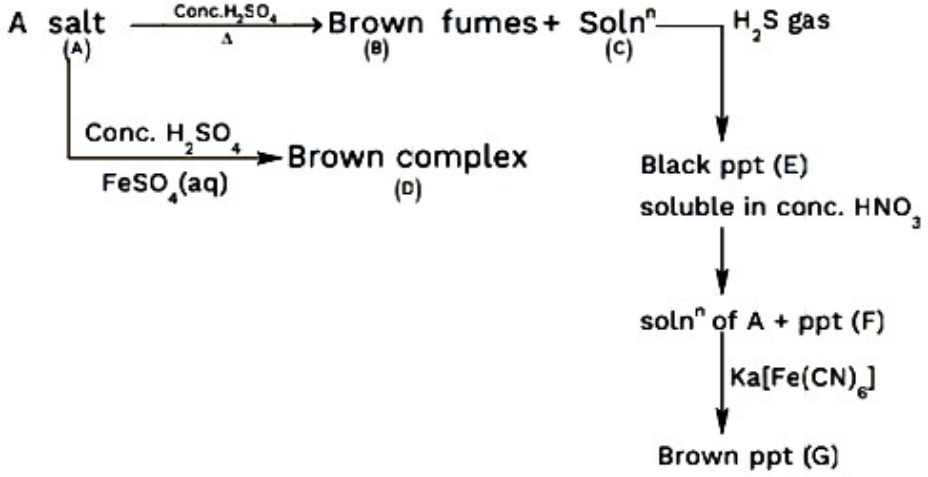
Reaction of II group




Co[Hg(SCN)4] ↓
Deep blue ppt
Co(SCN)2
HgS ↓ (black)
2MnNaS
HgS2 2− (soluble)
H2S(gas)
Hg2+
NH3
Imp
HgO.Hg(NH2)NO3 ↓
white
KI
SnCl2
NaOH
(Red) HgI2
KI (excess)
HgI4 −2
Hg2Cl2 ↓ white
SnCl2 (excess)
HgI black
HgO ↓ yellow


CuSCN + (SCN)2 ↓ white
Cu(SCN)2 ↓ Black
Cu3[Fe(CN)6]2 ↓
Green
K3[Fe(CN)6]
KSCN
black
CuS ↓ soluble in 50% W/W HNO3
Aslo soluble in KCN
Cu(CN)4
Cu2[Fe(CN)6] ↓
Chocolate brown
K4[Fe(CN)6]
H2S gas
(CN)2 ↑ +Cu(N) + Cu(CN)2 ↓
white
yellow
KCN(aq)
Bi3+
NH3 (aq.)
Cu(OH)2 ↓
Blue
excess
Cu(NH3)4 2
dsp2
KI soln
Na2CO3(aq)
NaOH (aq.)
excess(KCN)
K3[Cu(CNa)]
soluble complex
CuI + I2
Brown
Hypo soln
CuI ↓ + I−
white
CuCO3 ↓
Blue – Green
Δ
CuO ↓ +CO2
Black
Cu(OH)2 ↓ (blue)
Δ
CuO ↓ +H2O
Black

Q38

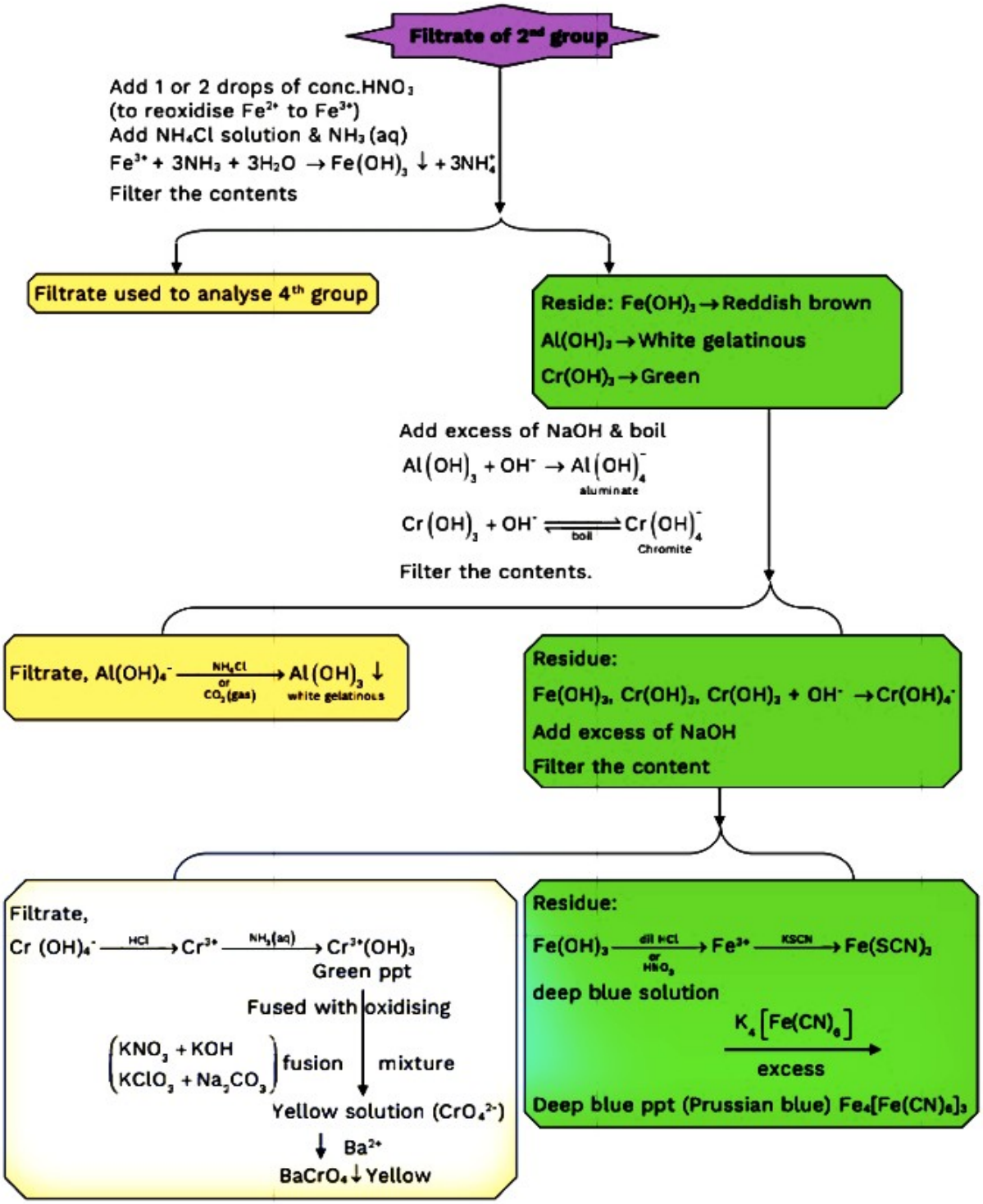


Sol: (A) → $Cu(NO_3)_2$
(B) → NO_2
(C) → $CuSO_4$
(D) → Brown ring test $[Fe(H_2O)_5(NO^+)]SO_4$
(E) → CuS
(F) → ↓
(G) → $Cu_2[Fe(CN)_6]$

Analysis of 3rd Group :

This group includes Fe^{3+}, Al^{3+} and Cr^{3+}. The group reagent is NH_4OH and in presence of NH_4Cl , the radicals are precipitated as their hydroxides. The function of NH_4Cl is to suppress the ionisation of NH_4OH so that only the IIIrd group radicals are precipitated, because the solubility product of IIIrd group hydroxides is less than that of IV and VI group hydroxides. Excess of NH_4Cl should be added, or else managanese will be precipitated in III group.

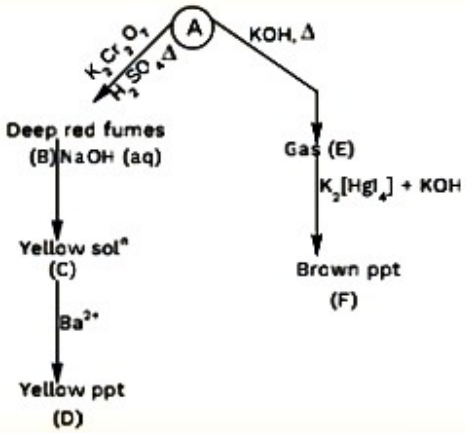
Cations → Fe^{3+}, Al^{3+}, Cr^{3+}
Group reagent → NH_3(aq) in presence of NH_4Cl


Filtrate of 2nd group
Add 1 or 2 drops of conc.HNO3
(to reoxidise Fe2+ to Fe3+)
Add NH4Cl solution & NH3 (aq)
Fe3+ + 3NH3 + 3H2O → Fe(OH)3 ↓ + 3NH4+
Filter the contents
Filtrate used to analyse 4th group
Reside: Fe(OH)3 → Reddish brown
Al(OH)3 → White gelatinous
Cr(OH)3 → Green
Add excess of NaOH & boil
Al(OH)3 + OH- → Al(OH)4- aluminate
Cr(OH)3 + OH- ⇌ Cr(OH)4- (boil) Chromite
Filter the contents.
Filtrate, Al(OH)4- —NH4Cl or CO2 (gas)→ Al(OH)3 ↓ white gelatinous
Residue:
Fe(OH)3, Cr(OH)3, Cr(OH)3 + OH- → Cr(OH)4-
Add excess of NaOH
Filter the content
Filtrate,
Cr (OH)4- —HCl→ Cr3+ —NH3(aq)→ Cr3+(OH)3
Green ppt
Fused with oxidising
(KNO3 + KOH / KClO3 + Na2CO3) fusion mixture
Yellow solution (CrO4 2-)
↓ Ba2+
BaCrO4 ↓ Yellow
Residue:
Fe(OH)3 —dil HCl or HNO3→ Fe3+ —KSCN→ Fe(SCN)3
deep blue solution
K4[Fe(CN)6] (excess)
Deep blue ppt (Prussian blue) Fe4[Fe(CN)6]3

Reaction of 3rd group cations

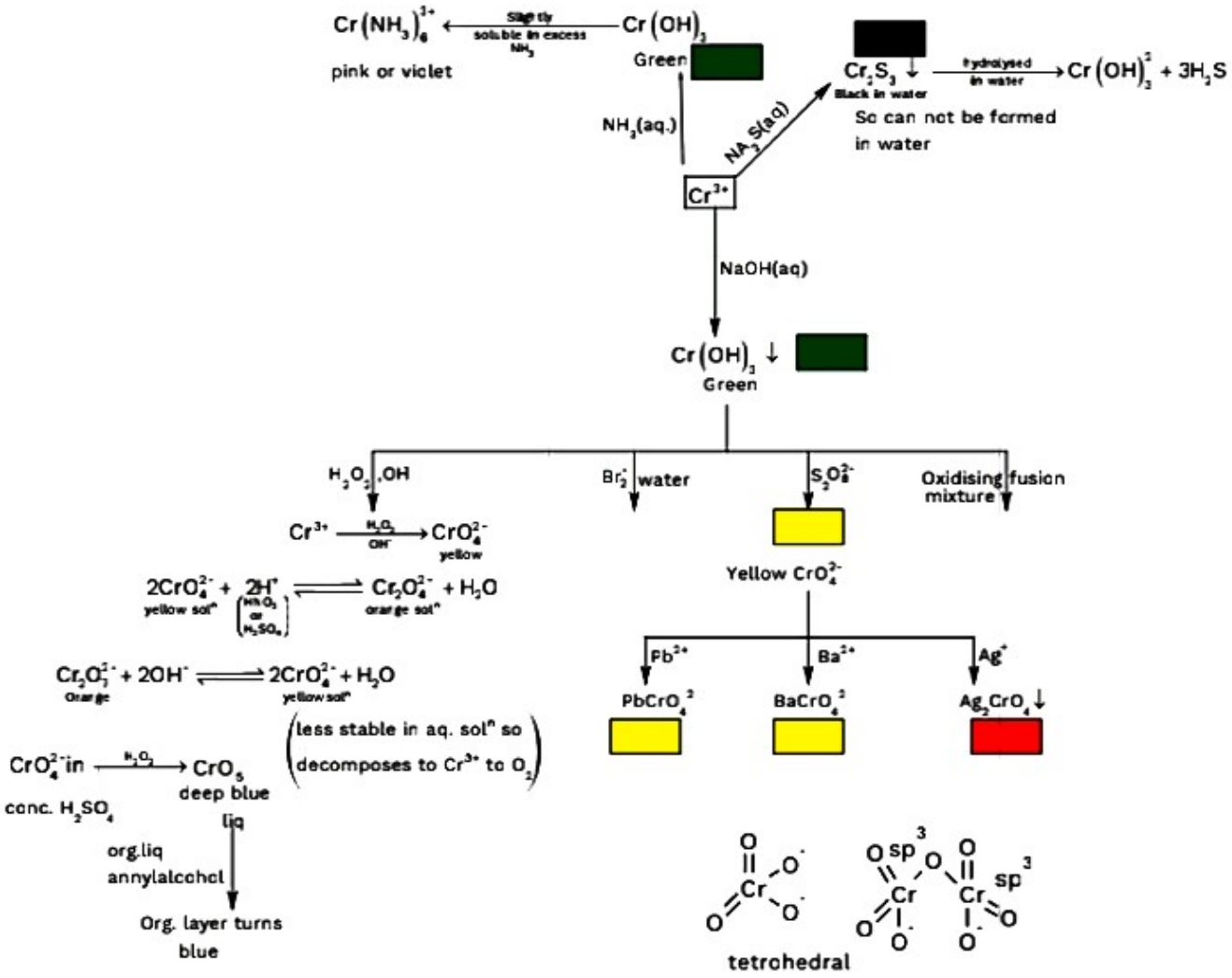
Ion	with	
$Al^{3+} \rightarrow$	1.Na_2S	$Al_2S_3 \downarrow \xrightarrow[\text{in water}]{\text{Hydrolysed}} \underset{\text{white}}{Al(OH)_3} \downarrow + 3H_2S \uparrow$ So in aq solution Al_2SO_3 cannot be formed
	2. $NH_3(aq)$	$NH_3(aq)$ $Al(OH)_3 \downarrow$ white gelatinous soluble in acids as well as excess NaOH
	3. NaOH(aq)	NaOH(aq) $Al(OH)_3 \downarrow \xrightarrow{\text{excess NaOH}} Al(OH)_4^-$ H^+(acid) → Al^{3+} NH_4OH → $Al(OH)_3$ white
	4. Na_2HPO_4	$AlPO_4$ white soluble in conc^n acid.
	5. CH_3COONa	In alkaline medium white ppt. of $Al(OH)_2(CH_3COO)$, basic aluminium acetate is formed.

Q39



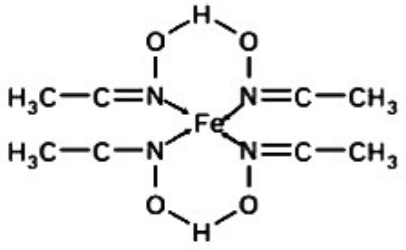
Sol: $(A) \rightarrow NH_4^+Cl^-$ $(B) \rightarrow CrO_2Cl_2$ $(C) \rightarrow Na_2CrO_2$

$(D) \rightarrow BaCrO_4$ $(E) \rightarrow NH_3$ $(F) HgO.Hg(NH_2)I$



Regent	Fe^{3+}	Fe^{2+}
1. H_2S gas	$Fe^{3+} + H_2S \rightleftharpoons Fe^{2+} + S\downarrow$	$Fe^{2+} + H_2S \rightleftharpoons FeS\downarrow + 2H^+$ (black) Alkaline medium
2. Na_2S or $(NH_4)_2S$	$Fe^{3+} + S^{2-} \rightarrow Fe_2S_3\downarrow$ black $\xrightarrow[\Delta]{O_2}, Fe(OH)_3$ Reddish brown	$Fe^{2+} + S^{2-} \rightarrow FeS\downarrow$ black $\xrightarrow[\Delta]{O_2} Fe_2O(SO_4)_2\downarrow$ Basic iron sulphate reddish brown

Regent	Fe^{3+}	Fe^{2+}
3. NH_3(aq)	$Fe(OH)_3 \downarrow$ reddish brown ppt. $\xrightarrow{\Delta} Fe_2O_3$ red oxide	In absence of air white ppt $Fe(OH)_2$. In presence of air it is converted to reddish brown $Fe(OH)_3$
4. NaOH	$Fe(OH)_3 \downarrow$ reddish brown	$Fe(OH)_2 \downarrow$ white
5. KCN(aq)	$Fe(CN)_3 \downarrow \xrightarrow[KCN]{excess} K_3[Fe(CN)_6]$ (Reddish brown) (ppt) yellow solution	$Fe(CN)_2 \downarrow \xrightarrow[KCN]{excess} K_3[Fe(CN)_6]$ (yellowish brown) (yellow solution)
6. $K_4[Fe(CN)_6]$ pot. Ferro cyanide or pot. Hexacyano ferrate (II)	$Fe^{3+} + Fe(CN)_6^{4-} \rightarrow Fe_4[Fe(CN)_6]$ (excess) deep blue ppt prussian blue HCl: $4Fe^{3+}$; $K^+ + Fe(CN)_6^{4-} + Fe^{3+}$ (excess) $\rightarrow KFe[Fe(CN)_6]$ Deep blue colloidal sol^n(soluble prussion blue) NaOH: $3Fe(CN)_6^{4-} + Fe(OH)_3\downarrow$ reddish brown	$Fe^{2+} + 2K^+ + Fe(CN)_6^{4-} \xrightarrow{\text{in absence of air}} K_2Fe[Fe(CN)_6]\downarrow$ white ppt In presence of air pale blue ppt. is formed.
7. $K_3[Fe(CN)_6]$ pot. hexacyano ferrate(III)	$Fe^{3+} + Fe(CN)_6^{3-} \rightarrow Fe[Fe(CN)_6]\downarrow$ Brown ppt.	$Fe^{2+} + Fe^{3+}(CN)_6^{3-}$ $\underset{\text{Redox}}{\rightleftharpoons} Fe^{3+} + Fe(CN)_6^{3-} \rightarrow Fe[Fe(CN)_6]\downarrow$ Prussian blue

Regent	Fe^{3+}	Fe^{2+}
8.KSCN or NH_4SCN	$Fe^{3+} + SCN^- \rightarrow Fe(SCN)_3$ $+Fe(SCN)_4^{-1} + [Fe(H_2O)(SCN)]^{2+}$ etc. Deep red (blood red) colour F^-; Hg^{2+} or oxdate ion bleach the colour due to the formation of colourless complexes. $Fe(SCN)_3 + 6F^- \rightarrow FeF_6^{3-} + 3SCN^-$ $Fe(SCN)_3 + Hg^{2+} \rightarrow Fe^{3+} + Hg(SCN)_2$ $+Hg(SCN)_4^{2-}$	No reaction
9. CH_3COO^- in alkaline medium	$[Fe_3(OH)_2(CH_3COO)_6]^+$ deep red solution $\xrightarrow{\Delta}$ $[Fe(OH)_2(CH_3COO)] \downarrow$ reddish brown ppt.	No reaction
10.NH_3(aq) dimethyl glyoxime	No reaction	$[Fe(dmg)_2]$ dsp^2 Red solution planar 

Analysis of 4th group

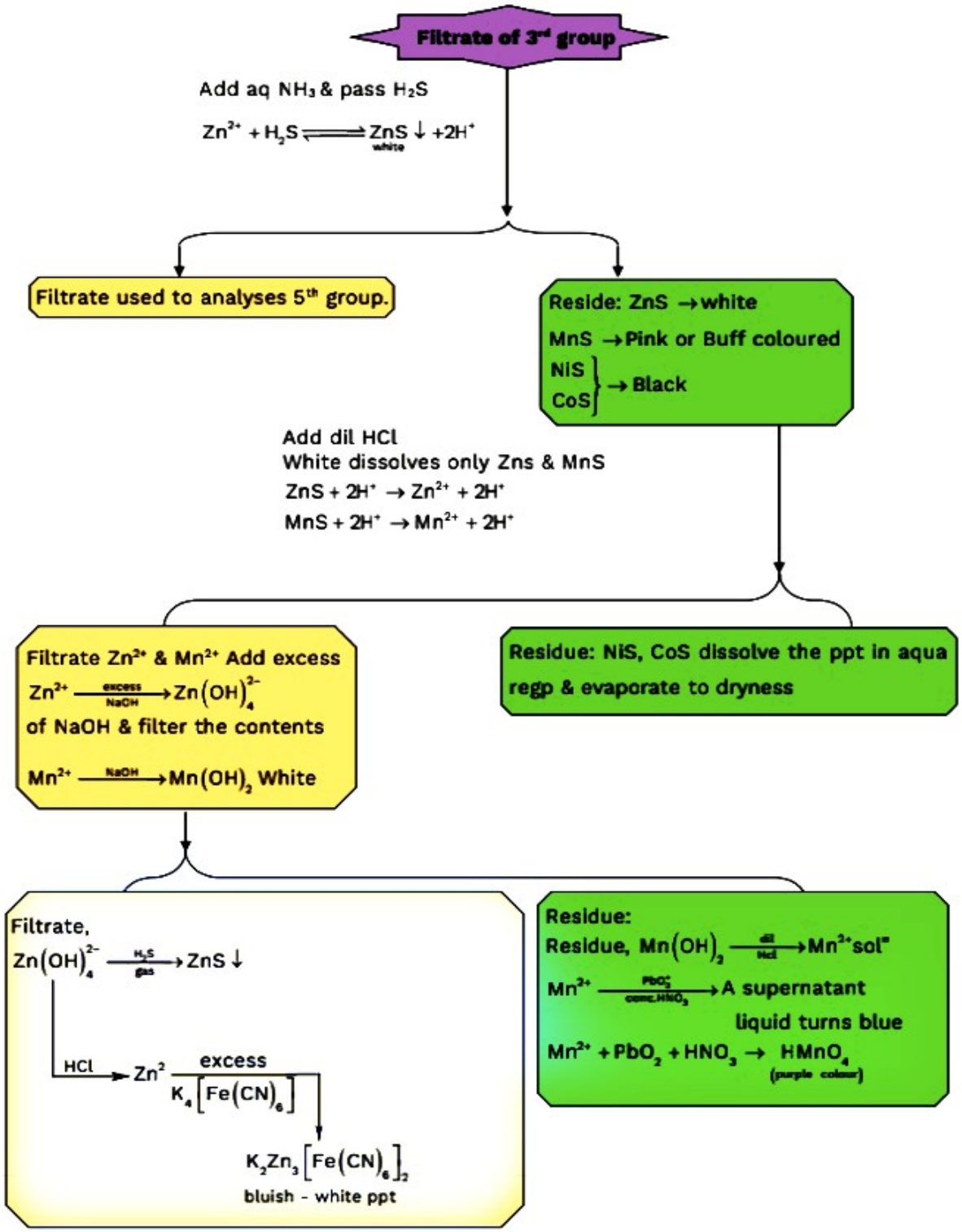
This group includes Co^{2+} Ni^{2+}, Zn^{2+} and Mn^{2+}. The group reagent is hydrogen sulphide in ammoniacal solution. Radicals are precipitated as sulphides which are insoluble in NH_4OH

CoS,NiS-Black MnS-Buff ZnS-White

The function of ammonium hydroxide is to increase the ionisation of H_2S

Thus, an excess of sulphide ions will be available and hence the ionic product of IV group sulphides exceeds their solubility product and precipitates will be obtained.

Cations $\rightarrow Zn^{2+}$, Mn^{2+}, Ni^{2+}, Co^{2+}


Filtrate of 3rd group
Add aq NH3 & pass H2S
Zn2+ + H2S ⇌ ZnS ↓ +2H+
white
Filtrate used to analyses 5th group.
Reside: ZnS → white
MnS → Pink or Buff coloured
NiS
CoS
→ Black
Add dil HCl
White dissolves only Zns & MnS
ZnS + 2H+ → Zn2+ + 2H+
MnS + 2H+ → Mn2+ + 2H+
Filtrate Zn2+ & Mn2+ Add excess
Zn2+ excess NaOH → Zn(OH)4 2-
of NaOH & filter the contents
Mn2+ NaOH → Mn(OH)2 White
Residue: NiS, CoS dissolve the ppt in aqua regp & evaporate to dryness
Filtrate,
Zn(OH)4 2- H2S gas → ZnS ↓
HCl → Zn2
excess
K4[Fe(CN)6]
K2Zn3[Fe(CN)6]2
bluish - white ppt
Residue:
Residue, Mn(OH)2 dil HCl → Mn2+sol"
Mn2+ PbO2 conc.HNO3 → A supernatant liquid turns blue
Mn2+ + PbO2 + HNO3 → HMnO4
(purple colour)

Residue: NiS, CoS dissolve the ppt in aqua regia & evaporate to dryness

Case – I

If dryness is yellow & turns to green on dilution with water $\Rightarrow Ni^{2+}$ is indicated.

$$sol^n \xrightarrow[NH_3(aq)]{dmg/p\&} \text{Rosy red ppt} \downarrow$$

$$Ni^{2+} + 2dmg + 2NH_3 \rightarrow [Ni(dmg)_2] \downarrow + 2NH_4^+$$

rosy red ppt.

→ dsp^2

→ sq. planer

→ intramolecular H-bonding

→ chelation

→ diamagnetic

Case – II

If dryness is blue turns to pink on dilution with water $\Rightarrow Co^{2+}$ is indicated

$$sol^n \xrightarrow[CH_3COOH]{KNO_2} \text{yellow ppt.}$$

$$Co^{2+} + NO_2^- + CH_3COOH + K^+ \rightarrow K_3[Co^{3+}(NO_2)_6] + NO$$

Yellow ppt

Reaction of 4th group cations

Zn^{2+}	1. H_2S & NH_3	$Zn^{2+} + H_2S \rightleftharpoons \underset{\text{white}}{ZnS} \downarrow 2H^+$ soluble in dil HCl or HNO_3
	2. $NH_3(aq)$	$Zn^{2+} + 2NH_3 + 2H_2O \rightleftharpoons \underset{\text{white}}{Zn(OH)_2} \downarrow + 2NH_4^+$ $Zn(OH)_2$ is soluble is acids NH_4Cl, excess of NH_2 & excess of NaOH $\xrightarrow{\text{excess NaOH}} \underset{sp^3}{Zn(OH)_4^{2-}}$ $\xrightarrow{\text{excess } NH_3} \underset{sp^3}{Zn(NH_3)_4^{2+}}$

	3. NaOH (aq)	$Zn(OH)_2 \downarrow$ (white) $\xrightarrow[NaOH]{excess} Zn(OH)_4^{2-}$ $\downarrow$ Hcl Zn^{2+}
	4. $K_4[Fe(CN)_6]$	A white ppt. of variable composition is formed in absence if air. In presence of air and in excess reagent a bluish white ppt. of $K_2Zn\ [Fe(CN)_6]_2$ is formed.
	5. $Hg(SCN)_4^{2-}$	$Zn[Hg(SCN)_4] \downarrow$ white ppt. In presence of small amount of Cu^{2+} ion a mixed ppt. of $\underbrace{Zn[Hg(SCN)_4] + Cu[Hg(SCN)_4]}_{\text{violet}}$
	6. Na_2HPO_4	$Zn_3(PO_4)_2 \downarrow$ white, soluble in excess NH_3 & excess NaOH $\xrightarrow{excess\ NaOH} Zn(NH_3)_4^{2+}$ $\xrightarrow{excess\ NH_3} Zn(OH)_4^{2-}$
	7. $HPO_4^{2-} + NH_4^+$	$Zn(NH_4)(PO_4) \downarrow$ white soluble in excess NH_3 & excess NaOH $\downarrow$ $\downarrow$ $Zn(NH_3)_4^{2+}$ $Zn(OH)_4^{2-}$
Mn^{2+}	1. $H_2S + NH_3(aq)$	$Mn^{2+} + H_2S \rightleftharpoons \underset{pink}{MnS} \downarrow + 2H^+$ This ppt is soluble in dil Hcl or HNO_3
	2. $NH_3(aq)$	$Mn^{2+} + 2NH_3 + 2H_2O \rightleftharpoons \underset{white}{Mn(OH)_2} \downarrow + 2NH_4^+$ $Mn(OH)_2$ is soluble in dil acids & NH_4Cl
	3. NaOH	$Mn(OH)_2 \downarrow$ white $\xrightarrow[or\ conc.HNO_3]{H_2O_2, OH^-} MnO(OH)_2$ or $MnO_2 . H_2O$ brown-black
	4. Oxidising fusion	$Mix(KNO_3 + KOH) \rightarrow K_2MnO_4$ greenness

	5.	$S_2O_8^{2-}$ or $\underset{\text{powerful oxiding agent}}{NaBiO_2 + H^+}$ $\underset{\text{pink colour}}{MnO_4^-}$
	6.	$\xrightarrow[\text{conc.}HNO_3]{PbO_2^*}$, $\underset{\text{purple colour}}{HMnO_4}$ We can be use Pb or in place of PbO_2 $(PbO + PbO_2)$ red lead

Reagent		**Ni^{2+}**	**Co^{2+}**
1.	H_2S + aq NH_3	NiS ↓ black soluble in aqua regia & in (conc HCl + $KClO_3$)	NiS ↓ black Soluble in aquaregia & in (Conc. HCl + $KClO_3$)
2.	NH_3(aq)	$\underset{\text{green}}{Ni(OH)_2}\downarrow \xrightarrow[NH_3]{\text{excess}} \underset{\text{deepblue}}{Ni(NH_3)_6^{2+}}$	$Co(OH)(NO_3)\downarrow \xrightarrow[\text{in } NH_3]{\text{soluble}} Co(NH_3)_6$ blue
3.	NaOH(aq)	$Ni(OH)_2\downarrow \xrightarrow[\Delta]{\text{in air}} Ni(OH)_3\downarrow$ Green black	$Co(OH)_2 \xrightarrow[\Delta]{\text{in air}} Co(OH)_3^2$ Pink black
4.	Dmg + NH_3(aq)	$[Ni(dmg)_2]\downarrow$ rosy red ppt. dsp^2 sq. planner	$[Co(dmg)_2]$ brown solution; dsp^2 sq. planner
5.	KSCN or NH_4SCN	No reaction	$Co(SCN)_4^{2-}$ blue colour ↓ Extracted in org layer (ammyl alcohol) $H_2[Co(SCN)_4]$ org layer turn blue
6.	KCN + Br_2, water	$\underset{\text{green}}{Ni(CN)_2}\downarrow \xrightarrow[KCN]{\text{excess}} \underset{\text{yellow sol}^n}{K_2[Ni(CN)_4]}$ dsp^2 ↓ Br_2 + water $Ni(OH)_2$ black	$\underset{\text{brown}}{CO(CN)_2}\downarrow \xrightarrow[KCN]{\text{excess}} \underset{\text{brown sol}^n}{K_4[Co(CN)_6]}$ Br_2 + water ↓ $\underset{\text{yellow sol}^n}{K_3[CO^{+3}(CN)_6]}$

Reagent		Ni^{2+}	Co^{2+}
7.	$NaHCO_3$ & Br_2 -water	$NiCO_3 \downarrow$ (green) $\xrightarrow{Br+H_2O} Ni(OH)_3 \downarrow$ (black)	$Co(HCO_3)_2 \xrightarrow[HCO_3^-]{\text{excess}} Na_4[Co^{+2}(CO_3)_3]$ $\downarrow Br_2 + \text{water}$ $Na_3[Co^{+3}(CO_3)]$ apple green solution

Note : Co^{2+} is stable in aq. solution but $3d^7$ in presence of complexing agent (i.e. strong ligangs) it is easily oxidised to CO^{3+}.

$$Co^{2+} \xrightarrow{NO_2^-} Co^{2+}(NO)_6^{4-}$$



$$Co^{2+} \xrightarrow{CN^-} Co(CN)_6^{4-}\ d^2sp$$

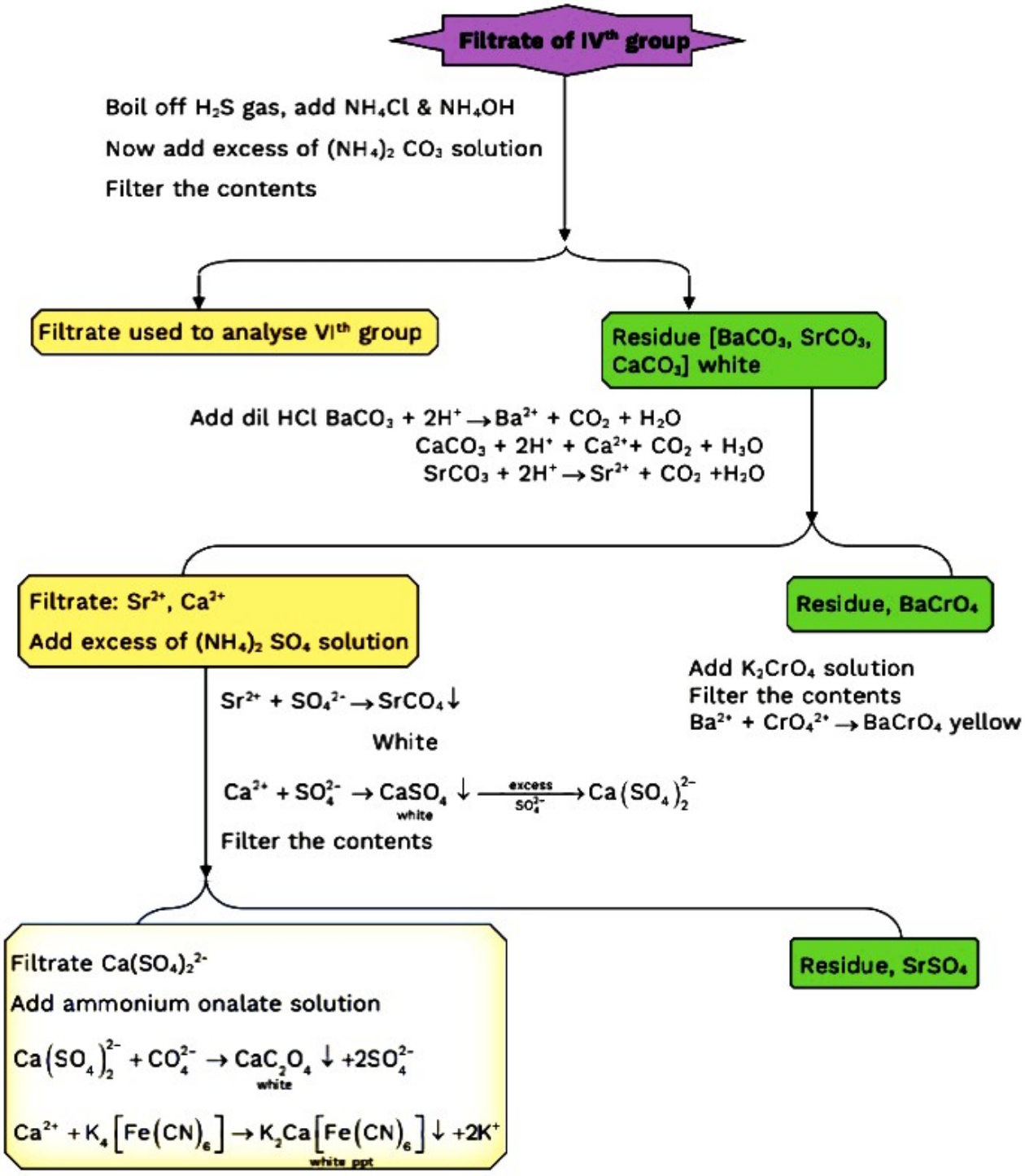
e.g $Co^{2+} \xrightarrow[CH_3COOH]{KNO_2} K_3[Co(NO_2)_6]\downarrow + NO$ (yellow ppt)

$$Co^{2+} \xrightarrow[CH_3COOH]{NaNO_2+NH_4Cl} (NH_4)_3[Co(NO_2)_6]\downarrow$$ (yellow ppt)

Analysis of Vth Group :

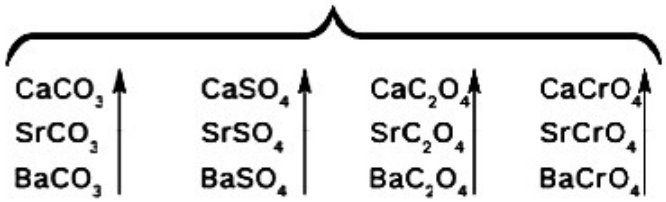
Cation $\rightarrow Ba^{2+}, Sr^{2+}, Ca^{2+}$

Group reagent $\rightarrow (NH_4)_2CO_3$ sol^n is presence of NH_4Cl & NH_4OH


Filtrate of IVth group
Boil off H_2S gas, add NH_4Cl & NH_4OH
Now add excess of $(NH_4)_2CO_3$ solution
Filter the contents
Filtrate used to analyse VIth group
Residue [$BaCO_3$, $SrCO_3$, $CaCO_3$] white
Add dil HCl $BaCO_3 + 2H^+ \rightarrow Ba^{2+} + CO_2 + H_2O$
$CaCO_3 + 2H^+ + Ca^{2+} + CO_2 + H_3O$
$SrCO_3 + 2H^+ \rightarrow Sr^{2+} + CO_2 + H_2O$
Filtrate: Sr^{2+}, Ca^{2+}
Add excess of $(NH_4)_2SO_4$ solution
Residue, $BaCrO_4$
Add K_2CrO_4 solution
Filter the contents
$Ba^{2+} + CrO_4^{2+} \rightarrow BaCrO_4$ yellow
$Sr^{2+} + SO_4^{2-} \rightarrow SrCO_4\downarrow$
White
$Ca^{2+} + SO_4^{2-} \rightarrow CaSO_4\downarrow \xrightarrow[SO_4^{2-}]{\text{excess}} Ca(SO_4)_2^{2-}$
white
Filter the contents
Filtrate $Ca(SO_4)_2^{2-}$
Add ammonium onalate solution
$Ca(SO_4)_2^{2-} + CO_4^{2-} \rightarrow CaC_2O_4\downarrow + 2SO_4^{2-}$
white
$Ca^{2+} + K_4[Fe(CN)_6] \rightarrow K_2Ca[Fe(CN)_6]\downarrow + 2K^+$
white ppt
Residue, $SrSO_4$

Solubility of oxo salts of alkaline earth metal cation decreases down the group.
Solubility increases in the direction of arrow.

Solubility increases in the direction of arrow.



Since, Solubility of axo salt of Ba^{2+} is least among Ba^{2+}, Sr^{2+}, & Ca^{2+} so first we have to precipitate out Ba^{2+}, then analyses Sr^{2+} & Ca^{2+}

Analysis of VIth group
Cation : $\rightarrow Mg^{2+}$

Group reagent $\rightarrow Na_4HPO_4 + NH_3(aq)$

Mg^{2+}	1. HCO_3^- :	No ppt in cold (because all HCO_3^- are soluble but white ppt. in hot $Mg^{2+} + 2HCO_3^- \xrightarrow{\Delta} \underset{\text{white}}{MgCO_3}\downarrow + H_2O + CO_3$
	2. NH_3(aq) :	$Mg^{2+} + 2NH_3 + 2H_2O \rightleftharpoons \underset{\text{white}}{Mg(OH)_2}\downarrow + 2NH_4^+$ $Mg(OH)_2$ is soluble in dil acids & NH_4Cl
	3. NaOH :	$\underset{\text{white}}{Mg(OH)_2}\downarrow$ soluble in acids
	4. Na_2COS_3 :	$2MgCO_3$, $Ng(OH)_2$, SH_2O white ppt. Soluble in dil Acids & in NH_4Cl
	5. Na_2HPO_4 + NH_3 (aq)	Heat & scratch the test tube $\rightarrow Mg(NH_4)PO_4\downarrow$ white
	6.	Titen yellow : Red ppt. red soly^n is obtained Ba^{2+} & CO^{2+} do not give ppt. but intensity the colour of above ppt.
	7.	Megneson-I $\rightarrow$ Blue adaired lake

Some important Reactions of Qualitative:
Analysis:

1. Reactions with KI :

$Pb^{2+} \xrightarrow{KI} \underset{\text{yellow}}{PbI_2} \underset{\text{dilution}}{\overset{\text{excess KI}}{\rightleftharpoons}} \underset{\text{colourless}}{PbI_4^{2-}}$

$Ag^{+} \xrightarrow{KI} \underset{\text{yellow}}{AgI} \downarrow$ soluble in KCN & HYPO

$Hg_2^{2+} \xrightarrow{KI} \underset{\text{green}}{Hg_2I_2} \downarrow \xrightarrow[KI]{\text{excess}} HgI_4^{2-} + Hg$

$Hg^{2+} \xrightarrow{KI} \underset{\text{red}}{Hg_2I_2} \downarrow \xrightarrow[KI]{\text{excess}} \underset{\text{colourless}}{HgI_4^{2-}}$

$Cu^{2+} \xrightarrow{KI} \underbrace{CuSI \downarrow + I_2}_{\text{brown}} \xrightarrow{\text{Hypo}} \underset{\text{white}}{CuI} \downarrow + I^{-}$

$Bi^{3+} \xrightarrow{KI} \underset{\text{black}}{BiI_3} \downarrow \underset{\text{dilute}}{\overset{\text{excess}}{\rightleftharpoons}} \underset{\text{Orange solu}}{K\{BiI_4\}}$

$\xrightarrow{H_2O,\,\Delta} \underset{\text{Orange ppt}}{BiO^{+}I^{-}}$

$Fe^{3+} + I^{-} \rightarrow Fe^{2+} + I_2$

$Sn^{4} + S^{-} \rightarrow Sn^{2+} + I_2$

$Pb^{4+} + \underset{\text{excess}}{I^{-}} \rightarrow \underset{\text{yellow}}{PbI_2} + I_2^{-}$

2. Reactions with $K_4[Fe(CN)_6]$

$Ag^{+} \xrightarrow{K_4[Fe(CN)_6]} \underset{\text{white ppt}}{Ag_4[Fe(CN)_6]} \downarrow$ soluble in KCN or hypo

$Cu^{2+} \xrightarrow{K_4[Fe(CN)_6]} Cu_2[Fe(CN)_6] \downarrow$ chocolate brown

$Cd^{2+} \xrightarrow{K_4[Fe(CN)_6]} Cd_2[Fe(CN)_6] \downarrow$ white or Bluish-white ppt.

$Fe^{2+} \xrightarrow{K_4[Fe(CN)_6]} \underset{\substack{\text{white ppt.in}\\ \text{absence of air}}}{K_2[Fe(CN)_6] \downarrow} \xrightarrow[\text{of air}]{\text{In presence}}$ blue ppt. is formed.

$\underset{\text{excess}}{Fe^{3+}} \xrightarrow{K_4[Fe(CN)_6]} Fe[Fe(CN)_6]_3 \downarrow$ deep-blue ppt prussian blue

$Fe^{3+} \xrightarrow[\text{excess}]{K_4[Fe(CN)_6]} K\ Fe[Fe(CN)_6]$ soluble prussian blue

$Zn^{2+} \xrightarrow{K_4[Fe(CN)_6]} K_2\ Zn_3[Fe(CN)_6]_2 \downarrow$ bluish white ppt.

$Ca^{2+} \xrightarrow[\text{excess}]{K_4[Fe(CN)_6]} K_3\ Ca[Fe(CN)_6] \downarrow$ white ppt.

3. Reactions with NaOH :

(A) Complexes are formed in excess NaOH :

$Pb^{2+} \xrightarrow[NaOH]{\text{excess}} Pb(OH)_4^{2-}$

$Al^{3+} \xrightarrow[NaOH]{\text{excess}} Al(OH)_4^{-}$

$Zn^{2+} \xrightarrow[NaOH]{\text{excess}} Zn(OH)_4^{2-}$

$Cr^{3+} \xrightarrow[NaOH]{\text{excess}} Cr(OH)_4^{-}$

$Sn^{2+} \xrightarrow[NaOH]{\text{excess}} Sn(OH)_4^{2-}$

(B) Precipitate is obtained

$Ag^{2+} \xrightarrow{NaOH} Ag_2O \downarrow$ brown

$Hg_2^{2+} \xrightarrow{NaOH} Hg_2O \downarrow$ black

$Hg^{2+} \xrightarrow{NaOH} HgO \downarrow$ yellow

$Cu^{2+} \xrightarrow{NaOH} \underset{\text{blue}}{Cu(OH)_2} \downarrow \xrightarrow{\Delta} \underset{\text{black}}{CuO} \downarrow + H_2O$

$Cd^{2+} \longrightarrow \underset{\text{blue}}{Bi(OH)_3} \downarrow \xrightarrow{H_2O,\Delta} \underset{\text{yellow}}{BiO^+}\ OH^- \downarrow$

$Bi^{3+} \xrightarrow{NaOH} \underset{\text{white}}{Bi(OH)_2} \downarrow \xrightarrow{H_2O,\Delta} \underset{\text{yellow}}{BiO^+}\ OH^+ \downarrow$

$Fe^{2+} \xrightarrow{NaOH} Fe(OH)_2$ white $\xrightarrow{air} \underset{\text{reddish brown}}{Fe(OH)_3} \downarrow$

$Fe^{3+} \xrightarrow{NaOH} Fe(OH)_3 \downarrow$

$Mn^{2+} \xrightarrow{NaOH} Mn(OH)_2 \downarrow$ white

$Ni^{2+} \xrightarrow{NaOH} Ni(OH)_3 \downarrow$ green

$Co^{2+} \xrightarrow{NaOH} Co(OH)_2 \downarrow$ pink

$Mg^{2+} \xrightarrow{NaOH} Mg(OH)_2 \downarrow$ white

4. Reactions with aqueous NH_3 solution

(A) Which form complexes with excess NH_3 :

$Ag^{+} \xrightarrow[NH_3]{excess} Ag(NH_3)_2^{+}$ sp

$Cu^{2+} \xrightarrow[NH_3]{excess} Cu(NH_3)_4^{2+}$ dsp^2 deepblue

$Cd^{2+} \xrightarrow[NH_3]{excess} Cd(NH_3)_4^{2+}$ sp^3

$Zn^{2+} \xrightarrow[NH_3]{excess} Zn(NH_3)_4^{2+}$ sp^3

$Co^{2+} \xrightarrow[NH_3]{excess} Co(NH_3)_6^{2+}$ sp^3d^2

$Ni^{2+} \xrightarrow[NH_3]{excess} Ni(NH_3)_6^{2+}$ sp^3d^2 deep-blue

(B) Which form ppt. with NH_3 :

$Pb^{2+} \xrightarrow[(aq)]{NH_3} Pb(OH)_2 \downarrow$ white

$Hg^{2+} \xrightarrow[(aq)]{NH_3} HgO.Hg(NH_3)(NO_3) \downarrow$ white

$Hg_2^{2+} \xrightarrow{NH_3} \underbrace{Hg + Hg(NH_3)(NO_3)}_{\text{Black}} \downarrow$

$Bi^{3+} \xrightarrow{NH_3} Bi(OH)_2(NO_3) \downarrow$ white

$Fe^{3+} \xrightarrow{NH_3} Fe(OH)_2 \downarrow$ reddish brown

$Cr^{3+} \xrightarrow{NH_3} \underset{\text{green}}{Cr(OH)_2} \downarrow \xrightarrow[\text{excess } NH_3]{\text{partially soluble in}} \underset{\text{pink or violet}}{Cr(NH_3)_6^{3+}}$

$Al^{3+} \xrightarrow{NH_3} \underset{\text{white gelatinoes}}{Al(OH)_3} \downarrow$

$Mn^{2+} \xrightarrow{NH_3} Mn(OH)_2 \downarrow$ white

$Mg^{2+} \longrightarrow Mg(OH)_2 \downarrow$ white

$Zn^{2+} \longrightarrow Zn(OH)_2 \downarrow$ white

} All are soluble in NH_4Cl

5. Reactions with KCN :

(A) Which form complexes with excess KCN :

$$Ag^{+} \xrightarrow{KCN} \underset{\text{white}}{AgCN\downarrow} \xrightarrow[KCN]{\text{excess}} Ag(CN)_2^{-}$$

$$Cu^{+} \xrightarrow{KCN} \underset{\text{yellow}}{Cu(CN)_2\downarrow} \xrightarrow[KCN]{\text{excess}} \underset{\text{white}}{CuCN\downarrow} \longrightarrow Cu(CN)_4^{3-} + (CN)_2\uparrow$$

$$Cd^{2+} \xrightarrow{KCN} \underset{\text{white}}{Cd(CN)_2\downarrow} \xrightarrow[KCN]{\text{excess}} K_2\left[Cd(CN)_4\right]$$

$$Fe^{2+} \xrightarrow{KCN} \underset{\text{yellow brown}}{Fe(CN)_2\downarrow} \xrightarrow[KCN]{\text{excess}} K_2\left[Fe(CN)_6\right]$$

$$Fe^{3+} \xrightarrow{KCN} \underset{\text{reddish brown}}{Fe(CN)_2\downarrow} \xrightarrow[KCN]{\text{excess}} \underset{\text{yellow sol}^{n}}{K_3\left[Fe(CN)_6\right]}$$

$$Zn^{2+} \xrightarrow{KCN} \underset{\text{white}}{Zn(CN)_2\downarrow} \xrightarrow[KCN]{\text{excess}} K_2\left[Zn(CN)_4\right]$$

$$Co^{2+} \xrightarrow{KCN} \underset{\text{brown}}{Co(CN)_2\downarrow} \xrightarrow[KCN]{\text{excess}} \underset{\text{brown sol}^{n}}{K_4\left[Co(CN)_6\right]}$$

$$Ni^{2+} \xrightarrow{KCN} \underset{\text{green}}{Ni(CN)_2\downarrow} \xrightarrow[KCN]{\text{excess}} \underset{\text{yellow sol}^{n}}{K_2\left[Ni(CN)_6\right]}$$

(B) Which do not form complexes:

$$Pb^{2+} \xrightarrow{KCN} Pb(CN)_2\downarrow \text{ white}$$

$$Hg^{2+} \xrightarrow{KCN} Mg(CN)_2\downarrow \text{Colourless}$$

$$Hg_2^{2+} \xrightarrow{KCN} \underset{\text{Black}}{Hg\downarrow} + Hg(CN)$$

6. Reactions with dmg H + NH_3 :

$$Ni^{2+} \longrightarrow \left[Ni(dmg)_2\right]\downarrow \text{rosy red ppt } dsp^3$$

$$Fe^{2+} \longrightarrow \left[Fe(dmg)_2\right]\downarrow \text{red colour, } dsp^2$$

$$Co^{2+} \longrightarrow \left[Co(dmg)_2\right]\downarrow \text{brown colour } dsp^2$$

$$Bi^{3+} \longrightarrow \left[Bi(dmg)_3\right]\downarrow \text{yellow ppt.}$$

7. Reactions in which Sulphides are formed:

$$Ag^{+} \xrightarrow{H_2S\,gas} Ag_2S\downarrow \text{black}$$

$$\left.\begin{aligned} Pb^{2+} &\xrightarrow{H_2S} PbS\downarrow \text{black} \\ Cu^{2+} &\xrightarrow{H_2S} CuS\downarrow \text{black} \\ Cd^{2+} &\xrightarrow{H_2S} CdS\downarrow \text{yellow} \\ Bi^{3+} &\xrightarrow{H_2S} Bi_2S_3\downarrow \text{Black} \end{aligned}\right\} \text{soluble in } 50\%\ \frac{w}{w}\ HNO_3$$

$Hg_2^{2+} \xrightarrow{H_2S} Hg + \underset{\text{black}}{HgS\downarrow}$

$Hg^{2+} \xrightarrow{H_2S} HgS\downarrow \text{black}$

} soluble in aquaregia

$As^{3+} \xrightarrow{H_2S} As_2S_3\downarrow \text{yellow}$

$Sb^{3+} \xrightarrow{H_2S} Sb_2S_3\downarrow \text{orange}$

$Sn^{2+} \xrightarrow{H_2S} SnS\downarrow \text{brown}$

$Sn^{2+} \xrightarrow{H_2S} SnS_2\downarrow \text{yellow}$

} are soluble in YAS

$Fe^{3+} \xrightarrow{S^{2-}} FeS\downarrow \text{black}$

$Zn^{2+} \xrightarrow{S^{2-}} ZnS\downarrow \text{white}$

$Mn^{2+} \xrightarrow{S^{2-}} MnS\downarrow \text{pink}$

} soluble in dil HCl or HNO_3

$Ni^{2+} \xrightarrow{H_2S \text{ or } S^{2-}} \underset{\text{Black}}{NiS\downarrow}$

$Co^{2+} \xrightarrow{H_2S \text{ or } S^{2-}} \underset{\text{Black}}{CoS\downarrow}$

} solubleinaqua regia

Q1 **$BaCl_2$ solution gives a white precipitate with a solution of a salt, which dissolves in dilute hydrochloric acid with the evolution of colourless, pungent smelling gas. The gas as well as the salt both are used as bleaching agent in the textile industries. The salt contains:**

(A) sulphite **(B) sulphide** **(C) acetate** **(D) carbonate**

Sol: **A**

$Ba^{2+} + SO_3^{2-} \longrightarrow BaSO_3 \downarrow$ (white)

$BaSO_3 + 2HCl \longrightarrow BaCl_2 + SO_2$(colourless pungent smelling gas) + H_2O

SO_3^{2-} and SO_2 both act as bleaching agent.

Q2 **Which of the following precipitate(s) does / do not dissolve in excess of ammonia solution ?**

(A) $Zn(OH)_2$ **(B) $Ni(OH)_2$**

(C) $Al(OH)_3$ **(D) (B) and (C) both**

Sol: **C**

(A) $Zn(OH)_2 + 4\ NH_3 \longrightarrow [Zn(NH_3)_4]^{2+}$ (colourless solution) + $2OH^-$

(B) $Ni(OH)_2 \downarrow + 6NH_3 \longrightarrow [Ni(NH_3)_6]^{2+}$ (deep blue solution) + $2OH^-$

(C) $Al(OH)_3 + NH_3 \longrightarrow$ No reaction.

Q3 **Chocolate brown precipitate is formed with :**

(A) Cu^{2+} ions and $[Fe(CN)_6]^{3-}$

(B) Cu^{2+} ions and $[Fe(CN)_6]^{4-}$

(C) Fe^{2+} ions and $[Fe(CN)_6]^{4-}$

(D) Fe 2+ ions and dimethylglyoxime

Sol: **B**

(A) $Cu_3[Fe(CN)_6]_2 \downarrow$ (green) ;

(B) $Cu_2[Fe(CN)_6] \downarrow$ (chocolate brown)

(C) $Fe_4[Fe(CN)_6]_3 \downarrow$ (Prussian blue) ;

(D) red solution of iron(II) dimethylglyoxime.

Q4 **Pink colour of acidified $KMnO_4$ is decolourised but there is no evolution of any gas. This may happen with the compound containing the following acid radical.**

(A) SO_3^{2-} **(B) NO_2^-**

(C) S_2^- **(D) All of these**

Sol: **D**

(A) $5SO_3^{2-} + 2MnO_4^- + 6H^+ \longrightarrow 2Mn^{2+} + 5SO_4^{2-} + 3H_2O$

(B) $2MnO_4^- + 5NO_2^- + 6H^+ \longrightarrow 2Mn^{2+} + 5NO_3^- + 3H_2O$

(C) $2MnO_4^- + H_2S + 6H^+ \longrightarrow 2Mn^{2+} + 5S \downarrow + 8H_2O$

Q5 **Which of the following gives a precipitate with $Pb(NO_3)_2$ but not with $Ba(NO_3)_2$?**

(A) Sodium chloride **(B) Sodium acetate**

(C) Sodium nitrate **(D) Disodium hydrogen phosphate**

Sol: **A**

(A) $Pb^{2+} + 2Cl^- \longrightarrow PbCl_2 \downarrow$ (white) ; $Ba^{2+} + 2Cl^- \longrightarrow BaCl_2$ (water soluble)

(B) $(CH_3COO)_2$ Pb and $(CH_3COO)_2Ba$ both are water soluble salts.

(C) Nitrates are mostly soluble in water

(D) $3Pb^{2+} + 2HPO_4^{2-} \longrightarrow Pb_3(PO_4)_2\downarrow$(white) + $2H^+$;$Ba^{2+}+HPO_4^{2-}\longrightarrow$BaH-$PO_4\downarrow$(white)

Q6 **Colour of cobalt chloride solution is :**

(A) pink **(B) black** **(C) colourless** **(D) green**

Sol: **A**

Anhydrous Co(II) salts are blue in colour while hydrated Co(II) salts are pink/ red.

Q7 **A red colouration or precipitate is not obtained when :**
(A) Fe^{3+} reacts with potassium thiocyanate
(B) Fe^{2+} reacts with dimethylglyoxime.
(C) Hg^{2+} reacts with potassium iodide.
(D) None

Sol: **D**
(A) $Fe^{3+} + 3SCN^- \longrightarrow Fe(SCN)_3$ (red solution)
(B) Red solution of iron(II) dimethylglyoxime.
(C) $Hg^{2-} + 2I^- \longrightarrow HgI_2 \downarrow$ (red).

Q8 **When H_2S gas is passed through an ammonical salt solution X, a slightly white precipitate is formed. The X can be :**
(A) a cobalt salt **(B) a lead salt**
(C) a zinc salt **(D) a silver salt**

Sol: **C**
$Zn^{2+} + H_2S \longrightarrow ZnS \downarrow$ (white) $+ 2H^+$

Q9 **Consider the following statement :**
S_1 : Cu^{2+} ions are reduced to Cu^+ by potassium iodide and potassium cyanide both, when taken in excess
S_2 : H_2S will precipitate the sulphide of all the metals from the solutions of chlorides of Cu, Zn and Cd if the solution is aqueous.
S_3 : The presence of magnesium is confirmed in qualitative analysis by the formation of a white crystal line precipitate of $MgNH_4PO_4$.
S4 : Calomel on reaction with potassium iodide gives red precipitate.
and arrange in the order of true /false.
(A) TTFF **(B) TFTF** **(C) TTTT** **(D) TTTF**

Sol: **D**
S_1, S_2 and S_3 are correct statements.
S_4 : $Hg^{2+} + 2I^- \longrightarrow Hg_2I_2 \downarrow$ (green)

Q10 **Statement - 1 : Addition of NH_4OH to an aqueous solution of $BaCl_2$ in presence of NH_4Cl (excess) precipitates $Ba(OH)_2$.**
Statement - 2 : $Ba(OH)_2$ is water soluble.
(A) Both Statement-1 and Statement-2 are true and Statement-2 is the correct explanation of Statement-1.
(B) Both Statement-1 and Statement-2 are true but Statement-2 is not correct explanation of Statement-1.
(C) Statement-1 is true but Statement-2 is false.
(D) Statement-1 is false but Statement-2 is true

Sol: **D**
Ba^{2+} ions does not give any precipitate with NH_4OH solution in excess of NH_4Cl because product formed, $Ba(OH)_2$ is soluble in water

Q11 **Statement - 1 : Sodium meta aluminate on boiling with ammonium chloride produces white gelatinous precipitate.**
Statement - 2 : Aluminium hydroxide is formed which is not soluble in water
(A) Both Statement-1 and Statement-2 are true and Statement-2 is the correct explanation of Statement-1.
(B) Both Statement-1 and Statement-2 are true but Statement-2 is not correct explanation of Statement-1.
(C) Statement-1 is true but Statement-2 is false.
(D) Statement-1 is false but Statement-2 is true

Sol: **A**
$$[Al(OH)_4]^- \xrightarrow[\text{boil}]{NH_4Cl} Al(OH)_3 \downarrow + OH$$

Q12 **Which of the following statement(s) is (are) incorrect?**
(A) Fe^{2+} ions give a dark blue precipitate with potassium hexacyanidoferrate (III) solution.
(B) Fe^{3+} ions give intense blue precipitate with potassium hexacyanoferrate (II) solution.
(C) Fe^{3+} ions give a brown colouration with potassium hexacyanoferrate (III) solution.
(D) Fe^{2+} ions give a deep red colouration with ammonium thiocyanate.

Sol: **D**

(A) $Fe^{2+} + [Fe(CN)_6]^{3-} \longrightarrow Fe^{3+} + [Fe(CN)_6]^{4-}$

$4Fe^{3+} + 3\,[Fe(CN)_6]^{4-} \longrightarrow Fe_4[Fe(CN)_6]_3$ (turnbull's blue)

(B) $4Fe^{3+} + 3[Fe(CN)_6]^{4-} \longrightarrow Fe_4[Fe(CN)_6]_3\downarrow$ (intense blue)

(C) $Fe^{3+} + [Fe(CN)_6]^{3-} \longrightarrow Fe[Fe(CN)_6]$ (brown colouration)

(D) $Fe^{3+} + 4SCN^- \longrightarrow Fe(SCN)_3$ (deep red colouration)

$Fe^{2+} + 4SCN^- \longrightarrow$ No reaction

Q13 **Which of the following pair (s) of ions would be expected to form precipitate when dilute solutions are mixed?**

(A) $NH_4^+, [Co(NO_2)_6]^{3-}$ **(B)** NH_4^+, CO_3^{2-}

(C) Fe^{3+}, OH^- **(D)** Ba^{2+}, SO_4^{2-}

Sol: **A,C,D**

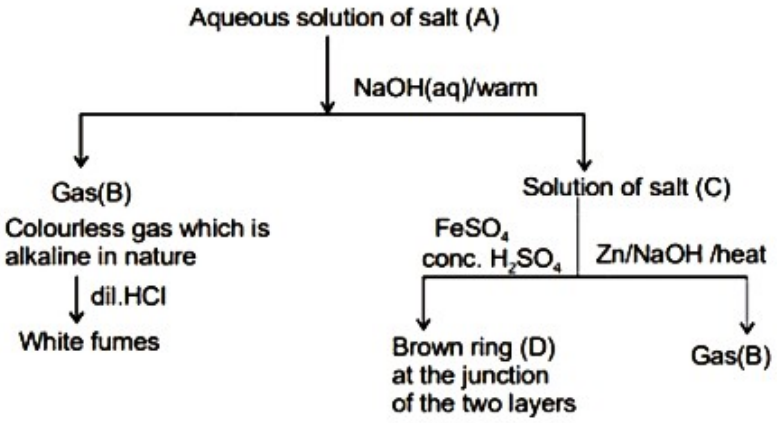
(A) $NH_4^+ + [Co(NO_2)_6]^{3-} \longrightarrow (NH_4)_3[Co(NO_2)_6]\downarrow$ (yellow)

(B) Ammonium and alkali metal carbonates are water soluble.

(C) $Fe^{3+} + OH^- \longrightarrow Fe(OH)_3\downarrow$ (reddish - brown)

(D) $Ba^{2+} + SO_4^{2-} \longrightarrow BaSO_4\downarrow$ (white)

Comprehension



Salt (A) on heating gives a colourless neutral gas which supports combustion. From the aforesaid, flow diagram, answer the following questions.

Q14 **The compound (A) contains the following acid radical.**

(A) NO_2^- **(B) NO_3^-** **(C) Br^-** **(D) SO_3^{2-}**

Sol: **B**

NO_3^- and NO_2^- both give brown ring test and reduction of NO_3^- and NO_2^- both give ammonia which with dilute HCl gives dense white fumes.

if $NH_4NO_3 \xrightarrow{\Delta} N_2O + 2H_2O$; N_2O supports the combustion

but $NH_4NO_2 \xrightarrow{\Delta} N_2 + 2H_2O$; Nitrogen does not supports combustion.

Hence, the anion is NO_3^-.

Q15 **The basic radical of salt (A) and gas B both gives brown precipitate with Nessleris reagent. The composition of the brown precipitate is :**

(A) $(NH_4)_2[HgI_4]$ **(B) $Hg(NH_2)NO_3$**

(C) HgO. Hg $(NH_2)I$ **(D) $(NH_4)_3[Co(NO_2)_6]$**

Sol: **C**

$NH_4^+ + 2[HgI_4]^{2-} + 4OH^- \longrightarrow$ HgO. Hg $(NH_2)I \downarrow + 7I^- + 3H_2O$

Hence the cation is NH_4^+.

Q16 **Which of the following statement is correct ?**

(A) Salt (A) gives yellow precipitate with chloroplatinic acid as well as with sodium cobaltinitrite.

(B) The brown ring is formed due to the formation of nitroso ferrous sulphate $[Fe(NO)]^{2+}SO_4^-$.

(C) Salt 'C' reacts with silver nitrate solution to form white precipitate.

(D) (A) and (B) both.

Sol: **D**

(A) $2NH_4^+ + [PtCl_6]^{4-} \longrightarrow (NH_4)_2[PtCl_6] \downarrow$ (yellow)

$3NH_4^+ + [Co(NO_2)]^{3-} \longrightarrow (NH_4)_3[Co(NO_2)_6] \downarrow$ (yellow)

(B) $2NO_3^- + 4H_2SO_4 + 6Fe^{2+} \longrightarrow 6Fe^{3+} + 2NO \uparrow + 4SO_4^{2-} + 4H_2O$

$SO_4^{2-} + Fe^{2+} + NO \longrightarrow [Fe(NO)]^{2+}SO_4^{2-}$

(C) $AgNO_3 + NaNO_3 \longrightarrow$ No reaction.

If the anion is NO_2^- then $Ag^+ + NO_2^- \longrightarrow AgNO_2 \downarrow$ (white)

Reactions :

$NH_4NO_3 + NaOH \longrightarrow NH_3\uparrow + NaNO_3$

(A) (B) (C)

$NH_3 + HCl \longrightarrow NH_4Cl\uparrow$ (White)

$NO_3^- + 4Zn + 7OH^- + 6H_2O \longrightarrow NH_3 + 4[Zn(OH)_4]^{2-}$